

INFORMAL SETTLER RELOCATION MANAGEMENT SYSTEM

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8.1 Abstract

The Informal Settler Relocation Management System (ISRMS) was developed as a GIS-integrated, web-based platform to improve the relocation management operations of the City Human Settlement and Urban Development Office (CHSUD) in Victorias City. The study addressed persistent challenges in informal settler relocation, including manual housing application processes, inefficient housing inventory tracking, limited transparency in beneficiary verification, and delays in housing allocation. The system was developed using the Rapid Application Development (RAD) methodology and integrated key features such as interactive GIS mapping for visualizing informal settlements and available housing units, a digital housing application and management module, real-time vacancy and allocation tracking, and a centralized housing inventory system.

The ISRMS was evaluated through white-box and black-box testing, which confirmed its functional reliability and operational stability. System quality assessment using the ISO/IEC 25010:2011 standard produced an overall mean score of 4.80 (Strongly Agree). Among the quality characteristics,



Functional Suitability and Performance Efficiency received the highest ratings (4.93), indicating that the system effectively met organizational and functional requirements. Security obtained the lowest mean score (4.60), although it still reflected a Strongly Agree interpretation, suggesting minor areas for enhancement. Usability evaluation based on the ISO 9241-11:2018 standard yielded an overall mean score of 4.57 (Strongly Agree). Satisfaction recorded the highest score (4.72), demonstrating strong user acceptance and ease of use, while Effectiveness received the lowest score (4.45), indicating opportunities for improving task execution efficiency. Overall, the results demonstrate that ISRMS significantly enhanced operational efficiency, data accuracy, and transparency in CHSUD, providing a scalable and sustainable solution for local government units facing similar urban housing challenges.

8.2 Rationale / Background of the Study

The relocation of informal settlers in the Philippines, particularly in urban areas like Victorias City, presented significant administrative and socio-economic challenges. The City Human Settlement and Urban Development Office (CHSUD) managed over 2,321 housing units and approximately 1,300 pending housing applications, relying heavily on manual, paper-based processes. This reliance led to delays, data duplication, lack of transparency, and inefficient tracking of housing allocations. The absence of a centralized digital system hindered effective urban planning and equitable housing distribution. In response, this study developed the Informal Settler Relocation Management System (ISRMS) to digitize and optimize CHSUD's operations. By integrating GIS mapping, automated workflows, and real-time monitoring, ISRMS aimed to enhance transparency, accuracy, and efficiency in managing informal settler relocation, supporting sustainable urban development and improved service delivery.

8.3 Summary

The study, entitled "Informal Settler Relocation Management System (ISRMS)," focused on designing, developing, and evaluating a GIS-integrated web-based platform to improve the relocation management processes of CHSUD in Victorias City. Using the Rapid Application Development (RAD) approach, the system was built to address key operational gaps, including manual application processing, inefficient tracking of housing units, and lack of spatial visualization. ISRMS featured an interactive GIS map for settlement and housing visualization, a digital housing application module, vacancy and allocation tracking, and a comprehensive housing inventory system. The system was rigorously tested



through white-box and black-box methods and evaluated against international standards for software quality (ISO/IEC 25010:2011) and usability (ISO 9241-11:2018).

The study aimed to answer the following questions:

1. What technical features are required to support CHSUD's informal settler relocation management needs?
2. How can the functionality of the developed system be validated through testing?
3. What is the level of system quality and usability based on ISO standards?
4. How can a user guide facilitate system adoption and effective use?

8.4 Findings

The development and implementation of the Informal Settler Relocation Management System (ISRMS) resulted in the following findings:

1. The ISRMS was successfully developed for CHSUD Victorias City with core technical features including an interactive GIS map, housing application module, vacancy tracking, and unit inventory management. These components effectively digitalized manual relocation processes, enhancing data clarity, accelerating decision-making, and ensuring real-time updates on housing status.
2. Comprehensive functionality testing using 94 black-box and white-box test cases confirmed the system's stability and reliability. All modules, including authentication, application submission, mapping, and database operations, passed validation without critical errors, ensuring the system met functional requirements for both applicants and administrators.
3. Quality evaluation based on ISO/IEC 25010:2011 and ISO 9241-11:2018 standards resulted in an overall system quality mean score of 4.80 and system usability mean score of 4.57, both rated "Strongly Agree." The system excelled in maintainability and performance efficiency, while users reported high satisfaction regarding effectiveness and ease of use.
4. A comprehensive User's Guide was completed to facilitate system adoption, providing detailed instructions for applicants, CHSUD personnel, and administrators. This documentation supports accurate system operation, aids in staff training, and ensures the long-term sustainability and maintainability of the ISRMS.



8.5 Conclusions

Based on the study's outcomes, the following conclusions were drawn:

1. The Informal Settler Relocation Management System (ISRMS) was successfully developed and implemented, effectively digitizing the relocation process of CHSUD Victorias City through integrated features such as GIS mapping, housing application management, vacancy tracking, and unit inventory management.
2. The ISRMS demonstrated high system stability and reliability, as confirmed by extensive functionality testing using black-box and white-box methods. All modules met the specified functional requirements without critical errors, ensuring dependable performance for both applicants and administrators.
3. The system achieved a high level of quality and usability, based on ISO 25010:2011 and ISO 9241-11:2018 standards evaluation results. With strong mean scores in system quality and usability, users expressed strong agreement regarding its effectiveness, performance efficiency, maintainability, and ease of use.
4. The completion of a detailed User's Manual ensures that the system can be effectively adopted and sustainably used by CHSUD staff and beneficiaries.

8.6 Recommendations

To enhance and sustain the effectiveness of ISRMS, the following recommendations are proposed:

1. Improve system speed by optimizing database queries, especially for modules handling large datasets such as housing inventory, GIS mapping, and reporting.
2. Strengthen access control mechanisms, implement more rigorous authentication procedures, and regularly review data security protocols to address the lower security rating identified in evaluations.
3. Keep the User's Manual updated to reflect system changes, improvements, and new processes, ensuring ongoing ease of use for all stakeholders.
4. Explore opportunities to integrate ISRMS with other local government systems and scale its adoption to other cities facing similar informal settler relocation challenge



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DEDICATION

This study aims to improve society and expand knowledge that encourages positive change. We hope its findings will lead to informed actions, responsible development, and real progress, both in Victorias City and beyond, as a small addition to global knowledge. The researchers would like to express deep gratitude to our adviser **Dr. Melanie T. Porquez**. Her kindness, humility, and steady guidance truly represent the spirit of mentorship. Her wisdom and support motivated us throughout this study.

This work is also dedicated to the researchers' faith and beliefs. They provided strength, direction, and perseverance in facing the challenges along this academic journey. Finally, this study is dedicated to ourselves for our resilience, discipline, and commitment to seeing this project through. This work reflects our growth, learning, and shared determination to make a meaningful contribution to society.

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CHAPTER I

INTRODUCTION

Moving informal settlers to other places was very hard in the Philippines, especially in cities. Many relocation efforts ignore the socio-economic requirements of the people involved, which leads to results that can't be sustained (Recio et. al, 2021).

In Baguio City, officials used geo-tagging and digital twin technologies to map homes in watersheds, fault lines, and other protected areas. This helped them make plans for housing and moving people (Zigzag Weekly, 2022). GIS helps cities and towns use spatial analytics to evaluate and carry out moves that make housing more efficient, bring people together, and encourage long-term urban growth (A. Muczyński et al., 2022).

Similarly, GIS was highlighted as a vital tool in reducing risk and supporting vulnerable communities during resettlement (P. Paudel et al., 2024). Due to the poor administration of informal settler data, Victorias City had trouble relocating people, duplicating beneficiary lists, and maintaining transparency in its operations. Without a centralized system, tracking families, managing housing units, and generating reports was slow and prone to errors. Local offices needed to update their operations (DHSUD, 2022). Digital technologies improved transparency, reduced government operational costs, and made it easier for individuals to access services (PIDS, 2023).



Previous research examined diverse relocation issues. Carrasco et al. (2022) found that relocated sites often lacked safety and accessibility, which forced residents to return to informal zones. Relocation could lead to gentrification and exclusion from services (L. Dai et al., 2022). Community profiling was conducted to achieve better outcomes. However, there was still a paucity of localized, GIS-based methods in the Philippines (Brown et al., 2022).

In this context, the project aimed to fill that gap through a GIS-integrated relocation management system tailored to Victorias City. The solution made planning easier through geotagging, digitized workflows, and simplified tracking of housing applications and inventory. This approach encouraged long-term urban growth, promoted transparency, and supported digital governance. It also served as a model for other cities addressing informal settler relocation challenges.

1.1 Background of the Study

The City Human Settlement and Urban Development Office (CHSUD) of Victorias City was responsible for addressing the city's housing and urban development challenges. The office played an important role in improving the quality of life of residents by implementing housing programs, promoting sustainable urban planning, and ensuring that the city's growth aligned with social and environmental goals. It managed urban expansion responsibly, supported urban development initiatives, and provided low-income families with access to affordable housing. One of the distinguishing features of CHSUD was its emphasis



on collaboration with national government agencies and private sector partners to develop long-term and sustainable urban solutions.

The primary functions of CHSUD included processing housing applications, urban planning, and ensuring that new housing developments met the demands of the city's growing population. As of April 2025, the city recorded a total of 2,321 housing units allocated for informal settler families. In addition, approximately 1,300 raw housing applicants had pending applications. The housing application process was largely manual. Applicants were required to complete physical forms and undergo income and residency verification to determine eligibility. In terms of urban planning, CHSUD was responsible for zoning areas for residential, commercial, and recreational development. These plans were developed using conventional procedures and relied on periodic evaluations, which often lacked access to real-time data. While these processes were functional, they were generally time-consuming and required modernization, particularly in leveraging technology to support decision-making and improve development management.

CHSUD faced several challenges that affected the efficiency of its operations. One major issue was the slow and cumbersome manual housing application system, which caused delays and inconvenience for both applicants and staff. The implementation of a digital housing application platform was identified as a solution to streamline the process by allowing applicants to submit information online, track application status in real time, and reduce the administrative workload of personnel.



1.2 Objectives of the Study

The main purpose of this study was to develop an Informal Settler Relocation Management System for the City Human Settlements and Urban Development Office to digitalize their manual processes and to make their work more efficient.

Specifically, this study aimed to:

1. Design and develop an Informal Settler Relocation Management

System with the following technical features:

1.1. Developed an interactive GIS map that shows informal settlements, available housing units, and relocation sites.

1.2. Developed a housing application and management module that allows applicants to check their application status and progress.

1.3. Created a vacancy and allocation tracking feature that helps assign housing units and update occupancy records.

1.4. Built a housing unit inventory management system that keeps records of all housing units.

2. Tested the functionality of the developed system.

3. Evaluated the quality of the system using the ISO 25010:2011 and ISO 92411:2018 Criteria.

4. Developed a User's Guide of the developed system.



1.3 Significance of the Study

The Informal Settler Relocation Management System was designed to be significant and beneficial to the following groups:

Informal Settlers / Housing Applicants. Informal settlers benefited from the system as it provided a more structured and accessible method for applying to housing programs, checking application status, and receiving real-time updates on the allocation process. As a result, applicants experienced improved access to affordable housing opportunities.

CHSUD Staff. The system helped CHSUD staff perform their tasks more efficiently by reducing the need for manual tracking of vacancies and applications. This allowed personnel to focus on more critical responsibilities and enhanced overall productivity. The system also improved communication between staff and applicants, which increased clarity and reduced errors in the housing allocation process.

City Human Settlement and Urban Development Office (CHSUD). The study improved the efficiency and transparency of housing management in Victorias City. Through the implementation of the Informal Settler Relocation Management System, CHSUD was able to streamline the housing application process, allocate housing units more effectively, and generate reports that assisted in decision-making and policy formulation.



Community Development. Overall, the study contributed to the long-term development of Victorias City by promoting more sustainable, efficient, and transparent housing management practices. These practices helped ensure that housing units were allocated to those who needed them most.

Researchers. The project provided researchers with valuable insights into the development and implementation of housing management systems and presented a practical case study on improving urban housing operations. It contributed to existing knowledge in urban development, information technology, and public administration and supported future research related to housing solutions for informal settlers.

Future Researchers. The findings of the study served as a useful reference for future researchers seeking to enhance systems related to informal settler profiling, relocation management, and community planning. The study offered insights into how structured digital processes, spatial analysis, and systematic documentation improved the accuracy and efficiency of relocation operations. Future researchers could build upon this work to develop more advanced, data-driven, and sustainable solutions aligned with social development goals and evidence-based decision-making in housing and urban development.



1.4 Scope and Limitation of the Study

This research focused on the development of an Informal Settler Relocation Management System with integrated GIS functionalities for the City Human Settlement and Urban Development Office of Victorias City. The primary objective was to facilitate the housing application and allocation process for informal settlers. The system enabled applicants to submit applications, track application status, and receive notifications or updates. For CHSUD personnel, the system supported the management of applicant records, monitoring of available housing units, tracking of vacancies and occupancy, assignment of housing units to qualified applicants, monitoring of post-ownership status, and generation of required reports.

The study included informal settler applicants from Victorias City and was conducted over a five-month period. It examined the system's functionality, ease of use for both applicants and CHSUD staff, and its capability to support urban planning through basic GIS integration. The scope covered the design and development of the system prototype and the evaluation of its efficiency, transparency, and accessibility within the institution.

Despite the objectives of the study, several factors limited the system's development and implementation. Resource constraints, such as limited budget and technology availability, restricted the full implementation of advanced features, particularly the more complex GIS functionalities initially planned. The five-month development timeline also posed a limitation, as it did not allow sufficient time for extensive testing, debugging, and system refinement prior to full deployment.



External factors beyond the researchers' control also affected the study. These included potential changes in local government regulations, challenges in obtaining accurate and up-to-date urban planning data, and technical difficulties related to data integration. Additionally, the project encountered issues related to user acceptance among informal settler applicants and CHSUD staff, who required adequate training to effectively use the system.

1.5 Definition of Terms

This section defines the essential ideas mentioned in the study. As a result, each term is defined operationally based on its usage in the study.

Certificate Generation. In this study, it referred to a system feature that automatically created and issued official relocation certificates for verified beneficiaries. It ensured that approved applicants received valid documentation of their relocation status for legal and administrative purposes.

Compatibility. In this study, it referred to the degree to which ISRMS could exchange information with other systems and perform its required functions while operating within the same hardware and software environment. It demonstrated the system's ability to co-exist with existing municipal systems through data export/import capabilities integration with external services and support for standard data formats for interoperability.

Effectiveness. In this study, it referred to the accuracy and completeness with which specified users achieved the primary goals of ISRMS within a specified context of use. It ensured that beneficiaries, CHSUD staff, and administrators



could successfully complete intended tasks such as submitting applications, verifying beneficiaries, assigning housing units, and generating certificates with minimal assistance and accurate data maintenance.

Efficiency. In this study, it referred to the resources expended in relation to the accuracy and completeness with which ISRMS users achieved specified goals. It optimized user effort by reducing processing time, preventing data entry errors through real-time validation, and enabling efficient workflows.

Functional Suitability. In this study, it referred to the degree to which ISRMS provided specified users with functions that supported task completion and achieved the objectives of the relocation management process. It ensured the system offered necessary features covering the full relocation lifecycle, including application submission, verification, housing assignment, utility coordination, property transfers, and certificate generation, with role-based functionality for administrators, staff, and beneficiaries.

GIS Mapping. In this study, it referred to a digital feature that allowed real-time visualization of informal settlements, relocation sites, and available housing units. It assisted planners and CHSUD staff in making location-based decisions for strategic urban development and housing allocation.

Housing Application and Management Module. In this study, it referred to a system component that enabled informal settlers to submit housing applications, monitor progress, and receive updates. It ensured a transparent, structured, and user-friendly application process.



Housing Unit Inventory Management. In this study, it referred to the system's capability to monitor and manage housing units categorized as available, occupied, or vacant. It helped CHSUD staff maintain accurate records and allocate housing more efficiently.

Informal Settler Relocation Management System (ISRMS). In this study, it referred to an integrated management system designed to streamline and modernize the relocation process of informal settlers. It combined application tracking, housing inventory management, GIS mapping, certificate generation, reporting, and analytics to enhance planning, transparency, and service delivery by CHSUD.

Maintainability. In this study, it referred to the ease and effectiveness with which ISRMS could be modified to correct defects, improve performance, or adapt to changing requirements.

Performance Efficiency. In this study, it referred to the degree to which ISRMS used time and computational resources appropriately relative to its level of use. It ensured acceptable response times for submissions, updates, and map queries.

Portability. In this study, it referred to the degree to which ISRMS could be transferred from one hardware, software, or operational environment to another. It supported deployment across different environments.

Reliability. In this study, it referred to the degree to which ISRMS performed specified functions under specified conditions for a specified period without failure.



Report Generation. In this study, it referred to the automated creation of reports using system-collected data, including application status summaries, relocation progress tracking, and housing allocation records, to support monitoring and planning by CHSUD staff.

Satisfaction. In this study, it referred to the comfort, acceptability, and positive perception experienced by users when interacting with ISRMS. It was enhanced through intuitive design, transparent processes, system responsiveness, and trust in data accuracy and security.

Security. In this study, it referred to the degree to which ISRMS protected information and ensured appropriate levels of access based on user roles. It safeguarded sensitive beneficiary data through multi-factor authentication, role-based access control, data validation, parameterized queries, CSRF protection, and activity logging.

Usability. In this study, it referred to the degree to which ISRMS could be used by specified users to achieve specified goals effectively, efficiently, and with satisfaction in a defined context of use.

User Experience. In this study, it referred to the overall perception, satisfaction, and ease of interaction experienced by users while navigating the system, including layout clarity, responsiveness, and functional coherence.

Vacancy and Allocation Tracking. In this study, it referred to a feature that maintained real-time records of available housing units and ensured the timely and accurate assignment of homes to qualified beneficiaries.



CHAPTER II

REVIEW OF RELATED LITERATURE/STUDIES

We often found new ideas, tools, and methods that helped us tackle challenges in our community. There were already many procedures and approaches in place to help make the process of moving informal settlers more organized and fairer. These tools included technologies, procedures, and tactics that helped with data management, tracking the progress of the transfer, and ensuring that people were treated well during the transition.

Researchers and planners studied how to improve relocation efforts, especially for people living in danger zones or on government-owned lands. Their purpose was to find better places to live for people who were relocating, while simultaneously protecting their rights. Using technology made it easier to keep track of records, check for updates, and ensure that the process was legal and ethical.

This chapter discussed several related studies and literature, both from the Philippines and other countries, that focused on relocation systems, housing programs, and community planning. These studies helped us understand how an Informal Settler Relocation Management System could be designed and used effectively.



2.1 Foreign Related Literature

Supporting tenant relocation in social housing management: Development of the dedicated geoinformation system in Poland

The study focused on addressing mismatches between tenants and available housing units in Poland through the development of a geoinformation system that is devoted. The goal was to create a GIS-based platform that combines spatial data and tenant information so that authorities may find incompatibilities, test relocation possibilities, and make choices based on facts. The researchers used business modeling, Unified Modeling Language (UML), and a geographic database powered by ArcGIS to put the system into action and test it in real life. The results showed that the system worked well for mismatch evaluation, relocation option visualization, and scenario simulation. This made social housing management much more coordinated, efficient, and open (R. B. Recio et. al., 2021).

Land Suitability Analysis Using GIS and MCDA for Settlement Relocation in Nepal

The research integrated Geographic Information Systems (GIS) and Multi-Criteria Decision Analysis (MCDA) to determine appropriate relocation sites for disaster-prone communities in Nepal. The method helped find safer places that lower risk and promote community health by looking at environmental, physical and social-economic factors (Paudel et al., 2024).



Rethinking urban relocation in the Global South

The study looked at how moving people around in quickly growing cities in the Global South affects their social and economic lives. It looked at gentrification, displacement, and unequal access to city resources. It looked at how moving projects affect neighborhoods that are already at a disadvantage and what they represent for the rights of people living in cities (L. Dai et al., 2022).

Community Profiling for Inclusive Urban Development

The study stressed how important it is to profile communities in order to make sure that relocation strategies are effective and include everyone. Their research concentrated on illustrating how comprehensive socio-economic data enhances planning results for displaced groups. The goals were to explain why socio-economic profile is important before moving and to help planners learn about the area's dynamics, weaknesses, and resource needs. The researchers looked at the traits of the afflicted groups by using community surveys, socio-economic assessments, and spatial mapping methods. The results show that relocation programs work better when planners know a lot about the specific needs of each community. This lowers the danger of misusing resources and The findings emphasize the importance of thorough community profiling, particularly in the Philippine context, where relocation programs are often planned using standardized approaches without enough supporting data (Brown et al., 2022).



2.2 Local Related Literature

Resettlement and the challenges of sustainable housing

The research published in the International Journal of Disaster Resilience in the Built Environment examined long-standing challenges faced by informal settlers relocated to new locations that fail to satisfy environmental and community requirements. It looked at how people are resettled around the world to figure out why many housing initiatives make people more open to harm than less so. The researchers found that relocation programs and the rules that go along with them often have problems that don't go away, such as sites that are bad for the environment, not enough community involvement, and a feeling alone in society (Carrasco et al., 2022).

Expropriation and the Margins: Urban Displacement in Manila

The research investigated the impact of urban development pressures in Metro Manila on the relocation of informal settlers to periphery regions characterized by inadequate services. It looked at the structural reasons for relocation and highlighted how Even though cities are growing swiftly, spatial inequity is still a problem. The results, obtained through qualitative research, policy analysis, and geographical mapping, demonstrated that non-inclusive planning and land expropriation remove disadvantaged individuals from jobs and important important things to do. Even if cities are growing quickly, this is still true. The results, which came from qualitative research, policy analysis, and geographical mapping, showed that planning that doesn't include everyone and taking property away from people makes it harder for marginalized groups (Ortega et al., 2020).



GIS-Driven Relocation Planning for Informal Settlers in Protected Areas

The project concentrated on formulating spatial methodologies to evaluate land suitability and devise equitable relocation strategies in ecologically sensitive regions. The major purpose was to use GIS techniques to look at environmental limits, community requirements, and legal limits to help with responsible settlement planning. The researchers produced relocation plans that fit with environmental protection policies by the researchers looked at government land in many ways, such as spatial analysis, suitability modeling, and classifications. The results showed that GIS provides reliable standards for finding the right places while minimizing harm to the environment (Garson et al., 2022).

Displacement and Resettlement Experiences of Informal Settlers in Metro Manila

This study examined the experiences of informal settlers in Metro Manila who were evacuated and then resettled via government housing initiatives. It intended to comprehend community reactions and sentiments around relocation initiatives, focusing on housing sufficiency, service accessibility, livelihoods, and site management. Researchers used interviews, questionnaires, and field observations to record the problems that Families that moved had problems like bad housing, not enough resources, and weak site governance. Results showed that communities have a hard time adjusting and keeping stable jobs after moving if they don't plan ahead and get help thereafter. The study stresses that relocation programs need to offer social support, easy-to-use services, and active community involvement to avoid failure and secondary displacement (Maningo, 2021).



GIS-Based Multi-Criteria Decision Analysis for Sanitation-Aware Relocation Planning

The study focuses on formulating a holistic, health-oriented strategy for determining relocation sites for informal settler families. The purpose was to include In a GIS-MCDA framework, we think about sanitation, environmental health threats, and what the community needs. Researchers used The study utilized spatial data, health metrics, and weighted criterion modeling to assess probable relocation sites. The results showed that adding sanitary standards made the site much more suitable and less likely to expose the population to health risks. The results show that GIS-based MCDA is a useful tool for planning based on data. It makes sure that moved communities get not only homes but also safe and healthy places to live (Pangilinan et al., 2024).

2.3 Foreign Related Systems

Mobile GIS Occupancy Audit for Housing Allocation

Hlalethu and Sithole (2021) implemented a mobile GIS system for auditing informal settlements in Ekurhuleni, South Africa, capturing real-time GPS, socio-economic data, and photos to qualify 56.86% of households for subsidized housing relocation. The web-linked platform automated verification and planning, boosting accuracy and integration like ISRMS's interactive mapping and allocation modules (Hlalethu & Sithole, 2021).



Geospatial Data Framework for Slum Upgrading

This systematic review identified GIS needs for informal settlement interventions globally, categorizing physical socio-economic boundary data collection via drones, satellites, mobile apps for relocation and monitoring. It stressed web-based platforms for comprehensive, up-to-date info to improve efficiency and equity. (Dovey et al., 2022).

GIS Slum Mapping and Tenability in Bhubaneswar

SPARC (2023) deployed a GIS geodatabase for 689 slums in Bhubaneswar, India, using high-res imagery and ground surveys to assess tenability, prioritize relocation, and plan slum-free habitats with vacancy/inventory tracking. The system enhanced transparency for urban poor management. (SPARC, 2023).

Digital Participatory Mapping in Slums

Da Silva et al. (2024) reviewed web-based participatory GIS mapping in global informal settlements (e.g., Caracas slums), that enabled resident-input spatial data for housing verification and relocation planning. It improved data accuracy over manual methods, with high user satisfaction, echoing ISRMS's ISO 9241 usability scores (da Silva et al., 2024).

GIS Modeling for Human Settlement Suitability

Wang et al. (2022) created a GIS-based model using RDLS, THI, LSWAI, and LCI indices for suitability assessment along China's Belt and Road, applicable to relocation by overlaying layers for optimal housing sites. The system supports real-time decision-making and scalability (Wang et al., 2022).



2.4 Local Related Systems

GIS-Based Multi-Criteria Site Selection for Informal Settlements

Raflores and Regmi (2024) developed a localized GIS-based multi-decision criteria approach for identifying suitable sites for informal settlement relocation in Davao City, Philippines, using AHP, slope analysis, distance buffers to hazards/amenities, and land cover data to prioritize safe housing areas. This supported ISRMS by providing spatial decision-making tools for housing allocation, reducing risks in urban planning and enabling efficient beneficiary placement amid growing slum populations (2.32% nationally per FIES 2018 data updated in study). The method integrates remote sensing for accurate mapping, enhancing transparency and data-driven relocation like your system's vacancy tracking (Raflores & Regmi, 2024).

Human settlements and informal settler families in Davao City: Implications for Sustainable Urban Development

This published study evaluates human settlements and informal settler families in Davao City, advocating an approved Local Shelter Plan (LSP) with GIS-enabled socialized housing programs and resettlement strategies to address inefficiencies in manual processes. It highlights the need for digital inventory and monitoring systems to manage applications and allocations, directly paralleling ISRMS's RAD methodology and ISO-evaluated quality for operational efficiency (Cuesta et al., 2025).



Rethinking Housing Provision for ISFs in Metro Cebu

Aota (2021) proposed a community-centered GIS-integrated approach for local governments in Metro Cebu to co-produce housing for informal settler families (ISFs), involving PPPs, slum upgrading, and spatial planning for equitable relocation. The framework digitizes workflows to replace paper-based tracking, improving transparency and sustainability, much like ISRMS's features for CHSUD operations (Aota, 2021).

New Urban Agenda Housing Implementation

The Philippine National Report details GIS-supported national programs for informal settler relocation under the New Urban Agenda, including digital platforms for housing inventory, beneficiary verification, and urban development planning across LGUs. It emphasized scalable web systems for real-time data to tackle delays and duplication, aligning with ISRMS's evaluation via ISO 25010:9241 standards (UN-Habitat Philippines, 2025).

Makati City Shelter Plan with GIS

Makati's Shelter Plan 2023 incorporated GIS mapping for informal settler identification, housing unit tracking, and relocation management, featuring digital applications and vacancy monitoring to enhance service delivery. This local system demonstrated high usability and efficiency in LGU contexts, supporting ISRMS's focus on Victorias City's CHSUD challenges (Cuesta et al., 2023).



2.5 Synthesis

Table 1
 Synthesis of Related Literatures

Related Literature	Features and Functionalities	Developed System
Supporting tenant relocation in social housing management	Spatial visualization; tenant database; automated matching; scheduling; feedback tracking	ISRMS has centralized beneficiary and housing records, relocation tracking, and dashboards, but lacks automated matching and feedback/dispute modules.
Land suitability analysis using GIS and MCDA for settlement relocation	MCDA; hazard and risk analysis; land suitability ranking; layered GIS data	ISRMS supports basic map visualization (Mapbox) only; no MCDA, risk analysis, or suitability ranking.
Resettlement, gentrification, and the right to the city	Community participation; policy analysis; social equity assessment	ISRMS manages applications and beneficiary data but does not support policy analysis or impact assessment.
Community profiling for inclusive urban development	Socio-economic profiling; needs mapping; stakeholder engagement	ISRMS collects basic beneficiary data but lacks advanced profiling, needs mapping, and engagement tools.
Socio-political dimensions of informal settler relocation: A GIS perspective	Socio-political analysis; participatory planning; GIS-based profiling	ISRMS handles beneficiary records and assignments only; no socio-political or participatory analysis.
Resettlement and challenges of sustainable housing	Post-relocation monitoring; service access; livelihood recovery	ISRMS tracks relocation and occupancy but does not monitor integration, services, or livelihoods.
Expropriation and the margins: Urban displacement in Manila	Legal frameworks; tenure conflicts; urban inequality mapping	ISRMS manages property and assignment records but lacks legal conflict or inequality mapping.
GIS-driven relocation planning for informal settlers in protected areas	Environmental zoning; buffer enforcement; site evaluation	ISRMS provides map-based visualization but no environmental zoning or site evaluation tools.
Examining urban poor voices in Metro Manila relocation	Community feedback; grievance mechanisms; impact evaluation	ISRMS does not include feedback, grievance, or impact assessment modules.
GIS-based MCDA for sanitation-aware relocation planning	Spatial analysis; health/sanitation risk; site ranking	ISRMS offers basic GIS viewing only; no health risk analysis or site ranking.



The Informal Settler Relocation Management System (ISRMS) represented a significant technological advancement aimed at improving the efficiency, accuracy, and quality of relocation services for informal settler communities. Based on a comprehensive review of recent studies and implementations, it was evident that systems like ISRMS addressed key operational challenges by reducing processing errors, standardizing relocation workflows, simplifying administrative tasks, and enhancing service delivery outcomes for local government units and affected households. By integrating organized data management, transparent monitoring, and streamlined coordination, ISRMS supported more equitable, timely, and sustainable relocation processes. In summary, the Informal Settler Relocation Management System provided a multifaceted solution for agencies and local governments seeking to improve operational efficiency, community engagement, data integrity, and overall service quality in the relocation process. The adoption of such systems served as a cornerstone for modernizing relocation and resettlement management, offering scalable, secure, and user-centric tools that aligned with the evolving demands of urban planning and disaster-risk reduction. Continued advancement and integration of ISRMS functionalities promised further improvements in relocation services, ultimately benefiting both implementers and informal settler families through streamlined processes, evidence-based decision-making, and enhanced transparency throughout the relocation cycle.



Table 2
 Synthesis of Related Systems

Related Literature	Features and Functionalities	Developed System
Mobile GIS Occupancy Audit for Housing Allocation	Mobile GIS; real-time GPS capture; socio-economic data and photo validation; automated verification; web-based planning	ISRMS supports mobile GIS capture; photo/GPS-linked occupancy and application updates
Geospatial Data Framework for Slum Upgrading	GIS data standards; drone and satellite imagery; mobile data collection; web-based monitoring for relocation planning	ISRMS supports standardized spatial layers and imagery for web-based monitoring
GIS Slum Mapping and Tenability in Bhubaneswar	GIS geodatabase; high-resolution imagery; ground surveys; tenability assessment; housing inventory and relocation prioritization	ISRMS supports imagery import, tenability scoring, and relocation prioritization
Digital Participatory Mapping in Slums	Participatory GIS; resident-generated spatial data; web-based mapping; housing verification; usability-focused design	ISRMS supports resident submitted spatial data linked to beneficiary records
GIS Modeling for Human Settlement Suitability	GIS suitability modeling; multi-index spatial analysis; layer overlay; decision-support for site selection	ISRMS does not include multi-index suitability and layer-overlay site analysis
GIS-Based Multi-Criteria Site Selection for Informal Settlements	GIS-MCDA; AHP; slope and hazard buffer analysis; land-cover mapping; site prioritization	ISRMS does not include MCDA/AHP-based site ranking using spatial criteria
Human Settlements and Informal Settler Families in Davao City	GIS-enabled local shelter planning; digital housing inventory; beneficiary monitoring and allocation management	ISRMS supports digital housing inventory, workflows, and monitoring dashboards
Rethinking Housing Provision for ISFs in Metro Cebu	Community-centered GIS planning; public-private partnerships; slum upgrading; digitized relocation workflows	ISRMS supports digitized relocation workflows and applicant tracking
New Urban Agenda Housing Implementation	National GIS platforms; housing inventory systems; beneficiary verification; real-time LGU data integration	ISRMS supports LGU data integration and beneficiary verification
Makati City Shelter Plan with GIS	GIS-based ISF identification; housing unit tracking; digital applications; vacancy and relocation monitoring	ISRMS supports unit-level tracking, vacancy monitoring, and digital applications



The Informal Settler Relocation Management System (ISRMS) was a valuable technology designed to improve the accuracy and efficiency of relocation services for informal settler groups. Studies indicated that systems like ISRMS could reduce processing errors, standardize workflows, and simplify administrative tasks. These improvements enhanced service delivery for local government agencies and affected households. ISRMS contributed to making relocation procedures fairer, faster, and more sustainable by organizing data management, increasing transparency in monitoring, and facilitating collaboration. Overall, ISRMS helped local governments modernize their approaches to urban relocation and settlement. Its implementation led to increased operational efficiency, improved data integrity, and better decision-making in urban planning and disaster risk reduction. As the system evolved, it was expected to become more transparent and to further improve service quality, benefiting both migrants and informal settler families. Additionally, ISRMS played a crucial role in planning, coordinating, and executing relocation efforts for informal settlements. Research demonstrated that such systems effectively addressed complex urban migration challenges by integrating spatial analysis, community data collection, and decision-support features. Key functions included identifying high-density or high-risk areas, analyzing land use patterns, and assisting residents in planning their relocations. These capabilities contributed to more organized and equitable relocation processes, ultimately supporting sustainable urban development and disaster preparedness.



CHAPTER III

METHODOLOGY

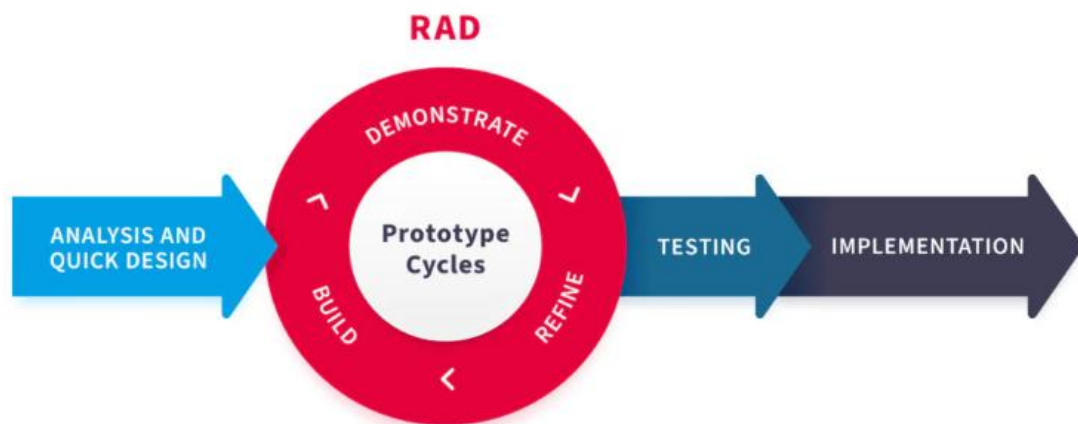
This chapter discussed the tools and methods used to build and test the study's system in a systematic fashion. This study utilized the Systems Development Life Cycle (SDLC) paradigm, a systematic approach for the planning, analysis, design, construction, testing, and implementation of the system. The chapter talked about the project's operational framework and shows important modeling tools, such as Data Flow Diagrams (DFD) with context and multi-level explosions to show how systems work, Entity Relationship Diagrams (ERD) to define data structure, and Unified Modeling Language (UML) Use Case Diagrams and scenarios to show how users interacted with the system.

A full Data Dictionary explained all the terminology and data items used in the document. The chapter also lists the hardware and software needed for implementation, gives a timeframe for the project, and talks about ways to test the system, like White-Box and Black-Box testing and the System Usability Instrument, to make sure it works and is easy to use. The section concluded by identifying the respondents involved in the study and addressing ethical considerations to protect participant rights and data integrity.

3.1 Systems Development Life Cycle

The Systems Development Method referred to the framework used to organize, plan, and manage the process of creating information systems. In this research, a mixed-methods approach combining descriptive and developmental research was employed. The descriptive component analysed the Informal Settler Relocation Management System (ISRMS) in terms of system acceptability, using ISO/IEC 25010:2011 for system quality evaluation, ISO 9241-11:2018 for usability evaluation, and test cases to validate system functionality. Developmental research was utilized during the planning, implementation, and testing phases, incorporating new techniques and tools to address challenges and enhance the management of the emergency response team. The system was developed using the Rapid Application Development (RAD) approach within the Software Development Life Cycle (SDLC) model for the project.

Figure 1
Rapid Application Development Model





As shown in the previous page, Figure 1 shows the RAD. The system development life cycle guided proponents in developing the Informal Settler Relocation Management System. This software development model involved requirement planning, user design, construction, and cutover phases. The method was suitable for developing the system, despite its progressive nature. Each phase produced a fully functional component that could be tested and evaluated. García et al. (2023) supported this approach, having applied a hybrid RAD-Agile SDLC to develop a web-based affordable housing allocation system in Spain. This system featured interactive mapping, real-time vacancy tracking, and applicant verification modules similar to ISRMS.

Phase 1 - Requirements Planning Phase

During this phase, researchers discussed ideas related to the study's goals and identified necessary requirements. They conducted face-to-face interviews with beneficiaries to gather data, observe operational processes, and collect relevant information. After analysing the data, they formulated the system's functional requirements and objectives. This phase lasted 7 days for requirements gathering and another seven days for analysis, as reflected in the Gantt Chart.

Phase 2 - User Design Phase

During this phase, researchers began by designing the database and then provided the system to users for testing and feedback. They considered suggestions and made necessary adjustments. Additionally, they developed the



context diagram, data flow diagram, entity-relationship diagram, and use case diagram. This phase involved 36 days of layout activities, as shown in the Gantt Chart.

Phase 3 - Rapid Construction Phase

At this stage, the system had come together. Researchers developed the system based on the previously agreed design. Coding the system components using PHP as the programming language. Testing was conducted on each module during development to ensure proper functionality. Test cases thoroughly verified that all features operated as intended, while ISO/IEC 25010:2011 was used to assess the system and ISO 9241-11:2018 usability from the users' perspective. During this phase, code reviews and debugging were performed, lasting 136 days to identify and resolve issues early in the process.

Phase 4 - Cutover Phase

During the Cutover phase, the system was prepared for live use, and users received training on how to operate the new system. After training, the system was deployed to the live environment. It was then made accessible to all intended users, and the documentation was finalized to support ongoing use and maintenance. This phase lasted 32 days, as reflected in the Gantt chart.

3.2 Operational Framework

Figure 2

Operational Framework of Informal Settler Relocation Management System

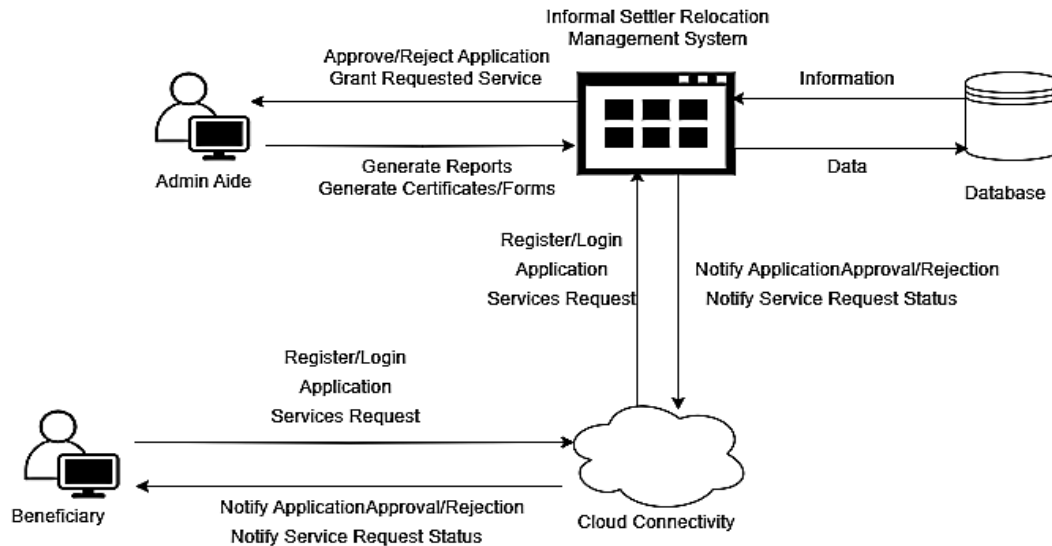


Figure 2 above showed how the Informal Settler Relocation Management System works by illustrating the system's operational framework, which is based on the Systems Development Life Cycle (SDLC). There are two primary types of users for the system: “Beneficiaries”, who sign up, submit requests for relocation, and ask for services; and “Admin Aides”, who examine requests, accept or deny them, handle services, and make papers. A layer in the cloud safely stores data, making it easy to process and update in real time.

3.3 Data Flow Diagram

Figure 3

Context Diagram (Level 0) of Informal Settler Relocation Management System.

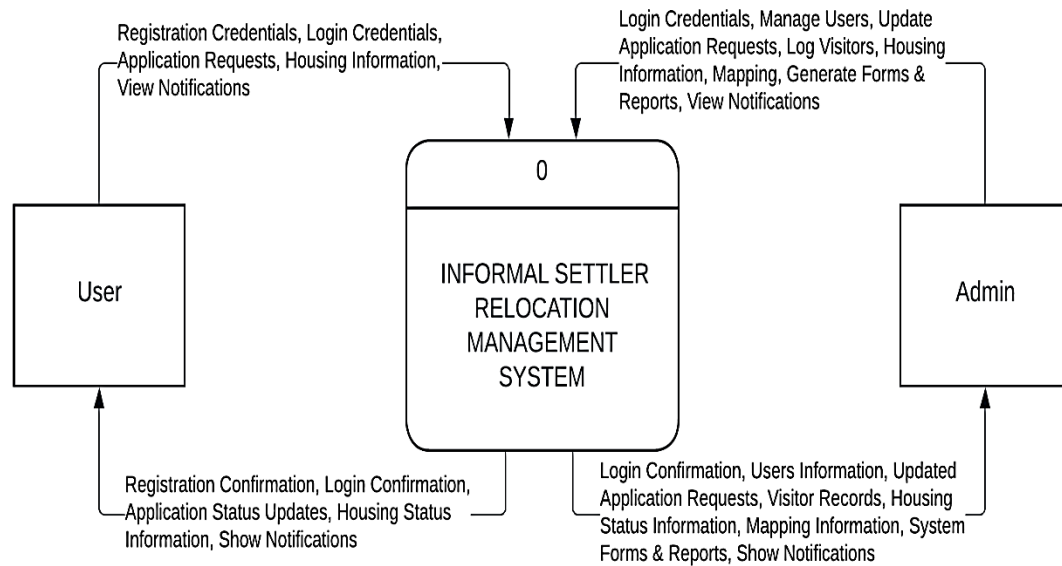
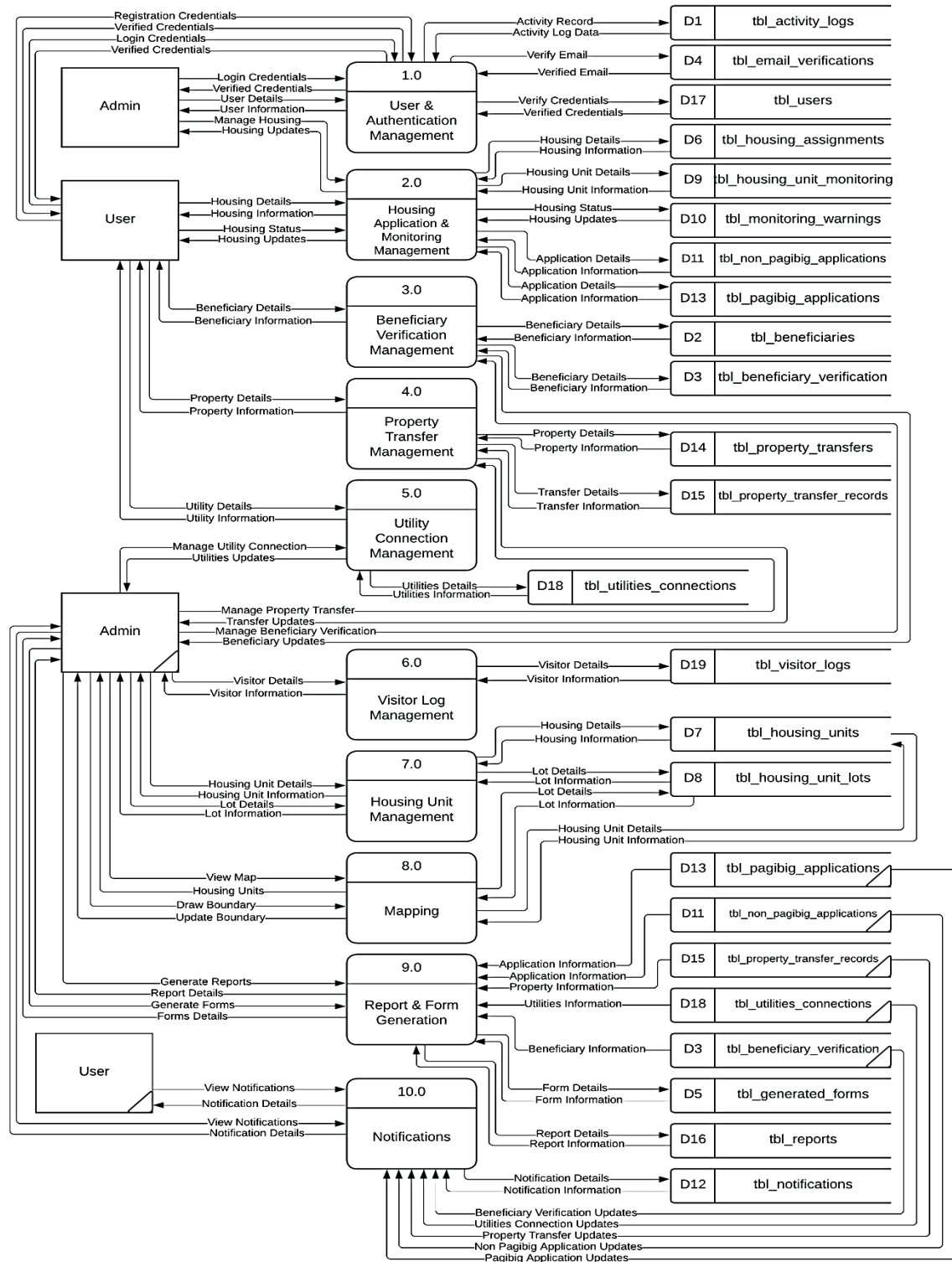


Figure 3 above showed the context diagram of the developed system, which showed the foremost data, and data saved into the system by the admin such as Housing Unit Data, Boundary Data, and various Status Updates and the output to be received by the customer, which included Registration Confirmation and the real-time ability to View Status regarding their housing application, property transfer, and utility connections.



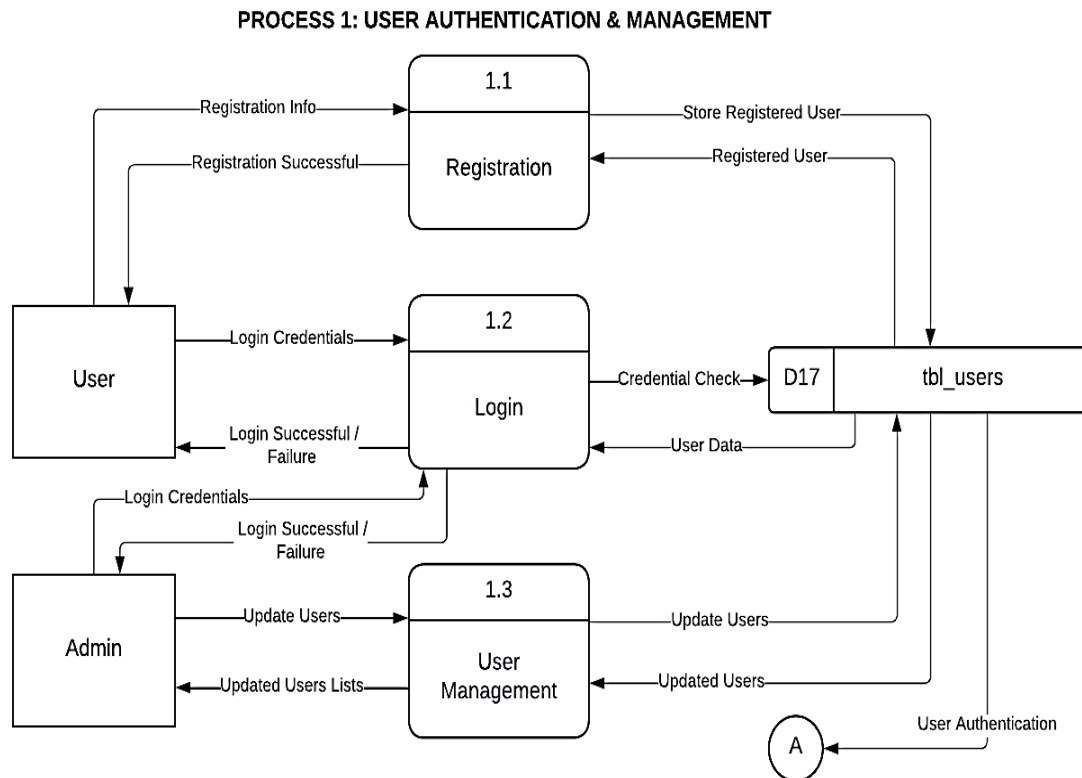
Figure 4
Data Flow Diagram Level 1 Explosion of ISRMS





As shown from the previous page, Figure 4 showed how the system worked with data. For example, tbl_applications stored housing requests, tbl_users stored user accounts, tbl_housing_units and tbl_lot_boundaries stored site info, tbl_beneficiaries stored verified users, tbl_property_transfer and tbl_utility_connections stored legal and utility issues, and tbl_generated_forms and tbl_documents stored forms and documents, and tbl_visitor_logs for reports, documents, and audit logs. All of these things worked together to make relocation management smooth, safe, and well-organized.

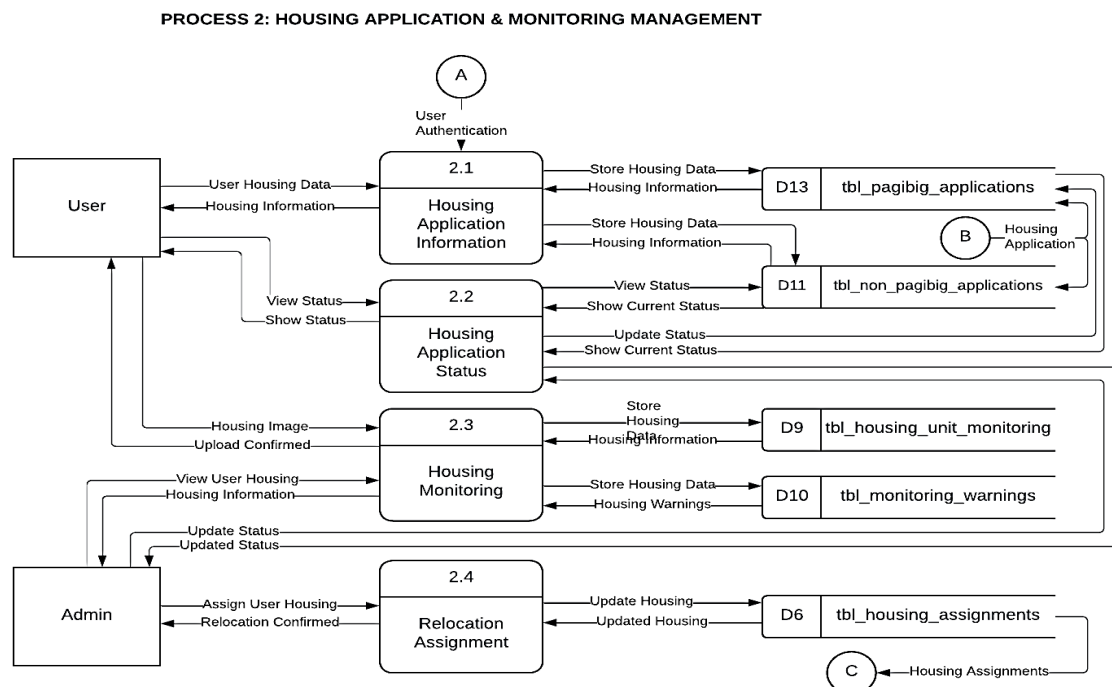
Figure 5
Data Flow Diagram Level 2 Explosion of User Authentication and Management





As shown from the previous page, Figure 5 showed the whole procedure of User Authentication and Management. Users first registered by giving their personal information and credentials, which were checked and saved in `tbl_users` (D17). Then they got a confirmation. After signing up, users could safely log in to the system. The system checked the user's username/email and password against `tbl_users` (D17). If the credentials matched an active account, they could log in. If they didn't, they couldn't. Process 1.3 let admins manage user accounts by letting them see, amend, suspend, or activate accounts. Changes were made to `tbl_users` and took effect right away. This kept all logins and account information safe, accurate, and up to current.

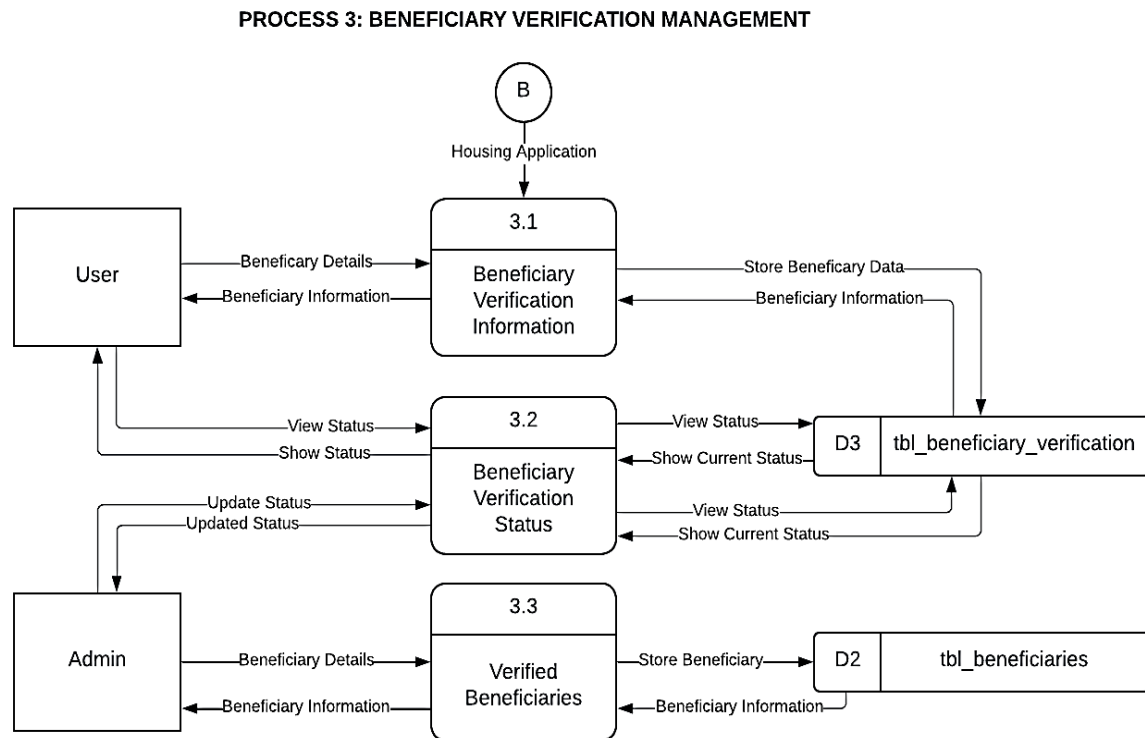
Figure 6
Data Flow Diagram Level 2 Explosion of Housing Application and Monitoring Management





As shown from the previous page, Figure 6 showed the entire process of applying for housing and keeping track of it. Users filled out the form with their personal and application information, submitted it, and got a confirmation. Through the monitoring module, they could get updates on housing and keep track of the status of their application in real time. Admins checked paperwork, made sure people were eligible, and assigned housing units. This changed the user's status right away and sent out messages. The technology made sure that users and admins could communicate and get updates in real time till the housing assignment was done.

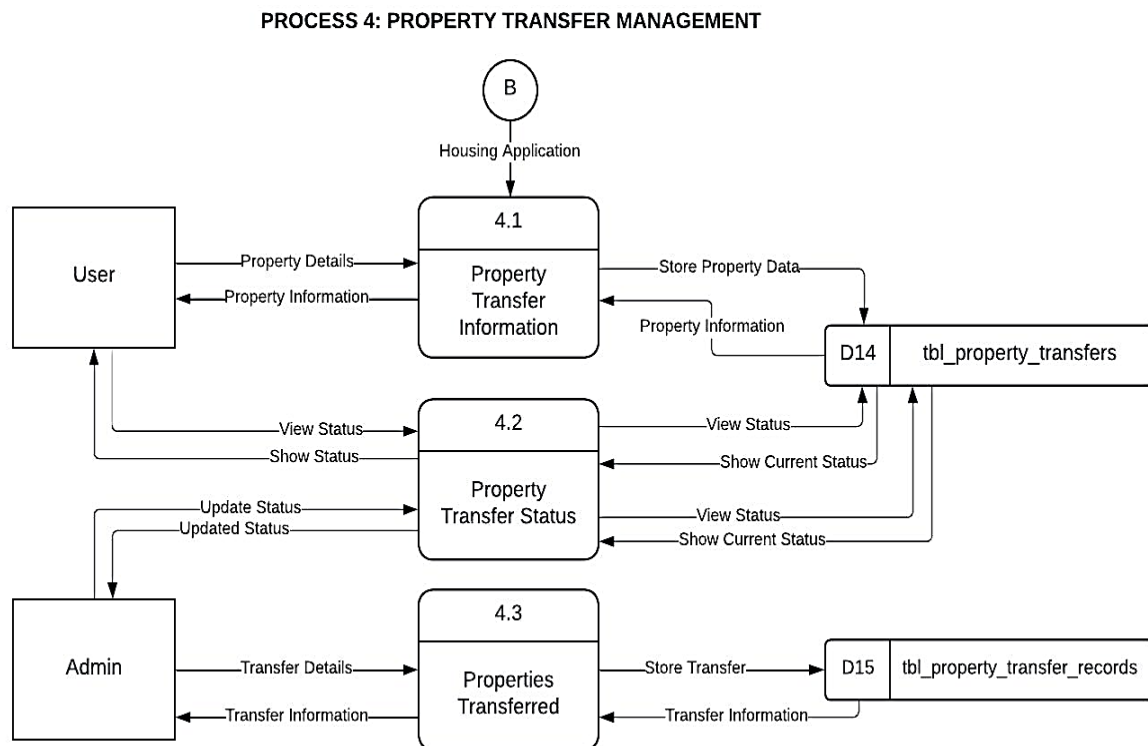
Figure 7
Data Flow Diagram Level 2 Explosion of Beneficiary Verification Management





As shown from the previous page, Figure 7 showed the whole procedure of managing beneficiary verification. Users sent in their information, got confirmation, and could check their verification status at any moment with real-time updates. Admins checked submissions, made sure they were eligible, and updated the status right away, letting users know right away. The system made sure that submissions got through smoothly, that tracking was always up to date, and that results were verified quickly.

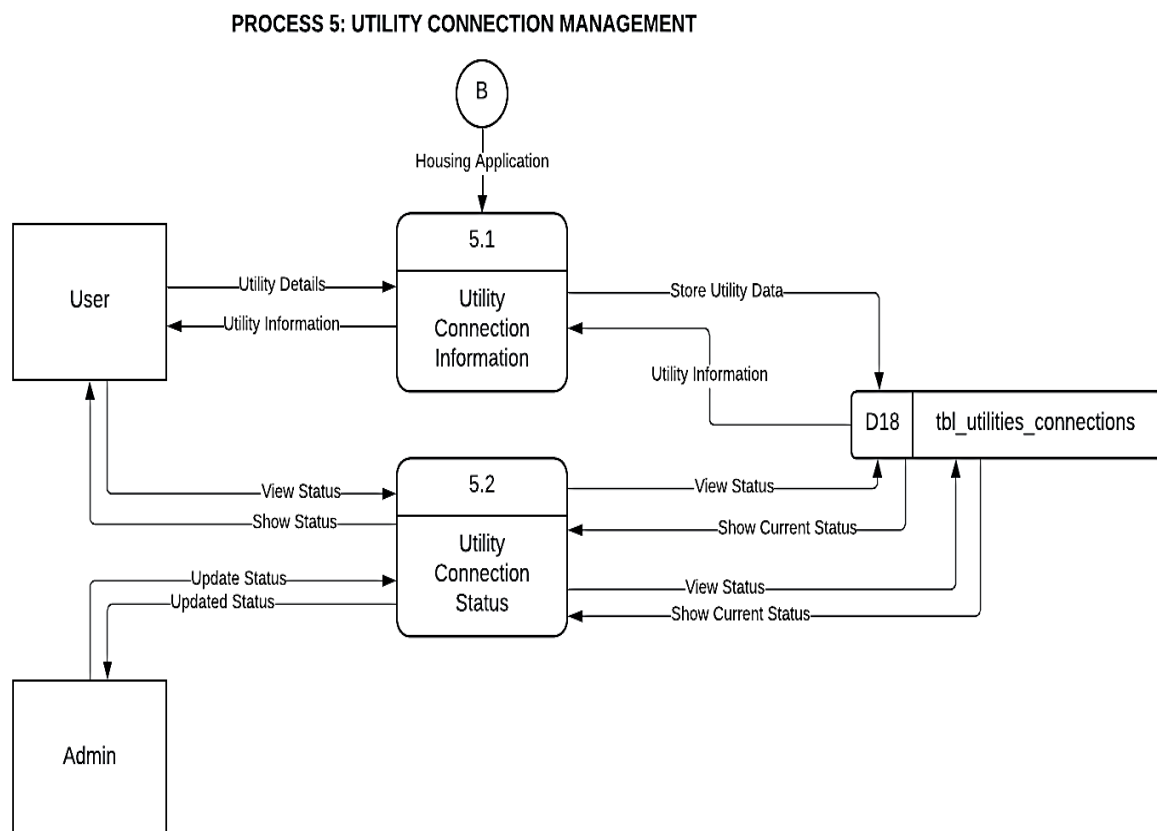
Figure 8
Data Flow Diagram Level 2 Explosion of Property Transfer Management





As shown from the previous page, Figure 8 showed Property Transfer Management process worked from start to finish. Users filled out property information, sent in their request, and got confirmation. The status module let them see how things were going in real time. Admins checked the transfer, processed it, and made changes. Then they sent final notifications to users. The system made sure that submissions went through smoothly, that changes happened all the time, and that property transfers were quick, clear, and easy to understand.

Figure 9
Data Flow Diagram Level 2 Explosion of Utility Connection Management

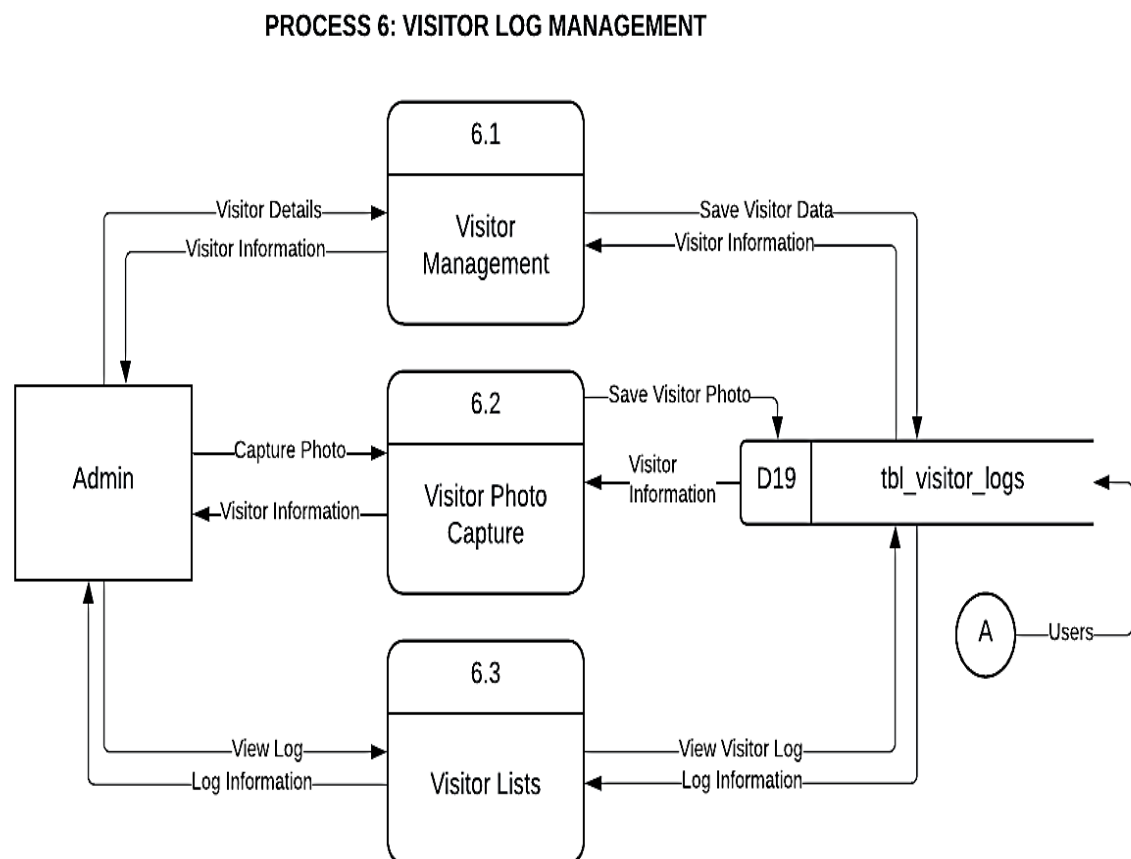




As shown from the previous page, Figure 9 showed the whole procedure of managing utility connections. Users sent in the necessary utility information and got a confirmation. They could see how things were going in real time, and admins could review, coordinate, and update the connection. When it was done, they sent a final notification. The system made sure that people who moved could submit their applications smoothly, keep track of their progress, and connect to utilities quickly.

Figure 10

Data Flow Diagram Level 2 Explosion of Visitor Log Management



As shown from the previous page, Figure 10 showed the whole procedure of managing visitor logs. The admin wrote down the visitor's information, took a picture, and saved the log for security and record-keeping when they arrived. Admins could look at records of visitors from the past and present at any moment.

Figure 11

Data Flow Diagram Level 2 Explosion of Housing Unit Management

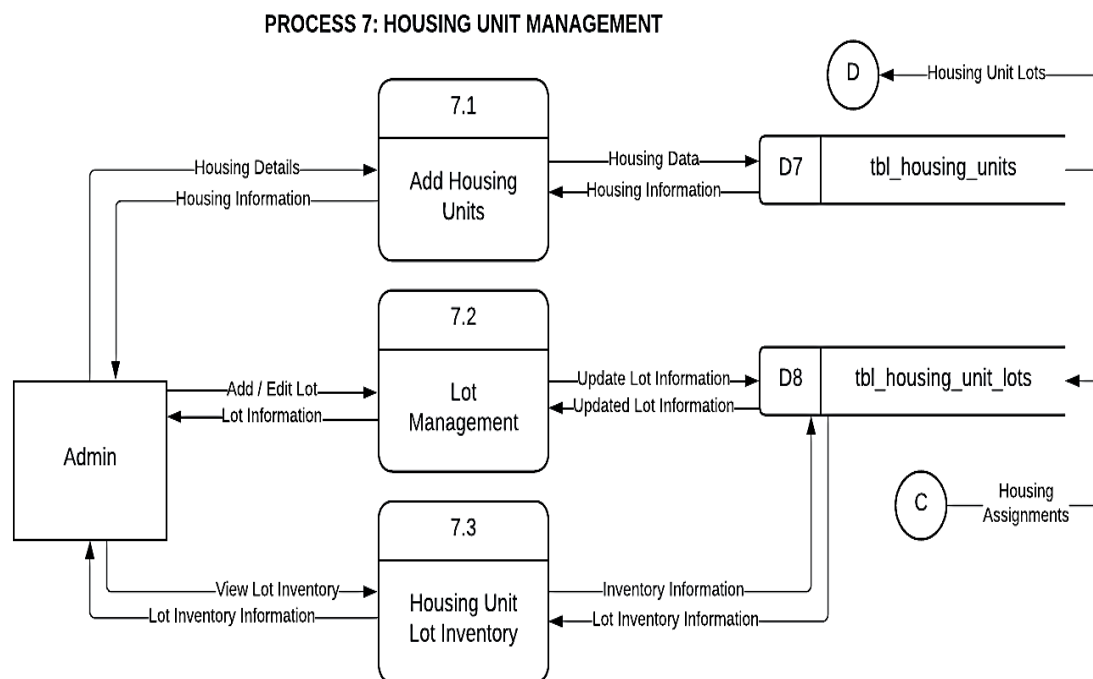


Figure 11 above showed the whole Housing Unit Management procedure, which was only done by the Admin. The admin created new housing units, kept track of and modified lots, registered new lots as they became available, and kept an eye on inventory in real time. This method made ensuring that housing units were well-maintained, tracked, and assigned throughout relocation assignments.

Figure 12
Data Flow Diagram Level 2 Explosion of Mapping

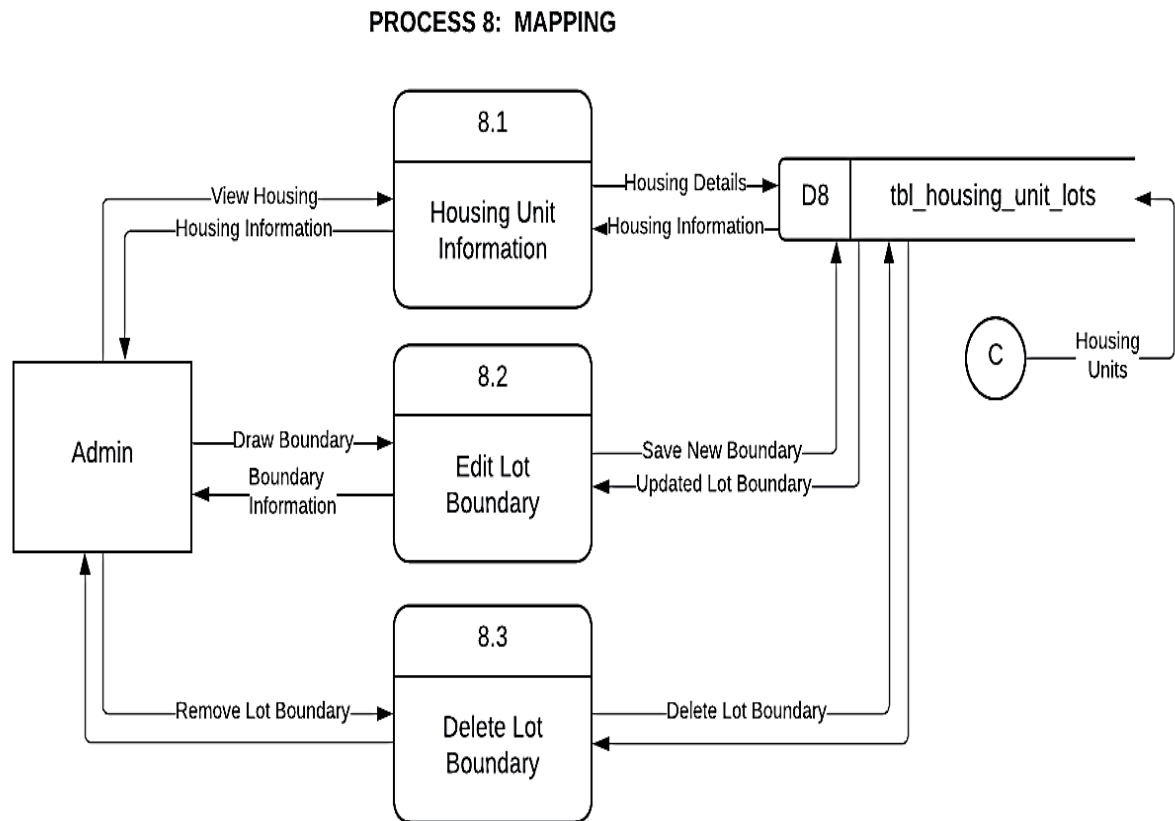


Figure 12 above showed the whole Mapping procedure, which was only done by the Admin. The admin could look at housing units, change or delete lot boundaries, and save changes. The solution let the admin see things in real time and with accuracy, so they could keep a digital map up to current for effective planning and housing assignments.

Figure 13
Data Flow Diagram Level 2 Explosion of Report and Form Generation

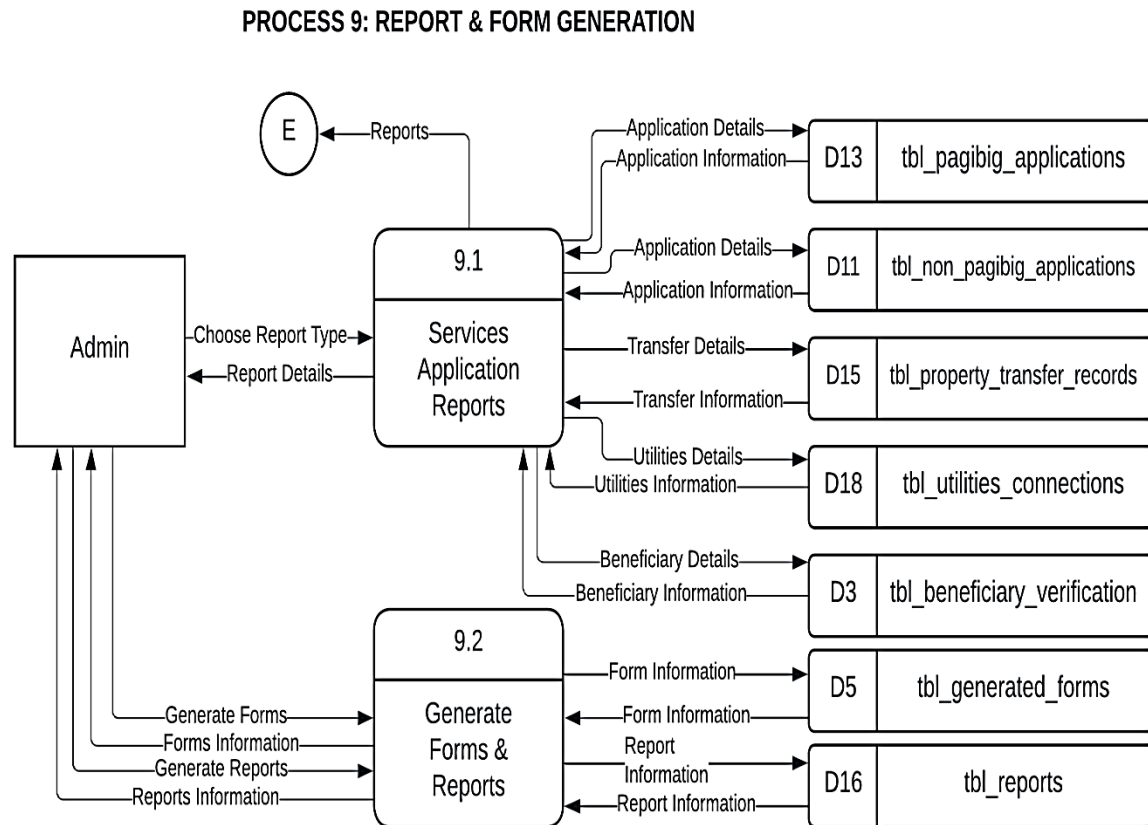


Figure 13 above showed the whole process of making reports and forms, which was only done by the Admin. The administrator chose a report type, including housing applications, property transfers, utility connections, or beneficiary verification. The system then got the right data. Then, reports and forms were made, saved automatically, and made available for printing, downloading, or sharing. This made sure that all of the paperwork for the housing allocation procedure was done quickly, correctly, and professionally.

Figure 14
Data Flow Diagram Level 2 Explosion of Notification

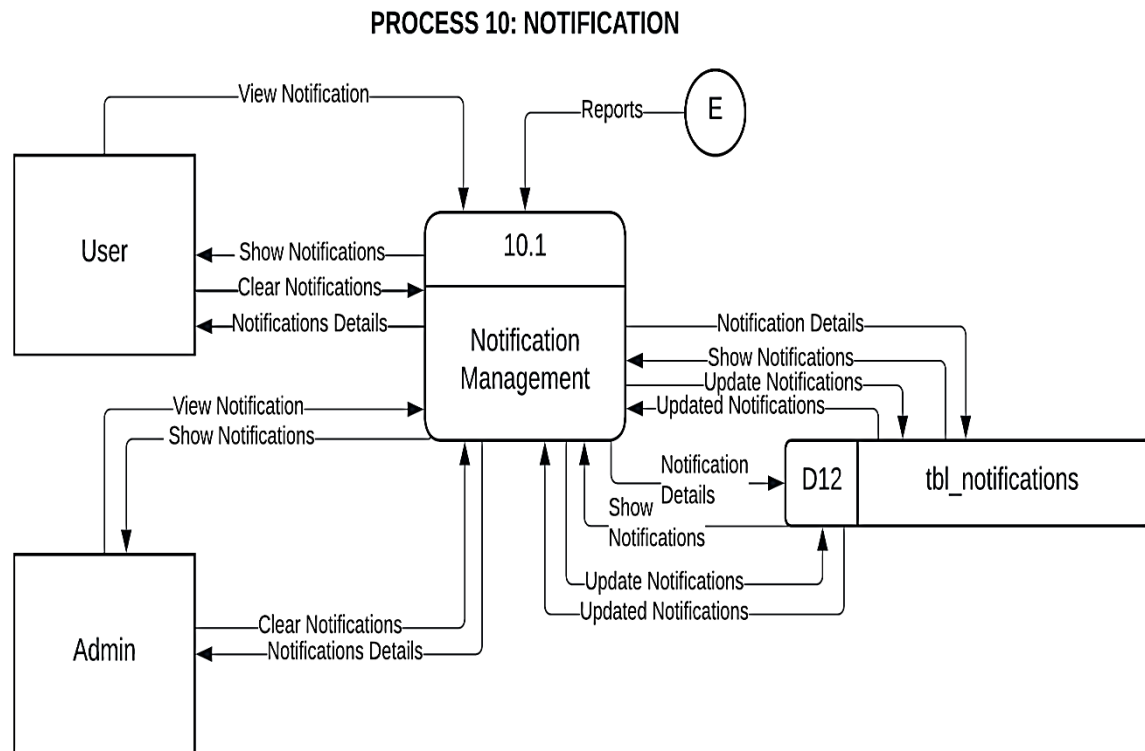
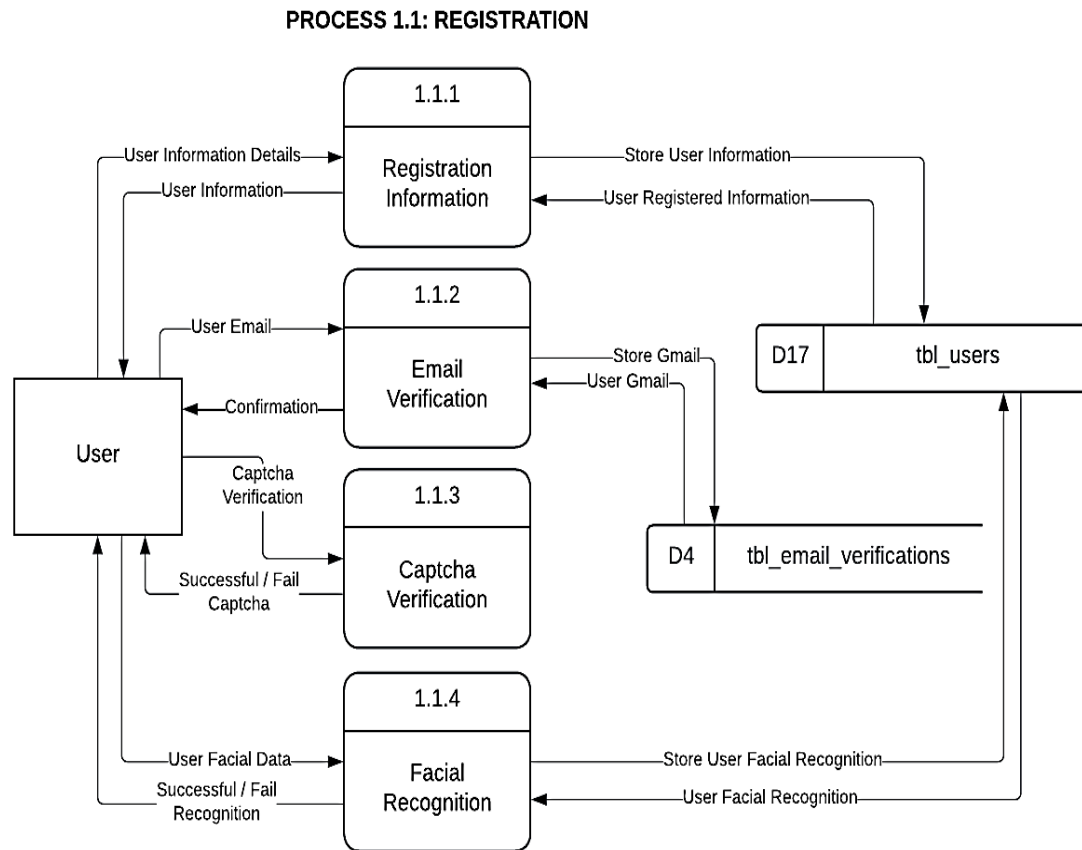


Figure 14 showed the complete workflow of the Notification process in the Housing Application Processing system. The User could access View Notification Icon (10.1) at any time to instantly see new or unread notifications about application status, approvals, assignments, or other updates. Behind the scenes, the Admin used Notify Users (10.2) to send important updates, which were immediately stored and pushed as Notification Info. The system also centrally handled all messages through Manage Notifications (10.2), ensuring reliable delivery and storage of every alert.

Figure 15
Data Flow Diagram Level 3 Explosion of Registration



As shown above in Figure 15, the registration process began when the user entered their details in Registration Information (Process 1.1.1). The system then performed Email Verification (Process 1.1.2) and stored the records in **tbl_email_verifications** (D4). Next, the user completed Captcha Verification (Process 1.1.3) and Facial Recognition (Process 1.1.4) for security purposes. Once validated, the user submitted the registration, and the information was saved in **tbl_users** (D17).

Figure 16
Data Flow Diagram Level 3 Explosion of Login

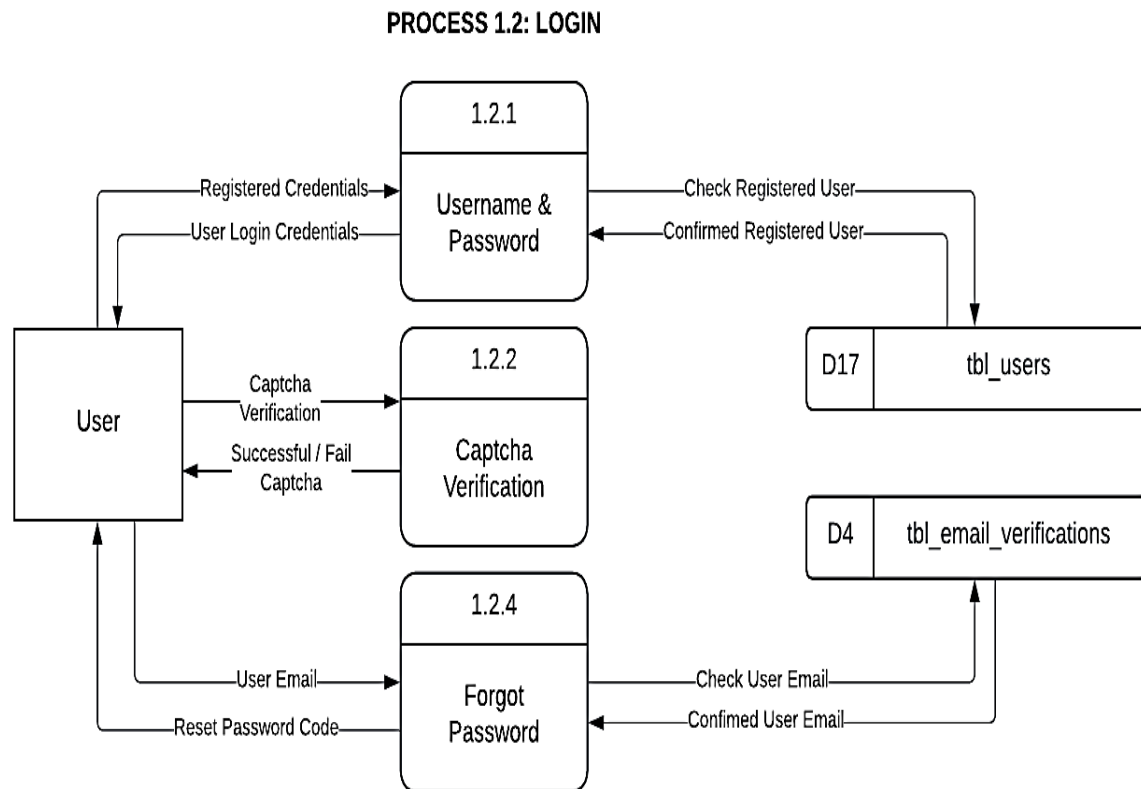


Figure 16 above showed the whole process of logging in to the Housing Application Processing system. The user started by inputting their credentials to get secure access to the platform. This was how the process worked: To log in, the user had to enter their Username and Password (Process 1.2.1). The system immediately checked the entered credentials against the tbl_users database table (D17) to make sure they corresponded to a real registered account. To make things even safer, the user had to pass Captcha Verification (Process 1.2.2).

Figure 17
Data Flow Diagram Level 3 Explosion of User Management

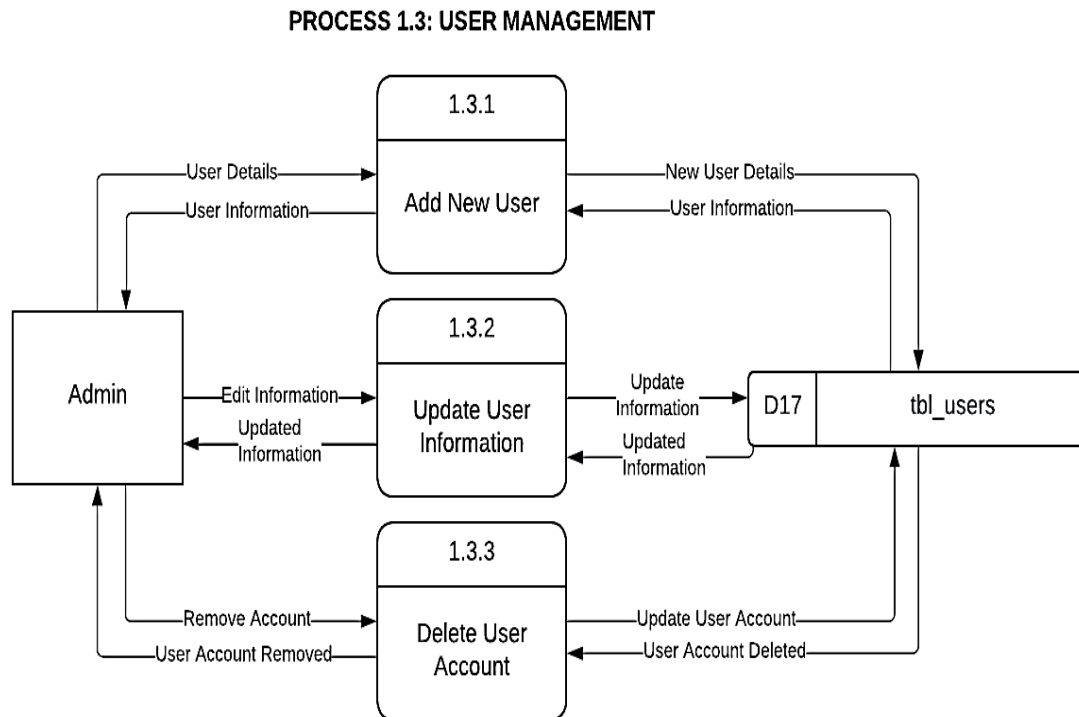


Figure 17 above showed the step-by-step process of the User Management sub-process, which was only done by Admin. The admin could create new users by filling up the relevant fields, edit or delete existing accounts, and save changes. These changes were immediately applied and stored. This sub-process let the admin safely and quickly make, change, and keep up with user accounts.

Figure 18
Data Flow Diagram Level 3 Explosion of Housing Application Information

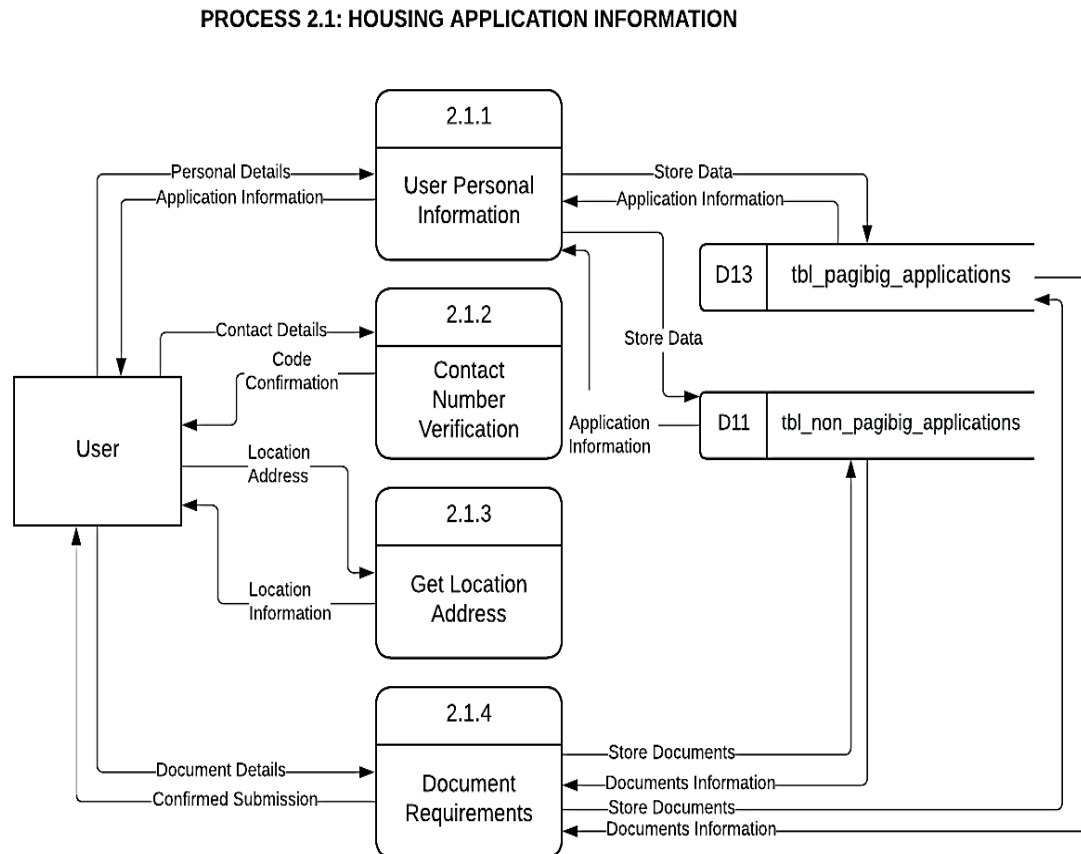


Figure 18 above showed the Housing Application Information sub-process worked in detail. The user filled out their personal information, confirmed their phone number, gave their current address, and uploaded all the necessary papers. After the application was finished, it was sent in and saved, and the user got a message saying that it was sent successfully. This sub-process checked that all the information was complete, correct, and confirmed before moving on to processing.

Figure 19
Data Flow Diagram Level 3 Explosion of Housing Monitoring

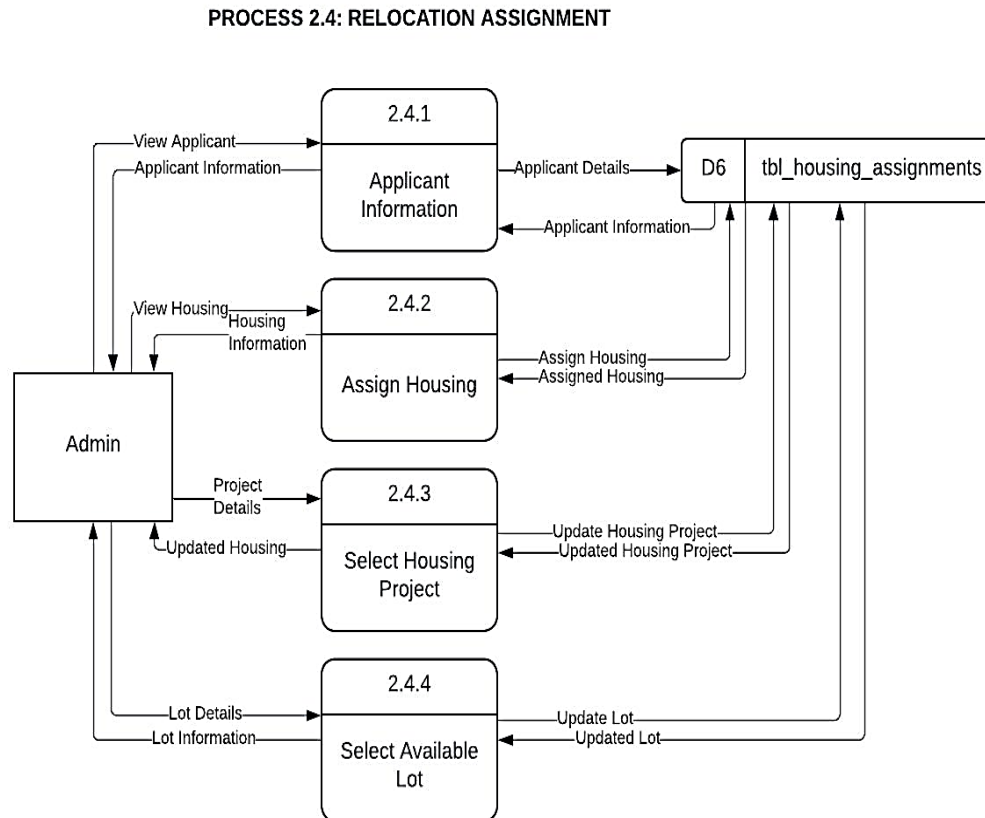


Figure 19 above showed the step-by-step method of the Housing Monitoring sub-process. The user took pictures of their home situation and sent them in, and they were promptly uploaded for inspection. Both the user and the admin could see the submissions right away. The admin could also see monitoring reports that were late or missing. This procedure made sure that living circumstances were regularly checked based on evidence and gave the admin a clear view of compliance.

Figure 20
Data Flow Diagram Level 3 Explosion of Beneficiary Verification Information

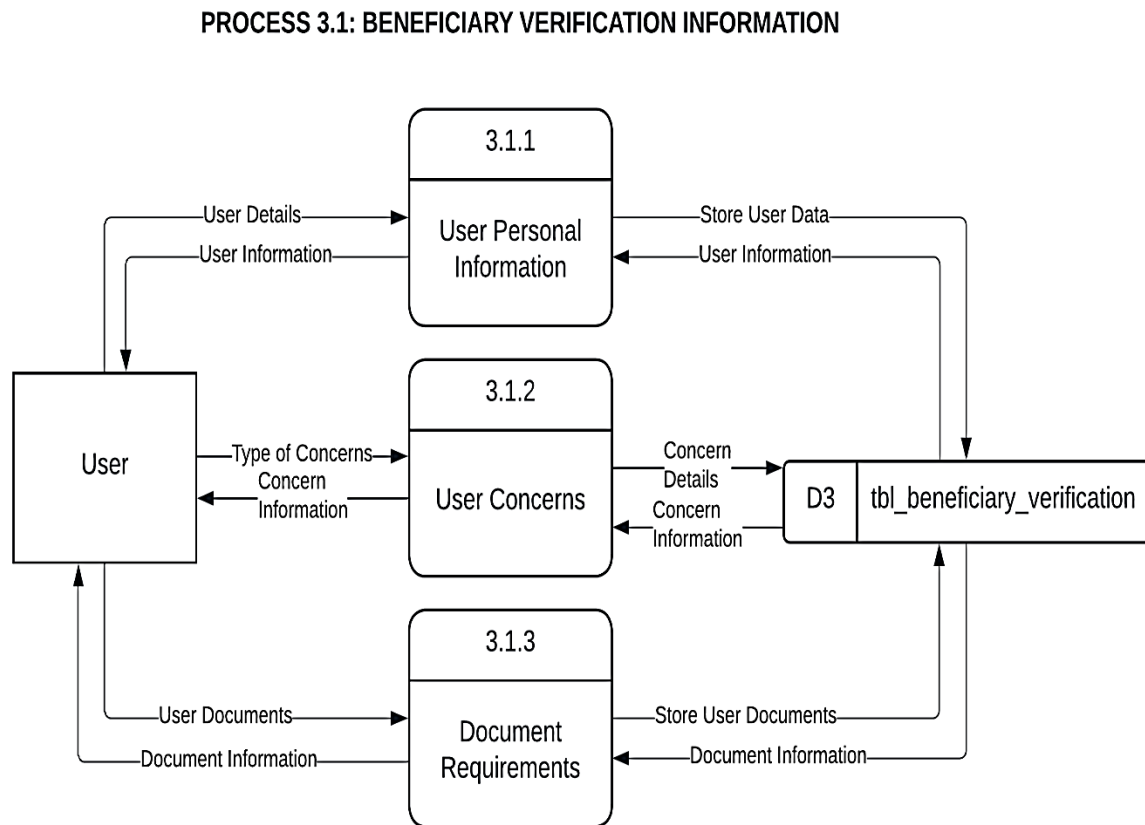


Figure 20 above showed the step-by-step process for the Beneficiary Verification Information sub-process. The user filled up their personal information, explained their worries, and uploaded all the necessary papers. After the user input the information, the system saved it and let the user know that it was successful. This method made sure that beneficiaries gave all the information and proof needed for correct and quick admin verification.



Figure 21

Data Flow Diagram Level 3 Explosion of Property Transfer Information

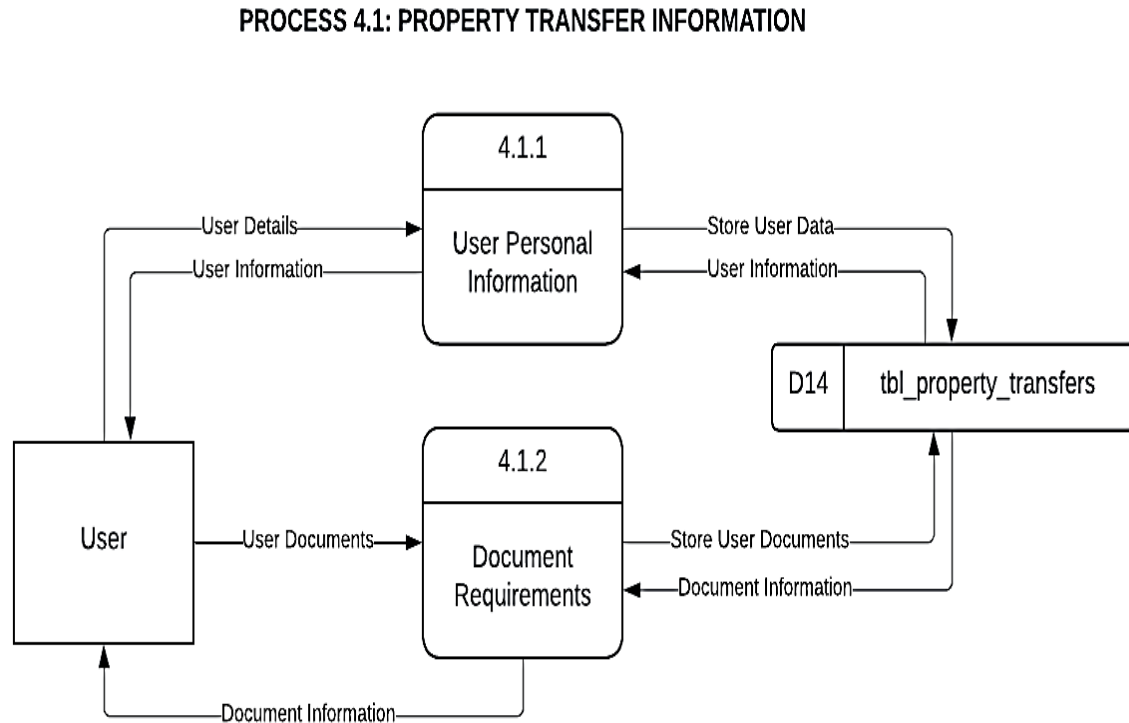


Figure 21 above showed the Property Transfer Information sub-process worked in detail. Before sending in the application, the user filled out all of their personal information, uploaded all of the necessary documents, and gave full details on the property. After it was checked, the system saved the request and certified that it was sent. All data was stored safely, which made it easy to process and review. This meant that users could send property transfer requests quickly and accurately.

Figure 22

Data Flow Diagram Level 3 Explosion of Utility Connection Information

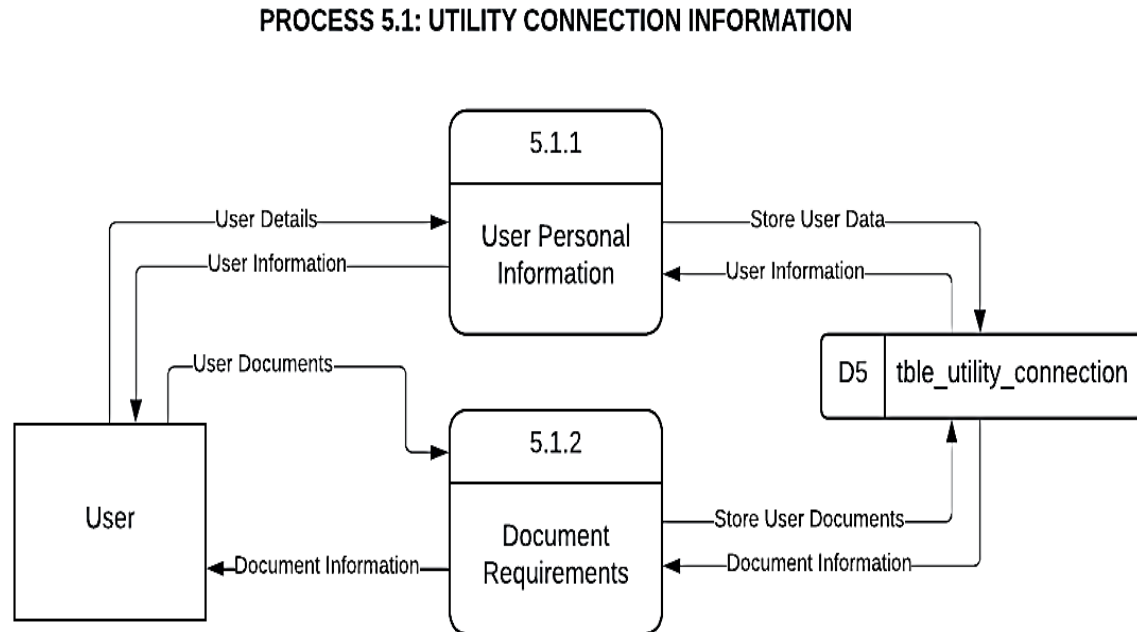


Figure 22 above showed the detailed workflow of the Utility Connection Information sub-process. The User started by completing User Personal Information (5.1.1) and uploading all required Document Requirements (5.1.2), such as proof of residency or identification. The system immediately validated the submission, stored all data, and displayed Submission Successful. This straightforward sub-process allowed relocated users to quickly request water, electricity, and other utility connections by providing verified personal details and documents in a single, efficient step.

Figure 23
Data Flow Diagram Level 3 Explosion of Visitor Lists

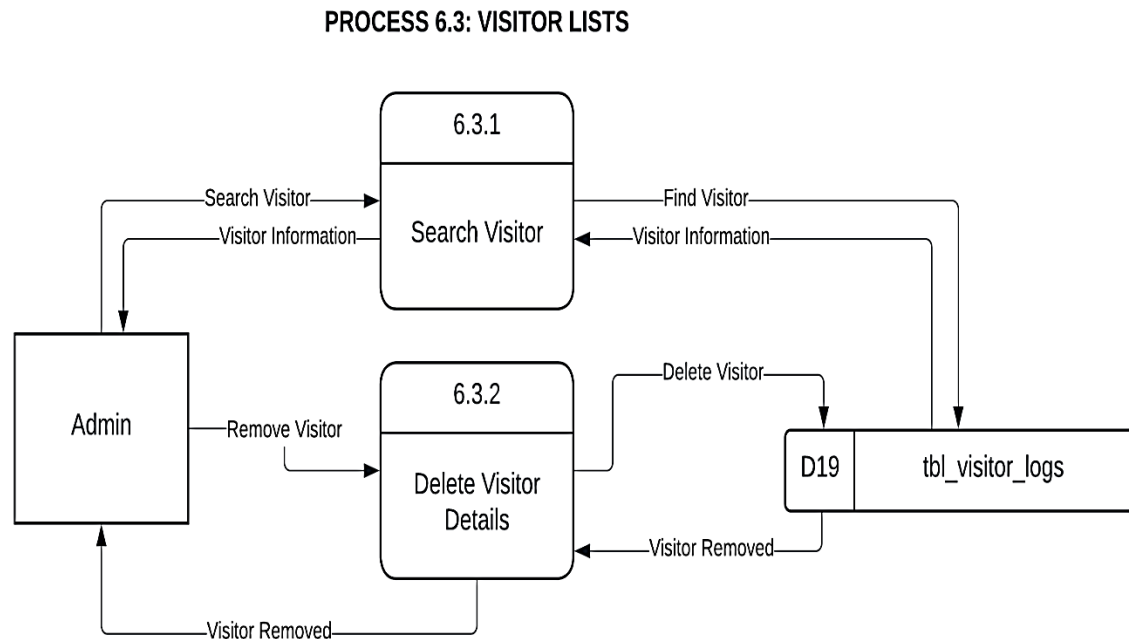


Figure 23 above showed the step-by-step process of the Visitor Lists subprocess, which the Admin was in charge of. The admin could look up visitors by name, date, or other criteria. They could see all the information, such as the time of entry, a photo, and the reason for the visit, and they could delete records if they needed to. This approach let the admin keep the database correct, up-to-date, and easy to use while also managing visitor logs.

Figure 24
Data Flow Diagram Level 4 Explosion of Forgot Password

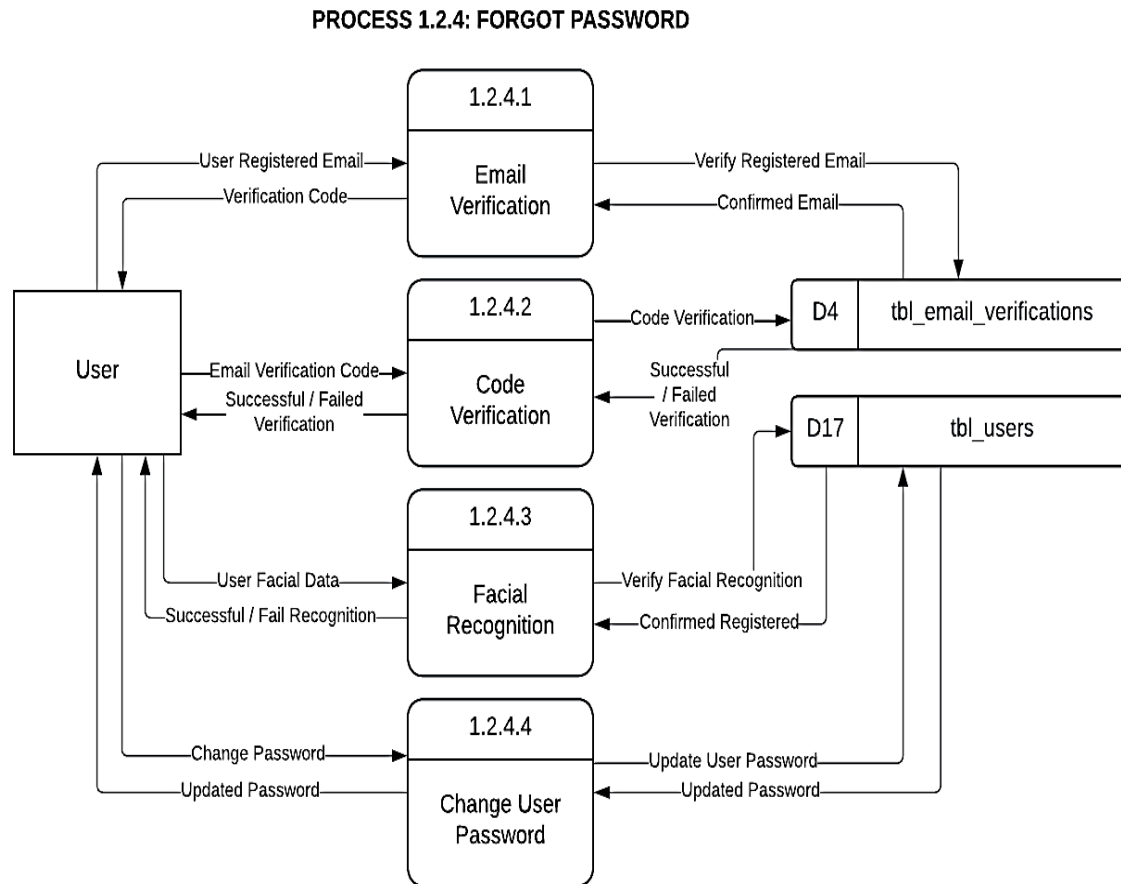
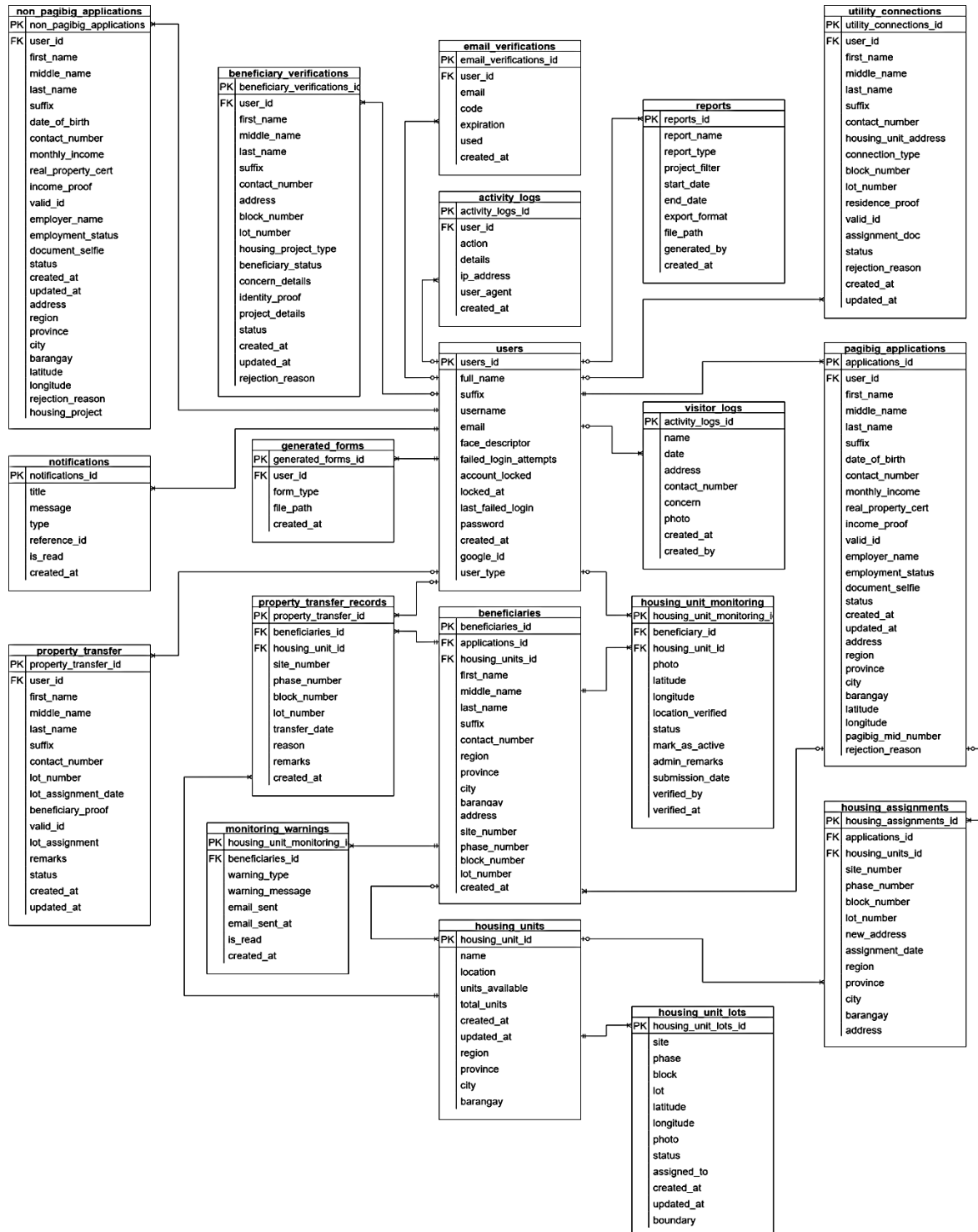


Figure 24 above showed the full process of the Forgot Password recovery sub-process. The user typed in their registered email address, and the system provided a verification code if the email address was valid. The system checked the code after it was entered and then used face recognition to confirm the user's identity. The user could then set a new password, which was stored securely, so they could get into their account right away.



3.4 Entity Relationship Diagram

Figure 25
Entity Relationship Diagram



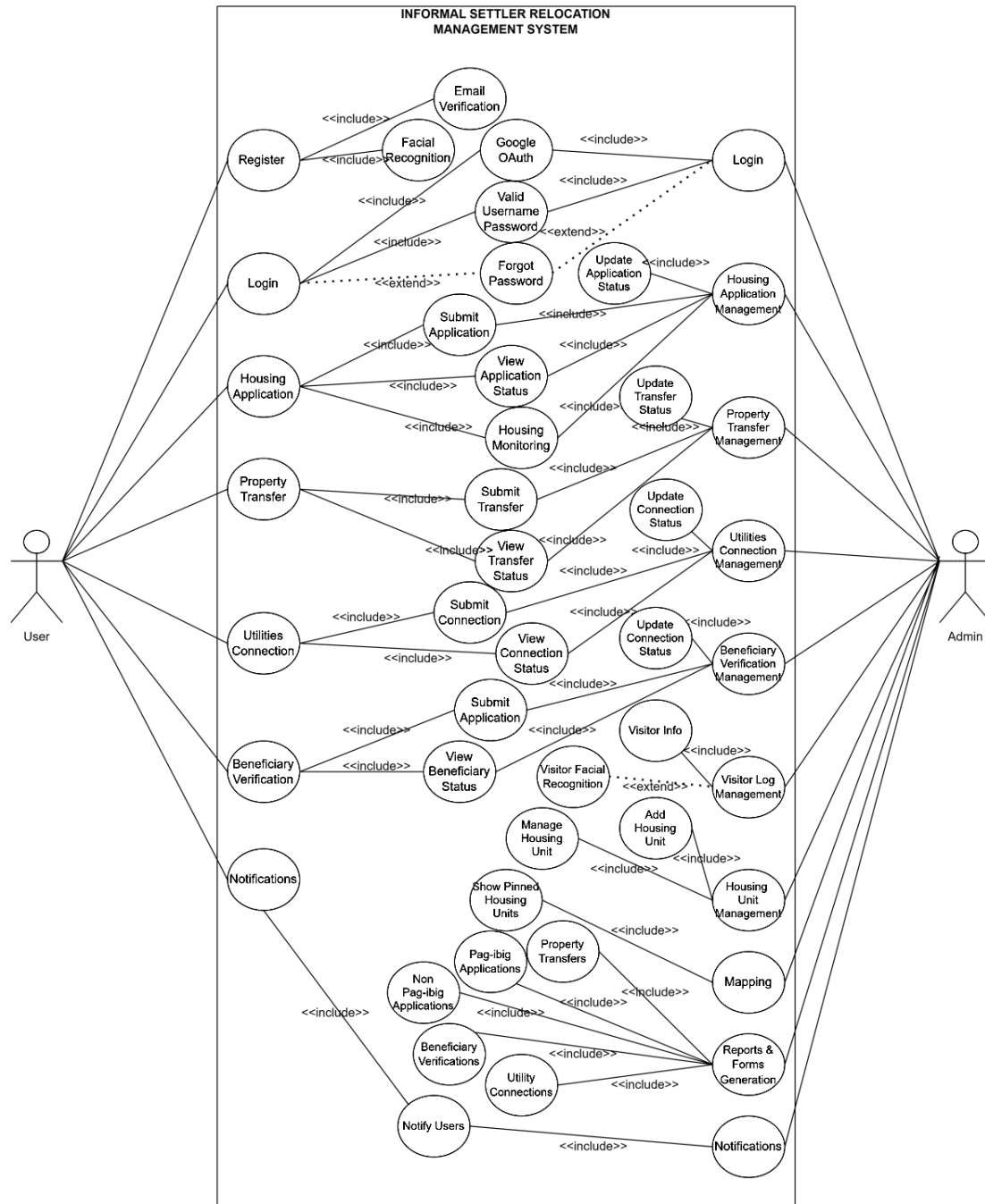


As shown from the previous page, Figure 25 showed the plan for the Informal Settler Relocation Management System's database structure. The model put all the important information into tables that were linked to each other. One set handled the main tasks of the system, like user accounts, activity logs, notifications, visitor logs, and forms that were generated. The main part was about beneficiaries and included demographic profiles, verification data, application statuses, reported problems, and submitted requirements. It also connected beneficiaries to housing units, lot details, and assignment records, and kept track of ongoing compliance by sending out warnings and monitoring. To make it apparent who owned what, property transfers were recorded. All data went into records for relocation, monitoring, and property transfer, which combined actions, choices, and status updates. This structure made sure that every stage, from verification to final assignment, could be tracked, connected, and precisely shown. This helped with transparency, accountability, and effective management during the relocation process.

3.5 UML Use Case Diagram

Figure 26

UML Use Case Diagram of Informal Settler Relocation Management System





As shown from the previous page, Figure 26 showed the UML use case diagram for the Informal Settler Relocation Management System. This Use Case Diagram presented the main system functions and how they were accessed by its two actors: the User and the Admin. The User represented applicants, beneficiaries, or individuals who used the system to request and track housing-related services. The Admin was responsible for managing, validating, and overseeing all operations within the platform. The User interacted with key processes such as Registration, Login, Housing Application, Property Transfer, Utilities Connection, and Beneficiary Verification. These functions allowed the User to submit requests, upload needed information, and monitor their progress through features like View Application Status, View Transfer Status, View Connection Status, and View Beneficiary Status. Additional features including Email Verification, Google OAuth, Facial Recognition, Forgot Password, and Valid Username and Password Checking ensured secure access to the system. The User also received Notifications that kept them updated on actions taken by the Admin. The Admin handled all backend and administrative responsibilities. This included Housing Application Management, Property Transfer Management, Utilities Connection Management, Beneficiary Verification Management, Visitor Log Management, and Housing Unit Management which covered adding, mapping, and monitoring housing units. The Admin also managed visitor information through facial recognition, and updated all status-related processes.



3.6 Use Case Scenario

Table 3

User & Authentication Management

Use Case 1: User & Authentication Management
Use Case Goal: To allow users to register an account, log in, and gain access to system services.
1. Successful log in to the system
Actor: User / Admin
Pre-condition: User has the system open and the system server is available.
Main Flow:
Step 1: The user accesses the system website.
Step 2: The user registers an account by completing required details and verification steps.
Step 3: The system validates and creates the user account.
Step 4: The user logs in using valid credentials.
Step 5: The system authenticates the user and displays the dashboard.
User Action: Enters registration info.
System Response: Validates input.
User Action: User submits registration.
System Response: Registers user.
User Action: Input login credentials.
System Response: Proceed to system dashboard.

Table 4

Housing Application

Use Case 2: Housing Application Processing
Use Case Goal: To allow users to submit housing applications and view status, and admin to update housing application.
1. Housing application is successfully recorded and viewable.
Actor: User / Admin
Pre-condition: User is logged in and the system server is available.
Main Flow:
Step 1: The user opens the Housing Application form.
Step 2: The user submits the housing application.
Step 3: The system validates and records the application.
Step 4: The admin updates the status for user viewing.
User Action: Selects housing application.
System Response: Loads form.
User Action: Inputs housing information.
System Response: Validates input.
User Action: Submits housing application.
System Response: Stores application.
Admin Action: Updates application status.
User Action: View housing application status anytime.



Table 5
Beneficiary Verification Management

Use Case 3: Beneficiary Verification Management
Use Case Goal: To allow users to submit beneficiary verification and view status, and admin to update beneficiary verification. 1. Beneficiary verification is successfully recorded and viewable.
Actor: User / Admin
Pre-condition: User is logged in and the system server is available, has submitted housing application.
Main Flow:
Step 1: User selects Beneficiary Verification from the dashboard.
Step 2: System displays the verification form.
Step 3: User enters beneficiary information and uploads required documents.
Step 4: User reviews and submits the application, and the system stores the data.
Step 5: Admin updates the status, and the user can view the verification status anytime.
User Action: Selects beneficiary verification.
System Response: Loads form.
User Action: Inputs beneficiary information.
System Response: Validates input.
User Action: Submits beneficiary verification application.
System Response: Stores beneficiary verification.
Admin Action: Updates beneficiary status.
User Action: View beneficiary verification status anytime.

Table 6
Property Transfer Management

Use Case 4: Property Transfer Management
Use Case Goal: To allow users to submit property transfer and view status, and admin to update beneficiary verification. 1. Property transfer application is successfully recorded and viewable.
Actor: User / Admin
Pre-condition: User is logged in and the system server is available, has submitted housing application.
Main Flow:
Step 1: User selects Property Transfer.
Step 2: System loads the form.
Step 3: User uploads documents and confirms details.
Step 4: User submits application and system stores it.
Step 5: Admin updates status and user views it.
User Action: Selects property transfer.
System Response: Loads form.
User Action: Inputs property transfer information.
System Response: Validates input.
User Action: Submits property transfer application.
System Response: Stores property transfer application.
Admin Action: Updates transfer status.
User Action: View property transfer status anytime.



Table 7
Utility Connection Management

Use Case 5: Utility Connection Management

Use Case Goal: To allow users to submit utility connection and view status, and admin to update utility connection.

1. Utility connection application is successfully recorded and viewable.

Actor: User / Admin

Pre-condition: User is logged in and the system server is available, has submitted housing application.

Main Flow:

Step 1: User selects Utility Connection.

Step 2: System loads the form.

Step 3: User enters information and uploads documents.

Step 4: User submits application and system stores it.

Step 5: Admin updates status and user views it.

User Action: Selects utility connection.

System Response: Loads form.

User Action: Inputs utility connection information.

System Response: Validates input.

User Action: Submits utility connection application.

System Response: Stores utility connection application.

Admin Action: Updates utility connection status.

User Action: View utility connection status anytime.

Table 8
Visitor Log Management

Use Case 6: Visitor Log Management

Use Case Goal: To manage and record visitor entries for monitoring and documentation.

1. Visitor entry is successfully recorded and viewable in the log.

Actor: Admin

Pre-condition: Admin is logged in and the system server is available.

Main Flow:

Step 1: Admin opens Visitor Log Management.

Step 2: System loads the visitor log interface.

Step 3: Admin enters visitor information and captures photo.

Step 4: System validates and stores the data.

Step 5: System displays the updated visitor log list.

Admin Action: Opens visitor log management.

System Response: Loads visitor log interface.

Admin Action: Inputs visitor information and photo capture.

System Response: Validates and store data.

Admin Action: Confirms entry.

System Response: Display updated visitor log list.



Table 9
Housing Lots & Housing Management

Use Case 7: Housing Lots & Units Management
Use Case Goal: To allow users to manage housing unit information, lot details, and boundaries.
1. Housing data is saved and viewable
Actor: Admin
Pre-condition: Admin is logged in and the system server is available.
Main Flow:
Step 1: Admin opens Housing Lots & Units Management.
Step 2: System displays lot and unit information.
Step 3: Admin adds or updates lot and unit details.
Step 4: System saves changes and confirms update.
Admin Action: Opens housing lots & units management.
System Response: Loads lot and unit management interface.
Admin Action: Creates or updates housing lot information and boundaries.
System Response: Validates data and stores.
Admin Action: Submits submit changes.
System Response: Updates records and displays confirmation.

Table 10
Report & Form Generation

Use Case 8: Report & Form Generation
Use Case Goal: To generate system reports and official forms related to visitors, lots, and housing units.
1. Report and form data is generated and viewable
Actor: Admin
Pre-condition: Admin is logged in and the system server is available.
Main Flow:
Step 1: Admin opens Report & Form Generation.
Step 2: System displays available reports and forms.
Step 3: Admin selects and generates a report or form.
Step 4: System creates and displays the output.
Admin Action: Opens report & form generation.
System Response: Displays available forms and report categories.
Admin Action: Selects report type.
System Response: Retrieves data from all data stores.
Admin Action: Generates form or report.
System Response: Creates output and stores forms.
Admin Action: Views generated report.
System Response: Displays report on screen.



Table 11
Notification & Communication

Use Case 8: Notification & Communication
Use Case Goal: To send and display system notifications to users. 1. Notifications are displayed and status is updated.
Actor: User / Admin
Pre-condition: User is logged in and the system server is available.
Main Flow:
Step 1: User/Admin opens the notification icon.
Step 2: System displays available notifications.
Step 3: User views the notification.
Step 4: System marks notifications as read and updates status.
Admin Action: Opens notification icon.
System Response: Show available notifications.
User Action: Opens notification icon.
System Response: Marks notifications as read and updates status in user records.

3.7 Data Dictionary

Data Dictionary shows detailed information about the system's data, like standard definitions of data elements, their meaning, and allowable values that are stored in the database tables.

Table 12
Activity Logs

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	0000000001
user_id	int	Integer	11	Foreign key to users table	0000000001
action	varchar	String	100	User action performed	"login", "update_profile"
details	varchar	String	255	Additional action details	JSON or descriptive text
ip_address	varchar	IP Address	45	User's IP address	"192.168.1.1", "2001:db8::1"



Table 13
Beneficiary

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	0000000001
application_id	int	Integer	11	Foreign key to pagibig_applications	0000000500
first_name	varchar	String	15	Beneficiary's first name	"Juan"
middle_name	varchar	String	15	Beneficiary's middle name	"Santos"
last_name	varchar	String	15	Beneficiary's last name	"Dela Cruz"
suffix	varchar	String	2	Name suffix	"Jr.", "Sr."
contact_number	varchar	String	12	Phone number	"09171234567"
region	varchar	String	25	Geographic region	"Region IV-A"
province	varchar	String	25	Province	"Negros Occidental"
city	varchar	String	25	City/Municipality	"Victorias City"
barangay	varchar	String	25	Barangay	"Barangay 6"
address	varchar	String	100	Complete address	"Malihao St., Estrella Village"
site_number	varchar	String	10	Housing site number	"Site A"
phase_number	varchar	String	10	Phase number	"Phase 1"
block_number	varchar	String	10	Block number	"Block 5"
lot_number	varchar	String	10	Lot number	"Lot 23"
housing_unit_id	int	Integer	11	Foreign key to housing_units	00000000025
created_at	timestamp	YYYY-MM-DD HH:MM:SS	-	Record creation time	"2024-01-15 10:30:00"



Table 14
Beneficiary Verification

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
user_id	int	Integer	11	Foreign key to users table	00000000150
first_name	varchar	String	15	Applicant's first name	"Pablito"
middle_name	varchar	String	15	Applicant's middle name	"Constantino"
last_name	varchar	String	15	Applicant's last name	"Doromal"
suffix	varchar	String	2	Name suffix	"Jr."
contact_number	varchar	String	12	Contact number	"09171234567"
address	varchar	String	100	Residential address	"Malihao St., Estrella Village"
block_number	varchar	String	10	Block number	"Block 10"
lot_number	varchar	String	10	Lot number	"Site 15"
housing	varchar	String	20	Type of housing project	"PAG-IBIG Housing"
beneficiary	varchar	String	7	Current beneficiary status	"Active", "Pending"
concern_details	varchar	String	150	Details of concern	"Need to verify ownership"
identity_proof	varchar	String	150	File path for ID proof	"uploads/ids/id123.jpg"
project_details	varchar	String	150	File path for project docs	"uploads/projects/project1.pdf"
status	varchar	String	8	Verification status	"Pending", "Approved", "Rejected"
created_at	timestamp	YYYY-MM-DD HH:MM:SS	-	Creation timestamp	"2024-01-15 10:30:00"
updated_at	timestamp	YYYY-MM-DD HH:MM:SS	-	Last update timestamp	"2024-01-16 14:20:00"
rejection	varchar	String	150	Reason for rejection	"Incomplete documents"



Table 15
Email Verification

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
user_id	int	Integer	11	Foreign key to users table	00000000200
email	varchar	Email format	50	Email to verify	"pablitdo@gmail.com"
code	varchar	String	6	Verification code	"ABC123"
expiration	datetime	YYYY-MM-DD HH:MM:SS	-	Code expiration time	"2024-01-15 23:59:59"
used	tinyint	Boolean (0/1)	1	Whether code was used	0 or 1
created_at	timestamp	YYYY-MM-DD HH:MM:SS	-	Creation timestamp	"2024-01-15 10:30:00"

Table 16
User Housing Assignments

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
user_id	int	Integer	11	Foreign key to users table	00000000250
form_type	varchar	String	50	Type of form generated	"Application Form", "Clearance"
file_path	varchar	String	150	Path to generated file	"forms/ app_123.pdf"
created_at	datetime	YYYY-MM-DD HH:MM:SS	-	Creation datetime	"2024-01-15 10:30:00"



Table 17
Admin Housing Assignments

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
application_id	int	Integer	11	Foreign key to pagibig_applications	00000000300
housing_unit_id	int	Integer	11	Foreign key to housing_units	00000000040
site_number	varchar	String	10	Site number	"Site B"
phase_number	varchar	String	10	Phase number	"Phase 2"
block_number	varchar	String	10	Block number	"Block 12"
lot_number	varchar	String	10	Lot number	"Lot 8"
new_address	varchar	String	150	New assigned address	"Barangay 13, Villa Victorias"
assignment	timestamp	YYYY-MM-DD HH:MM:SS	-	Assignment date	"2024-01-15 10:30:00"
region	varchar	String	25	Region	"Region VI"
province	varchar	String	25	Province	"Negros Occidental"
city	varchar	String	25	City	"Victorias City"
barangay	varchar	String	25	Barangay	"Barangay 6"
address	varchar	String	100	Complete address	"Barangay 13, Villa Victorias"

Table 18
Housing Units

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
name	varchar	String	50	Housing unit name	"Villa Victorias"
location	varchar	String	50	General location	"Victorias City"
units_available	int	Integer	11	Available units count	00000000050
total_units	int	Integer	11	Total units count	00000000100
region	varchar	String	50	Region	"Region VI"
province	varchar	String	50	Province	"Negros Occidental"
city	varchar	String	50	City	"Victorias City"
barangay	varchar	String	50	Barangay	"Barangay 6"



Table 19
Housing Unit Lots

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
housing_unit_id	int	Integer	11	Foreign key to housing_units	25
site	varchar	String	10	Site identifier	"Site A"
phase	varchar	String	10	Phase identifier	"Phase 1"
block	varchar	String	10	Block identifier	"Block 5"
lot	varchar	String	10	Lot identifier	"Lot 23"
latitude	decimal	Decimal (10,8)	-	GPS latitude	14.599512
longitude	decimal	Decimal (11,8)	-	GPS longitude	120.98422
photo	varchar	File patch	50	Photo filename	"lot_23.jpg"
status	enum	Available/ Assigned/ Occupied	-	Lot status	"Available"

Table 20
Housing Unit Monitoring

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
beneficiary_id	int	Integer	11	Foreign key to beneficiaries	00000000100
housing_unit_id	int	Integer	11	Foreign key to housing_units	00000000025
photo	varchar	File Path	150	Monitoring photo	"monitor_001.jpg"
latitude	decimal	Decimal (10,8)	-	GPS latitude	14.599512
longitude	decimal	Decimal (11,8)	-	GPS longitude	120.98422
location_verified	tinyint	Boolean (0/1)	1	Location verification	0 or 1
status	enum	Pending/Verified/Rejected	-	Monitoring status	"Pending"
marked	tinyint	Boolean (0/1)	1	Active status marker	0



Table 21
Non Pag-Ibig Applicants

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
user_id	int	Integer	11	Foreign key to users	00000000300
first_name	varchar	String	15	First name	"Pablito"
middle_name	varchar	String	15	Middle name	"Constantino"
last_name	varchar	String	15	Last name	"Doromal"
suffix	varchar	String	2	Name suffix	"Jr."
date_of_birth	date	YYYY-MM-DD	-	Birth date	"1985-05-15"
contact_number	varchar	String	12	Contact number	"09171234567"
monthly_income	decimal	Decimal (10,2)	-	Monthly income	25000.00
real_property	varchar	File path	150	Property cert file	"uploads/property.pdf"
income_proof	varchar	File path	150	Income proof file	"uploads/income.pdf"
valid_id	varchar	File path	150	Valid ID file	"uploads/id.jpg"
document_selfie	varchar	File path	150	Selfie with document	"selfie.jpg"
status	enum	Pending/ Approved/ Rejected/ Relocated	-	Application status	"Pending"
address	varchar	String	100	Complete address	"Malihao St., Estrella Village, Victorias City"
region	varchar	String	50	Region	"Region VI"
province	varchar	String	50	Province	"Negros Occidental"
city	varchar	String	50	City	"Victorias City"
barangay	varchar	String	50	Barangay	"Barangay 6"
latitude	decimal	Decimal (10,7)	-	GPS latitude	14.211575
longitude	decimal	Decimal (10,7)	-	GPS longitude	121.165641
rejection_reason	varchar	String	50	Rejection reason	"Incomplete requirements"
housing_project	varchar	String	50	Housing project name	"Villa Victorias"



Table 22
User Notification Table

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
user_id	int	Integer	11	Foreign key to users	00000000150
title	varchar	String	50	Notification title	"Application Approved"
message	text	Text	50	Notification message	"Your application has been approved."
type	enum	housing_application/utility_application/verification/property_transfer	-	Notification type	"housing_application"
reference_id	int	Integer	11	Related record ID	00000000200
is_read	tinyint	Boolean (0/1)	1	Read status	0
created_at	timestamp	YYYY-MM-DD HH:MM: SS	-	Creation timestamp	"2024-01-15 10:30:00"

Table 23
Admin Notification Table

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
user_id	int	Integer	11	Foreign key to users	00000000200
title	varchar	String	50	Notification title	"Application Recieved"
type	enum	housing_application/utility_application/verification/property_transfer	-	Notification type	"housing_application"
reference_id	int	Integer	11	Related record ID	00000000200
is_read	tinyint	Boolean (0/1)	1	Read status	0
created_at	timestamp	YYYY-MM-DD HH:MM: SS	-	Creation timestamp	"2024-01-15 10:30:00"



Table 24
Property Transfer Table

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
user_id	int	Integer	11	Foreign key to users	00000000150
first_name	varchar	String	15	First name	"Pablito"
middle_name	varchar	String	15	Middle name	"Constantino"
last_name	varchar	String	15	Last name	"Doromal"
suffix	varchar	String	2	Name suffix	"Jr."
contact_number	varchar	String	12	Contact number	"09171234567"
lot_number	varchar	String	10	Lot number	"Lot 15"
original_lot_assignment	date	YYYY-MM-DD	-	Original assignment date	"2020-05-10"
beneficiary_proof	varchar	File path	150	Beneficiary proof	"uploads/beneficiary.pdf"
valid_id	varchar	File path	150	Valid ID	"uploads/id.jpg"
lot_assignment	varchar	File path	150	Lot assignment doc	"uploads/assignment.pdf"
remarks	varchar	String	50	Additional remarks	"Transfer to family member"
status	varchar	String	10	Transfer status	"Pending", "Approved"
created_at	timestamp	YYYY-MM-DD HH:MM:SS	-	Creation timestamp	"2024-01-15 10:30:00"

Table 25
Property Transfer Record Table

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
from_beneficiary_id	int	Integer	11	Source beneficiary	00000000100
to_beneficiary_id	int	Integer	11	Destination beneficiary	00000000101
housing_unit_id	int	Integer	11	Housing unit ID	00000000025
site_number	varchar	String	10	Site number	"Site A"
phase_number	varchar	String	10	Phase number	"Phase 1"
block_number	varchar	String	10	Block number	"Block 5"



Table 26
Report Table

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
report_name	varchar	String	50	Report name	"Monthly Applications"
report_type	varchar	String	25	Report type	"Applications", "Housing"
project_filter	varchar	String	25	Project filter	"Pag-IBIG", "Non-PagIBIG"
start_date	date	YYYY-MM-DD	-	Report start date	"2024-01-01"
end_date	date	YYYY-MM-DD	-	Report end date	"2024-01-31"
export_format	varchar	String	5	Export format	"Excel"
file_path	varchar	File path	150	Report file path	"reports/jan2024.pdf"
generated_by	int	Integer	11	Generator user ID	00000000005
created_at	timestamp	YYYY-MM-DD HH:MM: SS	-	Creation timestamp	"2024-01-15 10:30:00"

Table 27
User Table

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
full_name	varchar	String	50	User's full name	"Pablito Constatino Doromal"
suffix	varchar	String	2	Name suffix	"Jr."
username	varchar	String	20	Username	"pablitodoromal"
email	varchar	Email format	50	Email address	"pablitodoromal@gmail.com"
face_descriptor	text	JSON format	500	Facial recognition data	"{descriptor: [...]}"
failed_login	int	Integer	11	Failed login count	00000000002
account_locked	tinyint	Boolean (0/1)	1	Account lock status	1
locked_at	datetime	YYYY-MM-DD HH:MM: SS	-	Lock timestamp	"2024-01-15 10:30:00"



Table 28
Utility Connection Table

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
user_id	int	Integer	11	Foreign key to users	00000000150
first_name	varchar	String	15	First name	"Pablito"
middle_name	varchar	String	15	Middle name	"Constantino"
last_name	varchar	String	15	Last name	"Doromal"
suffix	varchar	String	2	Name suffix	"Jr."
contact_number	varchar	String	12	Contact number	"09171234567"
housing_unit_address	varchar	String	100	Housing unit address	"Barangay 13, Villa Victorias"
connection_type	enum	Water/ Electricity	-	Utility type	"Water"
block_number	varchar	String	10	Block number	"Block 10"
lot_number	varchar	String	10	Lot number	"Lot 15"
residence_proof	varchar	File path	150	Residence proof	"uploads/ residence.pdf"
valid_id	varchar	File path	150	Valid ID	"uploads/id.jpg"
assignment_doc	varchar	File path	150	Assignment document	"uploads/ assignment.pdf"
status	varchar	String	10	Application status	"Pending", "Approved"
rejection_reason	varchar	String	50	Rejection reason	"Incomplete documents"

Table 29
Visitors Log Table

Field Name	Data Type	Data Format	Field Size	Description	Example
id	int	Integer	11	Primary key	00000000001
name	varchar	String	50	Visitor name	"Pablito Constantino Doromal"
date	datetime	YYYY-MM-DD:HH:MM:SS	-	Visit date/time	"2024-01-15 10:30:00"
address	varchar	String	150	Visitor address	"Barangay 13, Villa Victorias"
contact_number	varchar	String	12	Contact number	"09171234567"



3.8 Hardware and Software Requirements

Hardware and Software Requirements provided detailed minimum requirements needed to support the developed system's functions and executions.

Recommended Minimum Hardware Requirements

- Storage - 1TB.
- RAM - 16GB.
- Processor Speed: 3.00GHz.
- Processor – Intel Core 10th gen or newer or AMD Ryzen 7 and above.
- Internet Connectivity – At least 10Mbps.

Recommended Software Requirements

- Operating System - Windows 11, Ubuntu, and other Linux Distributions.
- Software development tool – XAMPP v3.3.0.
- Web Browser - Google Chrome, Mozilla Firefox.
- Text Editor/IDE - Visual Studio Code.

3.9 Project Timeline

The Gantt Chart displayed the project's timeframe, which spanned from April to August 2025. It divided work into four phases: planning, documentation, coding and designing, and testing. The chart showed when tasks started and concluded, their overlaps, and periods of peak activity. This helped the team monitor progress and maintain schedule adherence. The PERT/CPM diagram illustrated the sequence of tasks, their durations, start and end times, and slack.



It identified the critical route, indicating tasks that could not be delayed without impacting the overall project timeline. Additionally, it highlighted activities with permissible delays, revealing schedule flexibility. Together, the Gantt Chart and PERT/CPM provided a comprehensive view of the project timeline, supporting effective planning, coordination, and timely completion of the Informal Settler Relocation Management System project.

3.10 System Evaluation Tools

Block Box Testing

When we built the Informal Settler Relocation Management System, we used PHPUnit to run automated unit tests on each module to make sure it worked according to the system's requirements. By performing structured test cases that evaluated whether each part gave the right output, we were able to make sure that important functions including processing resident data, handling relocation documents, and upgrading the status worked as it should.

During the first round of testing, many features did not work as planned, especially the one that was supposed to handle uploading numerous documents. The development team found and fixed the issues that caused the system logic to be inconsistent after the tests failed. After making the changes, we ran the identical PHPUnit test scripts again to make sure that the problems were fixed and that no new errors came up as a result.



This re-testing process is essential for maintaining system reliability, which supports Sharma (2020) who noted that automated testing reduces defects early in development. After the corrections, all test cases passed, confirming the improved stability and functionality of the system, consistent with the findings of Cruz and Reyes (2020), Lopez (2020), and Sharma (2020).

White Box Testing

Pest PHP was a big part of our White-Box Testing method, which looks at how the system works and how it behaves on the inside to make sure that each part works correctly at the code level. Black-Box Testing only looks at results that users can see, but Pest PHP lets developers directly test the structure, logic flow, and expected outputs of particular functions using expressive and lightweight test cases.

According to the 2020 study "Lightweight Testing Approaches for Modern PHP Applications Using Pest PHP," Pest PHP lets developers design tests that are easy to read and clean, which makes code assessment faster and more accurate (Recio et. al., 2021). During our implementation, Pest PHP helped us find logical errors in the modules that handle relocation requests and process multiple documents. These problems would not have been easy to find through functional testing alone. Research shows that simplified automated testing methods help speed up debugging and maintenance (ZigzagWeekly, 2022).



The development team was able to find bugs quickly using the failed Pest tests. After fixing the problems, running the same set of tests again showed that the fixes worked and didn't cause any new problems. This method of testing over and over again is in line with what the literature says about how automated unit testing makes software more reliable and lowers long-term maintenance costs (Muczyński et. al., 2022). Through the integration of Pest PHP, the system achieved greater stability, security, and maintainability, ensuring it performs correctly both internally and externally.

Table 30

Likert Scale, Range, And Description For ISO 9241-11:2018 System Software Usability Evaluations

Range	Verbal Interpretation
4.21 - 5.00	Strongly Agree
3.41 - 4.20	Agree
2.61 - 3.40	Neutral
1.81 - 2.60	Disagree
1.00 - 1.80	Strongly Disagree

The usability test of the system employed the ISO 9241-11:2018 Software Evaluation Tool. According to the International Organization for Standardization (2018), ISO 9241-11:2018 provided a framework for understanding usability and its function when users interacted with systems, built environments, products, and services. It assessed the degree of usability of a website used by users to achieve specific goals effectively (effectiveness), efficiently (efficiency), and satisfactorily (satisfaction). Regarding usability measures based on ISO 9241-11:2018,



guidance included: (a) Effectiveness, which measured the accuracy and perfection in task completion; (b) Efficiency, which evaluated ease of use from a system time perspective in providing necessary data and information; and (c) Satisfaction, which gauged user satisfaction during system interaction to achieve objectives. Respondents rated the usability of the developed system on a 5-point Likert scale. The questionnaire results were analyzed to determine the influence of the three usability standards: satisfaction, effectiveness, and efficiency.

Table 31

Likert Scale, Range, And Description For ISO 25010:2011 System Software Usability Evaluations

Range	Verbal Interpretation
4.21 - 5.00	Strongly Agree
3.41 - 4.20	Agree
2.61 - 3.40	Neutral
1.81 - 2.60	Disagree
1.00 - 1.80	Strongly Disagree

Suryadi and Sulistiyanis (2022) study emphasized the importance of the ISO/IEC 25010:2011 standard as a comprehensive and internationally recognized method for assessing software quality. Their research demonstrated how the model provided a clear framework for evaluating whether a system met its functional requirements and adhered to high operational quality standards. The study showcased the application of ISO/IEC 25010 in assessing a real-world application's functionality, usability, and technical robustness. The standard outlined eight primary quality characteristics that collectively determined the



overall quality of software. Functional Suitability was used to verify if the system delivered the correct functions in the appropriate manner and timing. Performance Efficiency evaluated response times, processing speeds, and resource utilization. Compatibility ensured that systems or environments could operate together seamlessly. Usability measured how easily users could navigate the system and their overall satisfaction. Reliability assessed the system's consistency and its ability to maintain smooth operation over time. Security focused on protecting data privacy, integrity, and access control. Maintainability examined how simple it was to fix, update, or enhance the system. Portability evaluated the system's ability to function across different platforms.

3.11 Respondents of the Study

Table 32
Respondents of the Study

Respondents	f	%
Beneficiary	11	31.43%
Clients	19	54.29%
IT Experts	5	14.28%
Total	20	100%

The table above shows the respondents of the study which consisted of three major groups who play distinct roles in the informal settler relocation process and in the evaluation of the proposed system. The largest group were the Clients, representing 54.29% (19 individuals). These respondents are informal settlers who



provided critical insights based on their lived experiences with the relocation procedures. Their perspectives help determine how well the system addresses their needs, concerns, and expectations when moving. The second group was the Beneficiaries, which included 31.43% (11 people) of the beneficiaries. They are the major people who run relocation programs and use and benefit from the system that was created, thus their opinion is quite important. They manage documentation, resident profiles, case monitoring, and coordination with stakeholders. Understanding their workflow and challenges allows the system to be designed to reduce manual work, enhance data accuracy, and support more efficient service delivery. Lastly, IT Experts made up 14.28% (5 individuals). Their role in the study was to assess the system's technical quality, including functionality, usability, and alignment with software standards. Although the smallest group, their evaluations ensure that the system met essential technical requirements. Overall, the mix of informal settlers, CHSUD officials, and IT specialists gives a full and fair review. This wide range of answers makes sure that the system that was created is user-centered, works well in practice, and is technically sound. This will help both the CHSUD and the informal settler community.



3.12 Ethical Consideration

As researchers were developing an Informal Settler Relocation Management System, we recognized the importance of ethical considerations in this study. Our commitment to ethical conduct was integral to safeguarding the rights, privacy, and well-being of the individuals involved. Before conducting interviews with beneficiaries, we obtained informed consent. Participants were fully informed about the purpose, nature, and potential uses of the study. All information revealed during the interviews was kept confidential. We ensured that any private information shared was secure. We promised to use the information solely to develop a management system for CHSUD.



CHAPTER IV

RESULTS AND DISCUSSIONS

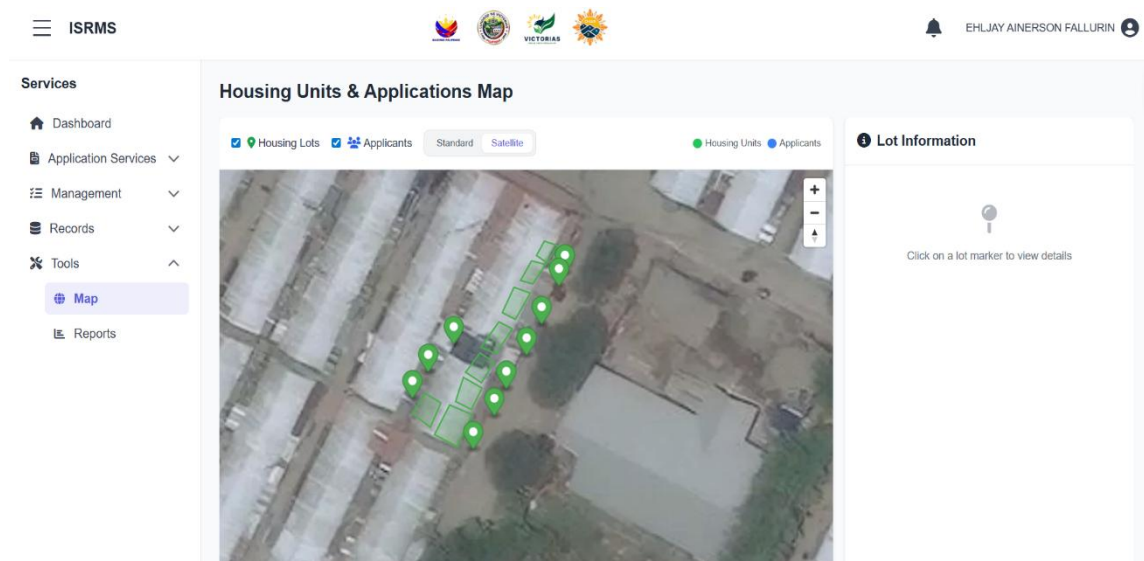
This chapter presented the results gathered from the development and testing of the Informal Settler Relocation Management System. The results was organized based on research objectives, covering the system's technical features, functional testing (alpha testing), and evaluation based on user and expert feedback. Screenshots, tables, and discussions was provided to support and explain the findings.

4.1 Results of the Study

The first objective of this study was to develop an interactive GIS map that shows informal settlements, available housing units, and relocation sites.

Figure 27

Objective 1.1 - Housing Units & Application Maps





As shown in the previous page, Figure 27 the interactive GIS map (Specific Objective 1.1) provided a visual and real-time overview of informal settlements, available housing units, and relocation sites within Victorias City. The system displayed each housing lot as a color-coded marker green for housing units and blue for applicants. When a marker was clicked, a detailed information panel appeared on the right side of the map, presenting the lot's exact location, status, unit details, and other relevant records. The map remained synchronized with the system's dashboard statistics, ensuring that any changes in lot availability or application updates were reflected immediately without requiring page reloads. Built-in filters enabled users to switch between housing lot views and application markers. This functionality allowed CHSUD staff to better visualize population distribution, identify available vacancies, and monitor relocation progress. The mapping application utilized satellite imagery from MapBox to accurately display communities and relocation sites, making navigation and planning easier for both staff and beneficiaries. Overall, this feature enhanced data clarity, accelerated decision-making, and supported the strategic management of relocation projects.



Figure 28
Objective 1.2 - Housing Application

ISRMS

Services

- Dashboard
- Application Services
- Management
- Records
- Tools

Housing Application

About this Service

CHSUD, in coordination with DHSUD, Pag-IBIG fund, and other relevant national and local housing agencies, endeavors to provide government employees and Pag-IBIG members who belong to informal settler families (ISFs).

Requirements

- Certificate of No Real Property
- Certificate of Net Pay OR any other proof of income
- Pag-IBIG MID Number
- Photocopy of valid ID with specimen signature

First Name Middle Name Last Name Suffix

Contact Number

Date of Birth

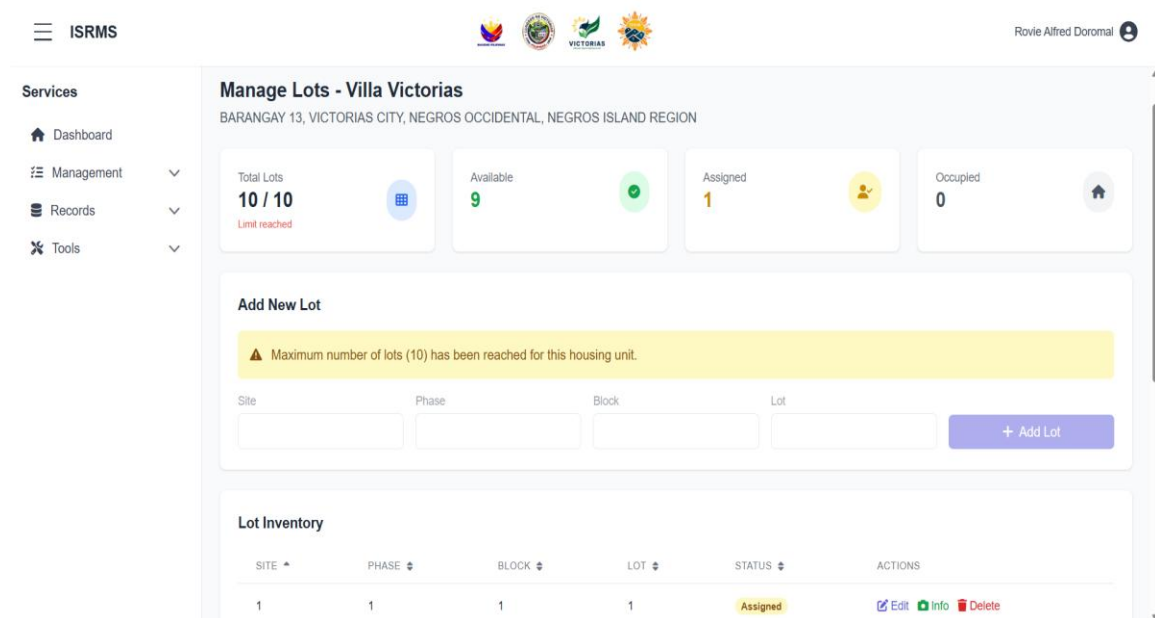
Region Province

As shown above, Figure 28, the housing application and management module (Specific Objective 1.2) enabled applicants to conveniently submit and track their housing applications through a dedicated interface within the system. Applicants were able to view the current status of their applications, including Pending, Approved, Rejected, or For Evaluation, along with progress indicators that showed the stage of the assessment process. This functionality enhanced transparency and informed beneficiaries of their position in the relocation process. The system dashboard and GIS map were automatically updated with each transaction, ensuring that statistics related to housing availability and relocation remained accurate and current. Additionally, the module maintained a complete history log for each application, allowing staff to review all communications,

modifications, and decisions made throughout the application process. By integrating applicant and staff functions into a single system, the module reduced paperwork, improved workflow efficiency, and supported faster and more organized decision-making.

Figure 29

Objective 1.3 - Vancy and Allocation Tracking Feature



As shown in the previous page, Figure 29, the vacancy and allocation tracking function (Specific Objective 1.3) assisted CHSUD staff in monitoring the availability of housing units and ensuring that qualified applicants were assigned to appropriate lots. The administrative interface provided a real-time summary of total lots, available units, assigned units, and occupied units for each relocation site. When a site reached its maximum lot capacity, the system prevented the addition of further lots and displayed an on-screen notification to inform the user.



This mechanism ensured data accuracy and compliance with site allocation rules. The Lot Inventory section listed all housing units along with their corresponding site, phase, block, and lot numbers, as well as their current status, categorized as Available, Assigned, or Occupied. Built-in actions such as Edit, Info, and Delete allowed staff to easily update lot information and status, ensuring that the database remained current and free from inconsistencies. When a lot was assigned to an approved application, the unit's status was immediately updated across the entire system. Both the GIS map and the system dashboard reflected these changes in real time, displaying updated availability figures. This integrated tracking process facilitated efficient housing allocation, prevented duplicate or incorrect assignments, and supported fair and transparent distribution of housing units across all relocation sites.

Figure 30
Objective 1.4 - Housing Unit Inventory Managment System

ISRMS

Services

- Dashboard
- Application Services
- Management
- Records
- Tools

Housing Units

Add Housing Unit

Region: NEGROS ISLAND REGION Province: NEGROS OCCIDENTAL

City/Municipality: VICTORIAS CITY Barangay: BARANGAY 13

Name: Total Units:

Add Unit

Search by name or location

ID	NAME	REGION	PROVINCE	CITY/MUNICIPALITY	BARANGAY	TOTAL UNITS	AVAILABLE UNITS
----	------	--------	----------	-------------------	----------	-------------	-----------------



As shown in the previous page, Figure 30 the housing unit inventory management system (Specific Objective 1.4) provided CHSUD staff with a centralized tool for creating, organizing, and maintaining detailed records of all relocation sites and their corresponding housing units. Staff used the "Add Housing Unit" interface to enter important location information, including the area, province, city/municipality, and barangay, as well as the name of the housing site and the total number of units. This enabled data to be entered in a uniform and organized manner, ensuring that all registered relocation areas in Victorias City were consistent. Once added, each housing site was displayed in an organized inventory table that listed its name, location classification, total units, available units, and a set of management actions. Staff could easily manage the site to configure or update its internal lots, edit site information if changes occurred, or delete outdated entries. The inventory automatically updated whenever lots were assigned, occupied, or modified, allowing the system to maintain accurate, real-time counts of available units. By consolidating all housing sites into a single, searchable record system, this feature improved administrative efficiency, reduced errors, and supported effective planning and resource allocation for the city's relocation programs. The second objective of this study was to test the functionality and reliability of the system's core features. A total of 94 test cases were designed and executed to evaluate how well the main components of the Informal Settler Relocation Management System (ISRMS) functioned. It demonstrated that the ISRMS was reliable and met the functional requirements of CHSUD's relocation management operations.



The table below presents the outcomes of test cases during execution:

Table 33
Black-Box Testing of the Alpha Testing of User Registration & Authentication

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-001	User Registration	User details and facial recognition stored; verification email sent; user can login after verification	User registration successful; information stored in database	Pass	Fully functional
TC-002	User Login	User successfully logged in; redirected to user system interface	User successfully logged in and entered system interface	Pass	Fully functional
TC-003	User Forgot Password	Code sent to email; facial recognition verified; password successfully reset; user can login with new password	Code sent successfully via Gmail; facial recognition scans and matches face descriptor; if verified, proceeds to password reset	Pass	Fully functional
TC-004	User Login with Invalid Credentials	Login rejected; error message "Invalid username or password" displayed; user remains on login page	Error message displayed; login failed as expected	Pass	Fully functional
TC-005	User Logout	User logged out; session cleared; redirected to login page; cannot access.	Session terminated successfully; redirected to login page	Pass	Fully functional



As shown in the previous page, Table 33 evaluated the User Registration and Authentication module to ensure that system access was secure, reliable, and functioning as intended. Both positive and negative test cases were executed using black-box testing during the alpha testing phase. Positive testing verified that users were able to successfully register accounts, with personal details and facial recognition data correctly stored in the database, verification emails properly sent, and access granted only after successful verification. The login functionality worked as expected, allowing verified users to access the system interface. The forgot password feature was also validated, ensuring that reset codes were sent via email, facial recognition was accurately verified, and users could securely reset and reuse their accounts. Overall, all test cases passed successfully, indicating that the authentication module is fully functional, secure, and dependable. The results confirm that the system can be trusted to manage user identities, protect sensitive data, and ensure controlled access to system features.



Table 34
Black-Box Testing of the Alpha Testing of User Application Submission

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-008	User Submit Housing Application	Application submitted successfully; status "Pending"; success message; form reset; trackable via Check Status	Application submitted successfully; status recorded as "Pending"; success message displayed; form cleared; user able to view and track via check status	Pass	Fully functional
TC-009	Housing Application Form Validation	Form prevents submission; red error messages appear below each required field; user cannot proceed without completing all required fields	Validation messages displayed for all required fields; submission blocked	Pass	Fully functional
TC-010	Housing Application Document Upload Validation	Invalid file types rejected; oversized files rejected; valid files accepted; preview shown; can remove uploaded files	Invalid files rejected with error messages; valid files accepted successfully	Pass	Fully functional
TC-011	Housing Application Cascading Location Dropdowns	Dropdowns cascade correctly; changing parent dropdown resets child dropdowns;	Cascading dropdowns work correctly; locations populate as expected	Pass	Fully functional



As shown in the previous page, Table 34 evaluated the User Application Submission module during the alpha testing phase using black-box testing to verify the accuracy, reliability, and usability of the application process. This module covered application workflows for Housing, Property Transfer, Utility Connection, and Beneficiary Verification, ensuring that users could submit and manage applications correctly. Positive testing confirmed that users were able to successfully submit applications, with submitted records accurately stored in the database and automatically assigned a “Pending” status. The system displayed appropriate success messages, reset input forms after submission, and allowed users to track their application status through the Check Status feature. User interface components such as cascading location dropdowns, date pickers, radio buttons, and document upload fields functioned correctly and enhanced ease of use. Overall, all test cases passed successfully, demonstrating that the Application Submission and Management module is fully functional, user-friendly, and robust. The results confirm that the system can reliably handle various application types while enforcing proper input validation and data integrity throughout the submission process.



Table 35
Black-Box Testing of the Alpha Testing of User Application Status Tracking

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-021	User Check Housing Application Status	Application status page loads; all user applications listed; status badges show correct current phase; progress bar shows current stage; view details shows complete application info	Application list displayed with correct status; progress tracker shows current phase; details viewable	Pass	Fully functional
TC-022	User Check Property Transfer Status	Status page loads; applications listed; status badges correct; progress tracker shows current stage; details viewable	Status page loaded successfully; application details visible	Pass	Fully functional
TC-023	User Check Utility Connection Status	Status page loads; applications listed with connection type; status badges correct; details viewable	Status page loaded successfully; connection type displayed correctly	Pass	Fully functional
TC-024	User Check Beneficiary Verification Status	Status page loads; applications listed; status badges correct; details viewable.	Status page loaded successfully; concern details visible	Pass	Fully functional



As shown in the previous page, Table 35 evaluated the User Application Status Tracking module during the alpha testing phase using black-box testing to verify the accuracy, clarity, and reliability of application progress monitoring across multiple service modules. This functionality was designed to allow users to independently monitor the status of their submitted applications for Housing, Property Transfer, Utility Connection, and Beneficiary Verification. Positive testing confirmed that the application status pages loaded correctly for each module and that all user-submitted applications were properly listed. Status badges accurately reflected the current processing phase, while progress trackers visually indicated the stage of each application within the workflow. Users were also able to view detailed application information, ensuring transparency and completeness of displayed data. All test cases passed successfully, indicating that the Application Status Tracking module is fully functional, user-friendly, and dependable. The findings confirm that the system effectively provides users with clear, accurate, and up-to-date information, supporting transparency and confidence in the overall application process.



Table 36
Black-Box Testing of the Alpha Testing of Housing Application Management

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-025	Admin View Housing Applications List	Applications table loads; all columns display; status badges colored correctly (Pending=Yellow, Verification/Processing=Blue, Approved/Relocated=Green, Rejected=Red) Search filters results in real-time; shows matching records only; partial matches work; case-insensitive search	Table loaded with all applications; status badges display correctly; pagination works	Pass	Fully functional
TC-026	Admin Search Housing Applications	Status filter works correctly; filtered results show only selected status; "All Status" shows all records	Search, filter and sorting functions work correctly	Pass	All search/filter/sort functions fixed and working
TC-027	Admin Filter Housing Applications by Status	Details modal opens; all application data displayed; documents viewable; modal closes properly	Filter function works correctly	Pass	Filter functionality fixed and working
TC-028	Admin View Housing Application Details	Application details modal shows complete info successfully		Pass	Fully functional

The test table above checked the Admin Housing Applications Management process to make sure that administrators can easily keep track of, update, and manage housing applications. Tests checked that the list loading, searching, and filtering were all correct, as well as the detailed application views, status updates, relocations, and relocation histories. All eight tests were successful.



Table 37
Black-Box Testing of the Alpha Testing of Porperty Transfer Management

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-033	Admin View Property Transfers List	Property transfers table loads; all columns display; pagination works; status badges colored correctly	Table loaded successfully; pagination functional	Pass	Fully functional
TC-034	Admin Search Property Transfers	Search filters results; matches name or email; partial matches work; case-insensitive	Search bar works correctly and filters results	Pass	Search functionality fixed and working
TC-035	Admin Update Property Transfer Status	Update modal opens; status updates successfully; remarks saved; status badge reflects change	Status updates successfully with all changes saved	Pass	Update functionality fixed and working
TC-036	Admin Delete Property Transfer	Confirmation dialog appears; cancel works; confirm deletes record; success message shown; record removed from list	Delete functionality works correctly with confirmation	Pass	Fully functional

The test table above checked the Admin Property Transfer Management process to make sure that administrators can easily monitor, search for, update, and manage property transfer records. Tests made sure that the list loaded correctly, the status badges were correct, the pagination worked, and the search,



update, and deletion processes were all dependable. All of the tests succeeded, which shows that the module is stable, easy to use, and completely effective for administrative purposes.

Table 38

Black-Box Testing of the Alpha Testing Utility Connection Management

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-037	Admin View Utility Connections List	Utility connections table loads; all columns display; connection types clear; pagination works	Table loaded successfully; connection types displayed correctly	Pass	Fully functional
TC-038	Admin Filter Utility Connections by Status	Status filter works; shows only selected status; "All" shows everything	Filter status works correctly	Pass	Filter functionality fixed and working
TC-039	Admin Update Utility Connection Status	Update modal opens; status updates; remarks saved; activity logged	Status updates successfully with all changes saved	Pass	Update functionality fixed and working
TC-040	Admin View Utility Connection Details	Details modal opens; all info displayed; documents viewable; modal closes properly	View button functional; details modal shows complete info	Pass	Fully functional

The test table above checked the Admin Utility Connection Management procedure to make sure that administrators can easily browse, filter, update, and review utility connection records. Tests confirmed that the list loading, status-



based filtering, update processes, notes logging, and detailed record viewing all worked well. All of the tests succeeded, which means that the module works, is accurate, and can be counted on for administrative tasks.

Table 39
Black-Box Testing of Alpha Testing of Beneficiary Management

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-041	Admin View Beneficiary Verifications List	Verifications table loads; all columns display; status badges correct; project types shown	Table loaded successfully; all data displayed correctly	Pass	Fully functional
TC-042	Admin Search Beneficiary Verifications	Search filters results; matches name or email; real-time filtering	Search bar works correctly and filters results	Pass	Search functionality fixed and working
TC-043	Admin Update Verification Status	Status updates successfully; remarks saved; status badges reflect changes; activity logged	Status updates successfully with all changes saved	Pass	Update functionality fixed and working
TC-044	Admin View Beneficiaries List	Beneficiaries table loads; all columns display; dates formatted correctly; records visible	Table loaded successfully; all beneficiaries displayed	Pass	Fully functional
TC-045	Admin Search and Filter Beneficiaries	Search filters by name/ID; date filter narrows to range; clear filters resets; filters work independently	Search and filter functions work correctly	Pass	Search and filter functionality fixed and working
TC-046	Admin Sort Beneficiaries Table	Column sorting works; toggles between ascending/descending; sort icons indicate direction	Sorting works correctly without redirecting	Pass	Sort functionality fixed and working



As shown in the previous page, Table 39 checked that the Admin Beneficiary Management module works, making sure that administrators can easily check applicants, manage beneficiaries, and find records. Tests checked that loading lists, status badges, search and filtering, verification updates, and sorting columns all worked. All scenarios passed, which means that the module works, is accurate, and can be trusted for administrative tasks.

Table 40
Black-Box Testing of the Alpha Testing of Housing Unit Management

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-047	Admin Add Housing Unit	Housing unit added successfully; appears in table; location cascades correctly; units count saved	Successfully added new housing unit; all action buttons functional	Pass	Fully functional
TC-048	Admin Edit Housing Unit	Edit modal opens; data pre-filled; updates save successfully; table reflects changes immediately	Edit button functional; updates save correctly	Pass	Fully functional
TC-049	Admin Delete Housing Unit	Confirmation dialog appears; cancel preserves record; confirm deletes record; table updates immediately	Delete button functional	Pass	Fully functional



As shown in the previous page the Table 40 checked the Admin Housing Unit Management module to make sure that administrators can correctly add, amend, and remove housing units. Tests confirmed that adding units worked, editing was dependable with rapid updates, and deleting was safe. All test scenarios passed successfully.

Table 41
Black-Box Testing of the Alpha Testing of Mapping and Reports

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-050	Admin View Interactive Map	Map loads; filters toggle markers; zoom controls work; navigation arrows move map; markers clickable; popups show details	Successfully integrated mapping; admin can navigate Philippines geography	Pass	Fully functional
TC-051	Admin View Map Statistics Panel	Statistics panel visible; all counts accurate; updates reflect database state	Statistics show accurate counts that update in real-time	Pass	Fully functional
TC-052	Admin Generate Custom Report	Report generates; appears in list; downloads as PDF; contains filtered data	Report generated and downloadable as PDF	Pass	Download functionality fixed and working
TC-053	Admin Generate Certificate Forms	Each form type generates correct PDF; data populated accurately; PDFs downloadable	All certificate forms generated successfully with correct data	Pass	Fully functional



As shown in the previous page, Table 41 checked the Admin Mapping and Reports module to make sure that administrators can see geographic data, get system statistics, and make reports and certificates correctly. Tests confirmed that interactive maps, real-time data, trustworthy report generating, certificate creation, and exporting to Excel and PDF worked. The module passed all of the tests, which means it is completely functional, accurate, and reliable for administrative reporting, visualization, and documentation.

Table 42
Black-Box Testing of the Alpha Testing of Document Viwer and File Handling

Test Case ID	Description	Expected Outcome	Actual Outcome
TC-058	Property Transfer Document Viewer	Document viewer opens; displays PDF with embed viewer or images with zoom controls; zoom buttons work; can close modal	Document viewer opens successfully; zoom controls functional
TC-059	File Upload Size Limit Validation	Files within limit accepted; oversized files rejected; clear error message states limit; file doesn't upload	File size validation works correctly
TC-060	File Upload Performance	Uploads complete successfully; progress indicator accurate; no timeouts; multiple uploads handled; server storage managed	File upload performance acceptable

As shown above the provided test table validated the Document Viewer and File Handling module, ensuring that administrators and users can reliably view, upload, and manage files with smooth performance and strict validation controls. The document viewer functioned as intended. All test cases passed successfully.



Table 43
Black-Box Testing of the Alpha Testing of Notification & Emails

Test Case ID	Description	Expected Outcome	Actual Outcome
TC-061	Email Notification on Application Submission	Email sent immediately; contains accurate info; tracking link works; professional formatting	Email notifications sent successfully
TC-062	Email Notification on Status Change	Email sent for each status change; contains relevant info; rejection email includes reason; approved email includes next steps	Status change emails sent correctly
TC-063	In-App Notification Bell	Bell shows unread count; dropdown lists notifications; clickable; marks as read; navigates correctly; timestamps accurate	In-app notifications work correctly

As shown above, the test table demonstrated that the Document Viewer and File Handling modules functioned effectively. This indicated that both users and administrators could easily read, upload, and manage files, with PDFs, photos, zoom controls, and uploads operating as intended. It also verified that the Notifications and Emails modules delivered emails and in-app notifications correctly, displayed properly, marked messages as read, and were easy to browse. All tests were successful, confirming that both modules were reliable, user-friendly, and kept users updated in real time, ensuring overall system dependability and usability.



Table 44
Black-Box Testing of the Alpha Testing of Security & Validation

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-064	Role-Based Access Control - Admin Features	Regular users see limited menu; admin pages blocked; manual URL access prevented; admins see all features	Role-based access control works correctly	Pass	Fully functional
TC-065	SQL Injection Prevention	Injection attempts fail; system not compromised; database intact; input sanitized; logged as security event	SQL injection prevention works correctly	Pass	Fully functional
TC-066	XSS (Cross-Site Scripting) Prevention	Scripts not executed; rendered as text; input sanitized; output encoded; no JavaScript execution	XSS prevention works correctly	Pass	Fully functional
TC-067	Email Format Validation	Invalid emails rejected; clear error messages; valid emails accepted; validates on blur and submit	Email validation works correctly for all formats	Pass	Fully functional
TC-068	Phone Number Format Validation	Invalid formats rejected; 11-digit requirement enforced; must start with 09; country code handled appropriately	Phone number validation works correctly	Pass	Fully functional



As shown in the previous page, Table 44 examined the Security and Validation module to ensure it had effective access control, input validation, and protection against security threats. Tests demonstrated that role-based access control functioned correctly, unauthorized URLs were blocked, and full admin rights were granted appropriately. The system was highly secure, successfully preventing SQL injection and XSS attacks. Input validation for essential fields, phone numbers, and emails operated flawlessly, ensuring all data was complete and accurate. The module passed all tests, confirming it was functional, robust, and secure. These results indicated the module was reliable and effective in maintaining security standards.

Table 45

Black-Box Testing of the Alpha Testing of UI/UX Navigation

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-070	Sidebar Navigation - Expand/Collapse	Sidebar toggles smoothly; icons visible when collapsed; tooltips show labels; content adjusts; submenu states persist Submenus toggle correctly; chevron indicates state; nested submenus work; states persist across navigation	Sidebar navigation works correctly	Pass	Fully functional
TC-071	Submenu Expand/Collapse	Submenu functionality works as expected		Pass	Fully functional



As shown in the previous page, Table 45 checked the UI/UX and Navigation module to make sure that users can use the system quickly and reliably on all of their devices. Tests confirmed that the sidebar and submenu navigation worked well, the layouts were responsive, the breadcrumb navigation was accurate, and the filtering was reliable with obvious reset options. All of the tests succeeded, which shows that the module works perfectly, is easy to use, and is set up for a smooth user experience.

Table 46
Black-Box Testing of the Alpha Testing of Performance and Optimization

Test Case ID	Description	Expected Outcome	Actual Outcome	Comments
TC-076	Page Load Time	Pages load <3 seconds on regular connection; <10 seconds on slow connection;	Page load times within acceptable limits	Passed
TC-077	Large Dataset Pagination	Pagination efficient; no lag with large datasets; page size configurable; navigation smooth;	Pagination works efficiently with large datasets	Passed
TC-078	Concurrent User Sessions	System handles concurrent users; no data corruption; race conditions prevented; performance acceptable;	System handles concurrent users correctly	Passed
TC-079	Database Query Optimization	Queries execute quickly; proper indexes used; no full table scans; efficient JOIN operations	Database queries optimized and perform well	Passed



As shown in the previous page, Table 46 checked the Performance & Optimization module to make sure that the system works well and reliably in a variety of situations. Tests confirmed that the pages loaded quickly, the pagination worked well with massive datasets, the user sessions stayed consistent, and the database queries were optimized. The module passed all of the tests, which shows that it is completely functional, responsive, and strong enough to provide a consistent user experience.

Table 47
Black-Box Testing of the Alpha Testing of Integration and API

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-080	Facial Recognition API Integration	Face recognition accurate; handles different conditions; matches stored descriptor; denies unauthorized faces; clear instructions	Facial recognition works correctly	Pass	Fully functional
TC-081	Mapbox API Integration	Map loads; tiles display; markers accurate; interactions smooth; API key valid; graceful offline handling	Mapbox integration works correctly	Pass	Fully functional
TC-082	Email Service Integration (SMTP)	Emails send successfully; delivered to inbox (not spam); headers	Email service integration works correctly	Pass	Fully functional



As shown in the previous page, Table 47 examined the Integrations & API module to ensure external services functioned correctly. Tests confirmed facial recognition was accurate, Mapbox mapping was interactive and precise, and SMTP email delivery was dependable. The module was fully operational, secure, and successfully integrated external services to enhance system capabilities. All tests were completed successfully.

Table 48
Black-Box Testing of the Alpha Testing of Housing Monitoring

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-083	User Access Monitoring Page	Only accessible to "Relocated" beneficiaries; others redirected or denied; dashboard shows correct due dates and status	Access control works correctly; dashboard displays accurate monitoring info	Pass	Fully functional
TC-084	User Submit Monitoring Update	Camera opens; photo captured; GPS location detected; submission successful; status updates to "Pending"	Camera and GPS integration works; submission recorded successfully	Pass	Fully functional
TC-085	User View Monitoring History	History table loads; past submissions visible; status badges correct; photos viewable in modal	Submission history displayed correctly with functional photo viewer	Pass	Fully functional



As shown in the previous page, Table 48 evaluated the Housing Monitoring module during the alpha testing phase using black-box testing to assess the accuracy, security, and effectiveness of post-relocation monitoring. This module was designed to ensure that only verified and relocated beneficiaries could access monitoring features, while unauthorized users were restricted from entry. Positive testing confirmed that access controls were correctly enforced, allowing only beneficiaries with a “Relocated” status to view the monitoring dashboard. The dashboard accurately displayed due dates, monitoring statuses, and compliance indicators. Users were able to submit monitoring updates by capturing photos through the device camera and automatically detecting GPS location data, with all submissions properly recorded and assigned a “Pending” status for administrative review. All test cases passed successfully, demonstrating that the Housing Monitoring module is reliable, efficient, and secure. The results confirm that the module effectively supports continuous oversight of relocated beneficiaries while ensuring data accuracy, accountability, and system integrity.



Table 49

White-Box Testing of the Alpha Testing of Authentication and Security Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result	Remarks
WB-001	login	Login action file exists	basePath ("")	expect(\$loginFile)	File is a valid file	PASS	Login is Valid
WB-002	login	Login requires POST method	file_get_contents (\$login)	expect(\$content)->toCont	Content contains REQUEST	PASS	Login POST Valid
WB-003	login	Login includes recaptcha verification	file_get_contents (\$login)	expect(\$content)->toCont	Content contains recaptcha response	PASS	Login verify valid
WB-004	register	Register action file exists	basePath ('register')	expect(\$registerFile)	File is a valid file	PASS	Register action valid.
WB-005	register	Register requires POST method	file_get_contents (\$register)	expect(\$content)->toCont	Content contains REQUEST	PASS	POST Method Secure

As shown in Table 49 above the Housing Monitoring module functioned effectively for both users and administrators. Users could submit updates with pictures and GPS data, review previous entries, and accurately track changes in status. Administrators had access to all submissions, could approve or reject updates, modify the status of active owners, identify overdue cases, and send automated warning emails. Filtering, searching, and GPS verification operated smoothly, and the system automatically calculated alert levels. The module successfully passed all tests, indicating it was fully operational, precise, and efficient in monitoring housing activities.



Table 50

White-Box Testing of the Alpha Testing of User Management Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result	Remarks
WB-031	add_user	Add user action exists	Base Path (s)	expect()->toBeFile	File is a valid file	PASS	Validates user creation function is working correctly. Confirms user profile editing function is available. Ensures user account deletion handler exists.
WB-032	edit_user	Edit user action exists	basePath(s)	expect()->toBeFile	File is a valid file	PASS	
WB-033	delete_user	Delete user action exists	basePath('delete_user')	expect()->toBeFile	File is a valid file	PASS	

As shown above, in Table 50 the white box test for the User Management Module confirmed that all of the backend scripts that let administrative users do their jobs were set up correctly and were working as they should. The tests showed that the key action files that create, update, and delete user accounts are present. This means that the system's administrative workflows are complete and working (WB-031, WB-032, WB-033). The retrieval handler for loading and showing user lists was also tested, which means that the system can correctly get and handle stored user data when it needs to (WB-034). All tests went well, showing that the module's basic user management functionalities are safely and reliably built into the system's backend.



Table 51

White-Box Testing of the Alpha Testing of Housing Application Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result	Remarks
WB-011	pagibig_housing_apply	Pagibig housing apply action exists	basePath ('pagibig_housing_apply')	expect()->toBeFile	File is a valid file	PASS	Validates Pag-IBIG housing application submission handler is present.
WB-012	get_application_details	Get application details action exists	basePath ('get_application_detail')	expect()->toBeFile	File is a valid file	PASS	Confirms application details retrieval function is available.
WB-013	update_application_status	Update application status action exists	basePath ('update_application_status')	expect()->toBeFile	File is a valid file	PASS	Verifies application status update handler exists for workflow.

As shown above, Table 51 is the white box tests for the Housing Application Module showed that all of the backend scripts needed to run the Pag-IBIG housing application process were set up correctly and could be found in the system. The evaluation confirmed that there was a submission handler (WB-011) in place to handle new housing applications and a script (WB-012) in place to get detailed information about applicants for review and tracking. The system also confirmed that the action file required to update application statuses during the review process was available, which showed that the workflow management features were working (WB-013). In addition, administrative controls such as deleting existing applications and relocating qualified applicants to available housing options were successfully validated.



Table 52
White-Box Testing of the Alpha Testing of Beneficiary Management Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result	Remarks
WB-021	beneficiary_verification	Beneficiary verification action exists	basePath ('beneficiary_verification')	expect()->toBeFile	File is a valid file	PASS	Validates beneficiary verification submission handler is present. Confirms beneficiary data retrieval function exists.
WB-022	get_beneficiary	Get beneficiary action exists	basePath ('get_beneficiary')	expect()->toBeFile	File is a valid file	PASS	Confirms beneficiary record deletion function is exists. Validates beneficiary verification deletion handler is present.
WB-023	delete_beneficiary	Delete beneficiary action exists	basePath ('delete_beneficiary')	expect()->toBeFile	File is a valid file	PASS	Ensures beneficiary status update functionality.
WB-024	delete_beneficiary_verification	Delete beneficiary verification exists	basePath ('delete_beneficiary_verification')	expect()->toBeFile	File is a valid file	PASS	
WB-025	update_beneficiary_status	Update beneficiary status action exists	basePath ('update_beneficiary_status')	expect()->toBeFile	File is a valid file	PASS	

As shown above, Table 52 tested the Beneficiary Management Module showed that all of the backend scripts work perfectly. Core verification handlers correctly handle submissions from beneficiaries (WB-021), and retrieval mechanisms can get to both general and detailed verification data (WB-022, WB-023). We also checked administrative tasks. All tests were passed, which shows that the module can reliably handle, track, and keep up with beneficiary data in the system.



Table 53
White-Box Testing of the Alpha Testing of Utility Connection Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result	Remarks
WB-039	water_electrical_connection	Water electrical connection action exists	basePath('water_electrical_connection')	expect()->toBeFile	File is a valid file	PASS	Validates utility connection request submission handler exists. Confirms utility connection details retrieval is available. Ensures utility connection status update functionality exists. Validates utility connection deletion handler is implemented.
WB-040	get_utility_details	Get utility details action exists	basePath('get_utility_details')	expect()->toBeFile	File is a valid file	PASS	
WB-041	update_utility_status	Update utility status action exists	basePath('update_utility_status')	expect()->toBeFile	File is a valid file	PASS	
WB-042	delete_utility_connection	Delete utility connection action exists	basePath('delete_utility_connection')	expect()->toBeFile	File is a valid file	PASS	

As shown above, Table 53 is the White Box testing of the Utility Connection Module showed that all of the backend scripts work as they should. The primary submission handler takes care of requests for water and electricity (WB-039), while the retrieval methods get comprehensive connection records (WB-040). We checked that administrative tasks like updating status (WB-041) and deleting records (WB-042) were working. All tests were passed, which shows that the module properly manages, tracks, and keeps utility connection data in the system.



Table 54
White-Box Testing of the Alpha Testing of Property Management Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result	Remarks
WB-047	assign_housing	Assign housing action exists	basePath('assign_housing')	expect()->toBeFile	File is a valid file	PASS	Validates housing unit assignment handler is present.
WB-048	get_available_lot	Get available lots action exists	basePath('get_available_lots')	expect()->toBeFile	File is a valid file	PASS	Confirms available housing lots retrieval functionality exists.
WB-049	get_lots_with_location	Get lots with location action exists	basePath('get_lots_with_location')	expect()->toBeFile	File is a valid file	PASS	Ensures lots with location data retrieval is available.
WB-050	update_lot_info	Update lot info action exists	basePath('update_lot_info')	expect()->toBeFile	File is a valid file	PASS	Validates housing lot information update handler is present.
WB-051	save_lot_bound	Save lot boundary action exists	basePath('save_lot_bound')	expect()->toBeFile	File is a valid file	PASS	Confirms lot boundary mapping save function is working.
WB-052	delete_lotbound	Delete lot boundary action exists	basePath('delete_lot_bound')	expect()->toBeFile	File is a valid file	PASS	Ensures lot boundary deletion handler exists for map.

As shown above, Table 54 is the white box test for the Reports & Documents Module made sure that .all of the backend scripts that were supposed to create, retrieve, and manage system reports and documents were properly set up and working as they should. All of the test cases passed, which shows that the module's ability to generate, delete, retrieve, create certificates.



Table 55
White-Box Testing of the Alpha Testing of Property Transfer Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result	Remarks
WB-053	property_transfer_current	Property transfer current action exists	basePath('property_transfer_current')	expect()->toBeFile	File is a valid file	PASS	Validates property transfer submission handler is available.
WB-054	get_property_transfer_details	Get property transfer details exists	basePath('get_property_transfer_details')	expect()->toBeFile	File is a valid file	PASS	Confirms property transfer details retrieval is functional.
WB-055	update_property_transfer_status	Update property transfer status exists	basePath('update_property_transfer_status')	expect()->toBeFile	File is a valid file	PASS	Ensures property transfer status update function exists.
WB-056	delete_property_transfer	Delete property transfer action exists	basePath('delete_property_transfer')	expect()->toBeFile	File is a valid file	PASS	Validates property transfer record deletion handler is present.

As shown above, Table 55 is the white box testing showed that all of the backend programs perform as they should. The core handler handles requests for property transfers (WB-053), and retrieval procedures get specific information about the transfers (WB-054). It was confirmed that administrative tasks like updating transfer status (WB-055) and removing entries when necessary (WB-056) were done correctly. All test cases passed, which shows that the module can reliably handle submissions, retrieval, status updates, and deletions for property transfer processes that work well.



Table 56
White-Box Testing of the Alpha Testing of Report and Document Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result	Remarks
WB-057	generate_report	Generate report action exists	basePath('generate_report')	expect()->toBeFile	File is a valid file	PASS	Confirms report generation functionality is implemented in the system. Validates generated report deletion handler is available.
WB-058	delete_report	Delete report action exists	basePath('delete_report')	expect()->toBeFile	File is a valid file	PASS	Ensures chart data retrieval for dashboard analytics exists. Confirms certificate generation functionality is implemented.
WB-059	get_chart_data	Get chart data action exists	basePath('get_chart_data')	expect()->toBeFile	File is a valid file	PASS	
WB-060	generate_certificate	Generate certificate action exists	basePath('generate_certificate')	expect()->toBeFile	File is a valid file	PASS	

As shown above, Table 56 presents the results of white-box testing conducted during the alpha testing phase for the Reports and Documents Module. This testing focused on validating the internal structure and implementation of backend scripts, ensuring that essential report-related and document-generation functionalities were properly implemented and accessible within the system. The results confirm that all critical backend action handlers required for report and document processing were present and functional.



Table 57
White-Box Testing of the Alpha Testing of Page and Interfaces

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result	Remarks
WB-016	applications	Applications page exists	basePath ('applications')	expect()->toBeFile	File is a valid file	PASS	Confirms applications listing page is exist in the system.
WB-017	pagibig_housing_apply	Pagibig housing apply page exists	basePath('pagibig_housing_apply')	expect()->toBeFile	File is a valid file	PASS	Validates Pag-IBIG housing application form page is available. Confirms application status tracking page is present for users.
WB-018	pagibig_housing_status	Pagibig housing status page exists	basePath('pagibig_housing_status')	expect()->toBeFile	File is a valid file	PASS	Verifies housing relocation assignment interface is available. Ensures applications page has proper PHP code structure.
WB-019	housing_relocation_assignment	Housing relocation assignment page exists	basePath('housing_relocation_assignment')	expect()->toBeFile	File is a valid file	PASS	Confirms beneficiaries listing page is present. Validates beneficiary verification form page is available.
WB-020	applications	Applications page contains php code	(basePath ('applications'))	expect(\$content)->toContain('<?php')	Content contains <?php tag	PASS	
WB-027	beneficiaries	Beneficiaries page exists	basePath ('beneficiaries')	expect()->toBeFile	File is a valid file	PASS	
WB-028	beneficiary_verification	Beneficiary verification page exists	basePath ('pages/beneficiary_verification')	expect()->toBeFile	File is a valid file	PASS	



As shown in the previous page, Table 57 is the white box test for the Pages and Interfaces module demonstrated that all front-end pages and interfaces were correctly configured, user-friendly, and integrated with backend services. We examined core pages for applications, housing forms, status tracking, relocation assignment, utility connections, reports, housing/property management, visitor logs, and interactive maps. Pages utilizing PHP successfully generated dynamic content, and both administrative and user-specific interfaces functioned as intended.

Table 58
White-Box Testing of the Alpha Testing of System Configuration and Infrastructure

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result	Remarks
WB-037	includes/ access_ control	Access control file exists	basePath ('includes /access_control')	expect()->toBeFile	File is a valid file	PASS	Ensures access control and permission system is present.
WB-038	includes/ check_ admin	Check admin file exists	basePath ('includes /check_admin')	expect()->toBeFile	File is a valid file	PASS	Confirms admin privilege verification mechanism is present.
WB-063	config	Config file exists and is readable	basePath ('config'), is_readable(\$configFile)	expect(\$configFile)->toBeFile()	File exists and is readable	PASS	Validates main configuration file exists and has proper read permissions.



As shown in the previous page, Table 58 is the white box test for the System Configuration & Infrastructure module showed that all of the configuration files, access controls, and system directories are set up and working properly. The access control and admin scripts work as they should. The module passed all tests, which means it is safe, well-organized, and helps the system run smoothly.

Table 59
White-Box Testing of the Alpha Testing of Database Testing

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result	Remarks
WB-073	PDO Connection	Can connect to database	getTestDatabaseConnection()	expect(\$pdo->toBeInstanceOf(PDO::class))	Returns PDO instance	PASS	Verifies successful database connection can be established.
WB-074	PDO Driver	Uses mysql driver	\$pdo->getAttribute(PDO::ATTR_DRIVER_NAME)	expect()->toBe('mysql')	Returns 'mysql'	PASS	Confirms MySQL database driver is properly configured and active.
WB-075	PDO Error Mode	Has error mode set to exception	\$pdo->getAttribute(PDO::ATTR_ERRMODE)	expect()->toBe(PDO::ERRMODE_EXCEPTION)	Returns PDO::ERRMODE_EXCEPTION	PASS	Ensures PDO throws exceptions for better error handling.
WB-076	users table	Users table exists	\$pdo->query("SHOW TABLES LIKE 'users'")	expect(\$stmt->rowCount())->toBe(1)	Row count equals 1	PASS	Validates users table is present in the database schema.
WB-077	pagibig_applications table	Pagibig applications table exists	\$pdo->query("SHOW TABLES LIKE 'pagibig_applications'")	expect(\$stmt->rowCount())->toBe(1)	Row count equals 1	PASS	Confirms Pag-IBIG applications table exists in database.



As shown in the previous page, Table 59 is the white box test for the Database Testing module showed that the database layer, schema, and PDO settings for the system are all working properly. PDO successfully connects to MySQL and handles problems with exceptions (WB-073 to WB-075). All of the important tables, such as those for users, Pag-IBIG applications, beneficiaries, housing units, utility connections, and property transfers, have columns that are set up correctly for authentication, workflows, and core data (WB-076). SQL prepared statements, query fetching, and transaction handling were validated to ensure reliable operations and data consistency.

Table 60
White-Box Testing of the Alpha Testing of House Monitoring and Mapping Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result	Remarks
WB-098	housing_monitoring_management	Housing monitoring management page exists	basePath('housing_monitoring_management')	expect()->toBeFile	File is a valid file	PASS	Validates admin interface for monitoring housing units is accessible.
WB-099	housing_unit_monitoring	Housing unit monitoring page exists	basePath('housing_unit_monitoring')	expect()->toBeFile	File is a valid file	PASS	Confirms user interface for submitting monitoring reports is available.
WB-100	submit_monitoring	Submit monitoring action exists	basePath('submit_monitoring')	expect()->toBeFile	File is a valid file	PASS	Validates monitoring report submission handler is present.



As shown in the previous page, Table 60 is the white box test for the Housing Monitoring & Mapping Module showed that all of the database structures, backend handlers, and interfaces work as they should. It was possible to get to both the admin monitoring interface (WB-C) and the user monitoring page (WB-099). We fully implemented backend operations for submissions, status updates, warnings, and data retrieval (WB-100). All test cases passed. Alpha testing using the white box approach was conducted to evaluate the internal logic, workflow structures, and code execution paths of the Informal Settler Relocation Management System (ISRMS). All test cases passed successfully, demonstrating that the system's backend processes are stable, logically sound, and functioning according to the intended design. This confirms that the internal implementation effectively supports key features such as housing application processing, GIS-based mapping, unit allocation, and inventory management. The third objective of this study was to use the eight characteristics of the ISO/IEC 25010:2011 Software Quality Model and the three high-level dimensions of the ISO 9241-11:2018 Usability Standard to evaluate the overall quality and usability of the Informal Settler Relocation Management System. The outcome of this objective was a robust affirmation of the system's technical quality and its efficacy in user interaction, attaining an Average Mean Score of 4.57 (Strongly Agree) for overall quality and an Average Weighted Mean of 4.80 (Strongly Agree) for overall usability.



Table 61

Mean Scores of the ISO 9241-11:2018 Software Evaluation Tool's Quantitative Outcome on the Effectiveness of the Informal Settler Relocation Management System

Items	Mean	Interpretation
Effectiveness		
Time to learn and use Tasks can be performed in a straight-forward manner	4.44	Strongly Agree
Data entry time I can effectively complete my work using this system	4.52	Strongly Agree
Task time Tasks can be performed in a short period of time using this system	4.32	Strongly Agree
Response time I can use it successfully every time	4.48	Strongly Agree
Time to install This system is fast to install	4.52	Strongly Agree
Mean Score	4.45	Strongly Agree
Efficiency		
Number of errors I can efficiently complete my work without spending too much time correcting things with this system	4.44	Strongly Agree
Completion rate I am able to complete my work quickly using this system	4.68	Strongly Agree
Mean Score	4.56	Strongly Agree
Satisfaction		
I am satisfied with the ease of completing the tasks using this system	4.68	Strongly Agree
I am satisfied with the amount of time it took to complete the tasks using this system	4.68	Strongly Agree
Overall, I am satisfied with this system	4.8	Strongly Agree
Mean Score	4.72	Strongly Agree
Overall Mean Score	4.57	Strongly Agree



As shown in the previous page, Table 61 shows the usability test that the system got its best score in Satisfaction (4.72) and its lowest score in Effectiveness (4.45). The ISO 9241-11:2018 standard says that usability is based on three main things: Effectiveness, Efficiency, and Satisfaction. The very high Satisfaction score shows that people are very happy with the system and find it easy and comfortable to use, which shows that it is well accepted. Effectiveness also got a high score, but it was the lowest of the three aspects, which means there are some little areas where it could be better. Overall, the results reveal that the system works quite well, and making Effectiveness better would make it even easier to use, according to ISO 9241-11:2018.

Table 62
ISO/IEC 25010:2011 Software Quality Model Characteristics

Criteria	Mean Score	Verbal Interpretation
Functionality Suitability Characteristics	4.93	Strongly Agree
Performance Efficiency Characteristics	4.93	Strongly Agree
Compatibility Characteristics	4.70	Strongly Agree
Usability Characteristics	4.73	Strongly Agree
Reliability Characteristics	4.85	Strongly Agree
Security Characteristics	4.60	Strongly Agree
Maintainability Characteristics	4.88	Strongly Agree
Portability Characteristics	4.80	Strongly Agree
Mean Score	4.80	Strongly Agree



As shown in the previous page, Table 62 shows Mean Scores of IT Experts on System Functionality Using ISO/IEC 25010:2011 Effectiveness of the Informal Settler Relocation Management System. The functionality test conducted using ISO/IEC 25010:2011 revealed that Functional Suitability, which assessed how well the system met specific standards, received a high score of 4.93, indicating a Strongly Agree rating. This finding was supported by the study of Susanto et al. (2018), which showed that Functional Suitability also achieved the highest mean score among the ISO/IEC 25010 criteria. This indicated that functionality was the most critical factor in determining whether a system provided value to consumers. Other quality traits that scored excellently included Performance Efficiency (4.93), Reliability (4.85), Maintainability (4.88), and Portability (4.80). These scores suggested that the system was reliable, easy to maintain, and compatible with other platforms. The results aligned with the findings of Susanto et al. (2018), which demonstrated that key ISO/IEC 25010 quality attributes also received high ratings. This consistency indicated that the system performed well and conformed to international software quality standards. Security scored the lowest at 4.60, yet it still reflected a Strongly Agree rating. This was consistent with Susanto et al. (2018), who found that Security had the lowest mean score among the quality characteristics.



CHAPTER V

CONCLUSION AND RECOMMENDATIONS

This chapter provides a comprehensive overview of the main conclusions, discoveries, and recommendations of the study. This chapter gives a summary of the results of the data collection and a rundown of the implementation of the Informal Settler Relocation Management System. The conclusions highlight the significance of these findings; the recommendations are intended to guide future advancements and handle any problems that may have come up during the study.

5.1 Summary of Findings

The Informal Settler Relocation Management System (ISRMS) was a digital platform developed to modernize and improve the efficiency of relocation operations managed by the City Human Settlement and Urban Development Office (CHSUD) of Victorias City. The system aimed to address longstanding issues in relocation management, such as manual housing application processes, inaccurate record-keeping, slow beneficiary verification, and difficulties in tracking housing inventory. It employed an integrated GIS-enabled system, which included an interactive GIS mapping tool for visualizing informal settlements and relocation sites, an automated housing application and management module, a tracking system for vacancies and allocations, and a central inventory of housing units.

The ISRMS underwent comprehensive evaluation to determine its effectiveness, efficiency, and user satisfaction through various testing methods, including black-box testing, white-box testing, and usability assessments based on



ISO 9241-11:2018 standards. The evaluation results indicated high scores across all usability factors, demonstrating that the system was user-friendly, intuitive, and capable of facilitating accurate completion of relocation activities. The system significantly accelerated application processing, improved data entry accuracy, and simplified installation and navigation processes. These outcomes confirmed that the ISRMS enhanced the operational efficiency of CHSUD and provided beneficiaries with a clearer, more organized, and accessible relocation experience.

5.2 Conclusion

The study successfully completed the design and development phases of the Informal Settler Relocation Management System (ISRMS). Its primary objective was to create a centralized, GIS-integrated digital platform that modernized and streamlined the relocation workflow for the City Human Settlement and Urban Development Office (CHSUD). The system met all design requirements by integrating key components such as GIS-based mapping, automated processing of housing applications, precise verification of beneficiaries, organized tracking of vacancies and allocations, management of housing unit inventories, and comprehensive reporting features.

These components demonstrated that the ISRMS was a comprehensive and operational solution capable of addressing longstanding issues associated with manual relocation procedures. Functionality testing of the ISRMS was also successfully conducted in accordance with Objective 2. Extensive white-box and



black-box testing revealed that all modules, action handlers, workflows, and database structures operated correctly and without issues.

The testing outcomes indicated that all cases functioned properly, confirming that the system consistently performed as intended. The system exhibited stable performance and produced no errors during processes such as user registration, housing allocation, monitoring, and report generation. These results verified that the system was ready for deployment and practical use in real-world scenarios. The evaluation of the system, based on international standards ISO 25010:2011 (Software Quality) and ISO 9241-11:2018 (Usability), yielded highly positive findings. The overall mean rating of 4.81 (Very Good) reflected that the ISRMS was a reliable, high-quality, and user-friendly platform.

The highest ratings were observed in Maintainability and Performance Efficiency, indicating that the system was robust, easy to upgrade, and efficient in data processing. Usability assessments showed that users found the system easy to operate, effective, and engaging. Although the Security attribute received the lowest mean score, it still garnered a Strongly Agree rating, suggesting that only minor enhancements were necessary to strengthen the system's protective measures. The final phase of the study involved developing and completing the User's Manual.

The manual provided clear, comprehensive, and step-by-step instructions for all system roles, enabling candidates, CHSUD personnel, and administrators to utilize the ISRMS effectively from the outset. This documentation facilitated



easier system adoption, supported users during operation, and contributed to the platform's long-term sustainability.

Overall, the study concluded that the Informal Settler Relocation Management System (ISRMS) fulfilled its objectives concerning design, functionality, quality, and documentation. It proved to be an effective, accurate, and current system that simplified relocation management, increased transparency, and supported data-driven decision-making for sustainable urban growth.

5.3 Recommendation

Based on the system's development results, testing, and evaluation, numerous improvements are suggested to enhance the Informal Settler Relocation Management System (ISRMS) and ensure its long-term functionality. First, the system's speed is targeted for improvement by optimizing database queries, especially in modules handling large datasets such as housing inventory, vacancy tracking, GIS mapping, and reporting. These enhancements aim to maintain smooth and rapid system performance as more records and data about relocations accumulate over time.

Second, although the overall quality and user-friendliness of the software receive high marks, security, which scores the lowest, is prioritized for improvement. To safeguard critical information during daily operations, access control mechanisms are strengthened, more rigorous authentication procedures



are implemented, sensitive relocation records are protected, and data security protocols are reviewed regularly.

Third, the system documentation is maintained to reflect any changes, improvements, or new processes introduced in the future. This ensures that CHSUD staff, administrators, and end users operate the system accurately and efficiently, regardless of version updates or staff changes.

Lastly, it is recommended that CHSUD adopt the ISO 25010:2011 and ISO 9241-11:2018 standards for annual system evaluations. These regular assessments aim to identify areas needing improvement, sustain the system's initial performance, usability, and reliability, and support continuous enhancements. Overall, these suggestions facilitate ongoing system improvements, protect sensitive relocation data, and ensure the ISRMS remains effective, high-quality, and reliable over time.



REFERENCES

- A. A. C. Ortega, "Exposing necroburbia: Suburban Relocation, Necropolitics, and Violent Geographies in Manila," *Antipode*, vol. 52, no. 4, pp. 1175–1195, 2020. [Online]. Available: <https://doi.org/10.1111/anti.12629>
- A. Brown and R. Eyenghe, "Community profiling for inclusive urban development," *Journal of Urban Management*, vol. 11, no. 3, pp. 287–296, 2022. [Online]. Available: <https://doi.org/10.1016/j.jum.2022.05.002>
- A. M. Muczyński, A. Zwirowicz-Rutkowska, and J. N. da Costa, "Supporting tenant relocation in social housing management: Development of the dedicated geoinformation system in Poland," *Cities*, vol. 120, p. 103431, 2022. [Online]. Available: <https://doi.org/10.1016/j.cities.2021.103431>
- C. C. R. Canlas, R. D. Dela Fuente, and E. D. A. Mercado, "Kapam-Pamangan: A Cross-Platform Application for Kapampangan Recipes," *MCC-IT Research Journal*, Mabalacat City College, Philippines, 2023–2024.
- C. Garson and E. Julian, "To resist is to survive: Lived experiences of the untenured migrants in the watershed protected areas," *International Journal of Innovative Science and Research Technology*, pp. 1573–1577, 2024. [Online]. Available: <https://doi.org/10.38124/IJISRT/IJISRT24MAR679>
- Department of Human Settlements and Urban Development, "DHSUD urges local offices to modernize operations," 2022. [Online]. Available: <https://dhsud.gov.ph>



International Organization for Standardization, ISO 9241-11: Ergonomics of human-system interaction – Part 11: Usability: Definitions and concepts, Geneva, Switzerland: ISO, 2018.

Kutoyi, “Land readjustment strategy in slum redevelopment using GIS: Case study — Mzambarauni Slum in Mtwapa, Kilifi County,” University of Nairobi E-Repository, 2021.

L. Dai, Q. Liu, and H. Zhou, “Resettlement, gentrification and the right to the city: Rethinking urban relocation in the Global South,” *Cities*, vol. 124, p. 103608, 2022. [Online]. Available: <https://doi.org/10.1016/j.cities.2022.103608>

M. A. Uy, C. M. D. Cuesta, S. J. N. Cahandig, and S. U. Lamalan, “Human Settlements and Informal Settler Families in Davao City: Implications for Sustainable Urban Development,” *International Journal of Engineering Technology Research & Management (IJETRM)*, vol. 9, no. 5, May 2025. [Online]. Available: <https://www.ijetrm.com/>

M. K. Navarro and C. R. Almaden, *The Cost of Misguided Urbanization: The Case of Informal Settlements in Butuan City, Philippines*, Xavier University, Butuan City, Philippines, Sep. 2014.

MaybeWorks, “Rapid App Development: Full RAD Methodology Overview and Practical Advice,” Dec. 9, 2024. [Online]. Available: <https://maybe.works/blogs/rapid-application-development-rad>

N. Pangilinan, J. Bautista, L. Mabaquiao, and V. K. Aldrian, “A Localized GIS-Based Multi Decision Criteria Approach in Determining Suitable Sites for Community Toilet Facilities in BASECO Compound,” *Int. Arch. Photogramm.*



- Remote Sens. Spatial Inf. Sci., vol. XLVIII-5, pp. 75–81, 2024. doi: 10.5194/isprs-archives-XLVIII-5-2024-75-2024
- P. Paudel and B. Sherchan, “Land Suitability Analysis for Relocation of Settlement Based on Natural Hazard Issues: A Case Study of the Budhigandaki Hydropower Project in Nepal,” *Journal of Engineering and Sciences*, vol. 3, no. 2, pp. 92–106, 2024. [Online]. Available: <https://nepjol.info/index/jes2/article/view/72197>
- Philippine Institute for Development Studies, “Digital systems enhance transparency, cut administrative costs, and improve public service delivery,” 2023. [Online]. Available: <https://pids.gov.ph>
- R. B. Recio and K. Dovey, “Forced Eviction by Another Name: Neoliberal Urban Development in Manila,” *Planning Theory & Practice*, vol. 22, no. 5, pp. 806–812, Jul. 2021. [Online]. Available: <https://www.tandfonline.com/doi/full/10.1080/14649357.2021.1958537>
- R. Carrasco and T. Egbelakin, “Resettlement and the challenges of sustainable housing: A review of practices,” *International Journal of Disaster Resilience in the Built Environment*, vol. 13, no. 2, pp. 123–138, 2022. [Online]. Available: <https://doi.org/10.1108/IJDRBE-06-2021-0068>
- R. Garcellano, L. Japson, and S. Wy, “Water quality of Puerto Princesa Bay in relation to the presence of informal settlers in its coastal areas,” *PalawanSci*, vol. 14, no. 1, pp. 84–99, 2022. [Online]. Available: <https://doi.org/10.69721/tps.j.2022.14.1.10>



- R. Maningo, "Examining Urban Poor Voices: Displacement and Resettlement of Informal Settlers in Metro Manila," *Social Transformations: Journal of the Global South*, vol. 9, no. 1, Art. 7, 2021. doi: 10.13185/2799-015X.1122
- Suryadi, D. A., & Sulistiyani, E. (2022). Evaluation of information quality using ISO/IEC 25010: 2011 (Case research: Menu harianku application). *International Journal of Innovation in Enterprise System*, 6(2), 143-156.
- T. D. Susanto, R. Hidayat, and E. S. Nugroho, "Title of the paper," *J. Phys.: Conf. Ser.*, vol. 974, p. 012043, 2018.
- Zigzag Weekly, "City maps informal settlers for housing needs," Oct. 16, 2022. [Online]. Available: <https://www.zigzagweekly.net/city-maps-informal-settlers-for-housing-needs>



APPENDICES

“Appendix A”

ISO/IEC 25010:2011 Software Quality Requirements and Evaluation Questionnaire

Characteristics	1	2	3	4	5
Functional Suitability					
Functional Completeness The feature is complete that covers the user specified tasks.					
Functional Correctness The system provides correct results with the needed degree of precision.					
Functional Appropriateness The system facilitate the accomplishment of user specified tasks.					
Performance Efficiency					
Time-behavior The system response and processing time when performing its functions, meet the requirements.					
Resource Utilization The amount and type of resources used by a system, when performing its functions, meet the requirements.					
Capacity The maximum limits of the system parameter, meet the requirements					
Compatibility					
Co-existence The system can perform its required functions efficiently while sharing a common environment and resources.					
Interoperability Multiple systems components can exchange information.					
Usability					
Appropriateness Recognisability The users can recognize the appropriateness of the system according to their needs.					
Learnability The system enables the user to learn how to use it with effectiveness, efficiency in emergency situations.					
Operability The system is easy to operate.					
User error protection The system protects users against making errors.					
User Interface Aesthetics					



The system user interface is pleasing and satisfying for the user.					
Accessibility The system can be used with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.					
Reliability					
Maturity The system is reliable under normal operation.					
Availability The system is operational and accessible when required for use.					
Fault tolerance The system operates as intended despite the presence of hardware or software faults.					
Recoverability The system can recover the data in the event of an interruption or a failure.					
Security					
Confidentiality The system ensures that data are accessible only to those authorized to have access.					
Integrity The system prevents unauthorized access to, or modification of, computer programs or data.					
Non-repudiation The system actions or events can be proven to have taken place, cannot be repudiated.					
Accountability The system actions is traceable.					
Authenticity The system can identity of a subject or resource.					
Maintainability					
Modularity The system is composed of discrete components such that a change to one component has minimal impact on other components.					
Reusability The system asset can be used in more than one or in building other assets.					
Analyzability The effectiveness and efficiency to assess the impact on a system of an intended change to one or more of its parts, or to diagnose a system for deficiencies or causes of failures.					
Modifiability The system can be effectively and efficiently modified without introducing defects or degrading existing product quality.					
Testability The system effectiveness and efficiency can be performed to determine that test criteria are met.					
Portability					
Adaptability					



The system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments.					
<i>Installability</i> The system effectiveness and efficiency in the installation and/or uninstallation in a specified environment.					
<i>Replaceability</i> The system can replace another specified software product for the same purpose in the same environment.					



“Appendix B”

**ISO/IEC 9241-11:2018 Software Quality Requirements and Evaluation
Questionnaire**

Effectiveness	1	2	3	4	5
<i>Time to learn and use</i> Tasks can be performed in a straight-forward manner					
<i>Data entry time</i> I can effectively complete my work using this system					
<i>Task time</i> Tasks can be performed in a short period of time using this system					
<i>Response time</i> I can use it successfully every time					
<i>Time to install</i> This system is fast to install					
Efficiency	1	2	3	4	5
<i>Number of errors</i> I can efficiently complete my work without spending too much time correcting things with this system					
<i>Completion rate</i> I am able to complete my work quickly using this system					
Satisfaction	1	2	3	4	5
I am satisfied with the ease of completing the tasks using this system					
I am satisfied with the amount of time it took to complete the tasks using this system					
Overall, I am satisfied with this system					



“Appendix C”

Letter to Beneficiary



Carlos Hilado Memorial State University
Alijis Campus • Binalbagan Campus • Fortune Towne Campus • Talisay (Main) Campus
A leading GREEN institution of higher learning in the global community by 2030
(Good governance, Research-oriented, Extension-driven, Education for Sustainable Development, and Nation-building)

College of Computer Studies

April 24, 2025

MA. JOSEFA VAFLOP-PADARNAL
Assistant Department Head
City Human Settlement and Urban Development Office
Victorias City, Negros Occidental 6115

Dear Mrs. Padernal:

Greetings!

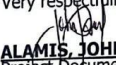
We are third-year Bachelor of Science in Information Systems students from Carlos Hilado Memorial State University, currently enrolled in the ISCAP1 – Capstone course. As part of our capstone project, our group is developing an **Informal Settler Relocation Management System** in relation to the Housing Program of Victorias City. We have selected the **City Human Settlement and Urban Development Office** as our project reference and intended beneficiary.

In connection with this, we are respectfully requesting your approval to allow our group to access various forms and documents relevant to the Housing Program, specifically:

- **Application Forms** used by individuals applying to be part of the Housing Program.
- **Certificates of Allocation or Occupancy** issued to applicants who have been successfully allocated a unit.
- **Transfer Forms or related documents** used when an applicant intends to transfer their allocated unit to another party.
- **Any other official forms** that are part of the housing program process, including those used for status updates, monitoring, or verification.
- **Official headers and footers** used in these documents, so we can reflect the accurate and formal structure within our system's design.

Access to these materials is essential for us to properly analyze and replicate the actual processes and documentation standards used by your office. It will greatly help us ensure that our system is functional, reliable, and aligned with your operational workflows. We guarantee that any data or documents provided will be handled with strict confidentiality and will be used exclusively for academic purposes related to our capstone project. If you require any additional information, please do not hesitate to contact us at **09924341723** or via email at **rsdoromal.chmsu@gmail.com**. We sincerely appreciate your consideration of our request and your support in the successful completion of this project. Thank you very much for your time and attention.

Very respectfully yours,


ALAMIS, JOHN PAUL C.
Project Documentation and Support


DOROMAL, ROVIE ALFRED S.
Project Leader and Programmer


VILLARUZ, JOHN EDWIN B.
Project Analyst

Noted:


MELANITE, PORQUE, PhD
Capstone Adviser


DANES P. TUMABING, MIT
Program Chairperson, BS Information Systems


RUBY MAE MORANTE-BERMEJO, PhD
Capstone Project 1 Instructor



“Appendix D”

Letter to Capstone Project Adviser

**Carlos Hilado Memorial State University**
Alijis Campus • Binalbagan Campus • Fortune Towne Campus • Talisay (Main) Campus
A leading GREEN institution of higher learning in the global community by 2030
(Good governance, Research-oriented, Extension-driven, Education for Sustainable Development, and Nation-building)

College of Computer Studies

CAPSTONE PROJECT ADVISER ACCEPTANCE FORM

Date: 04/04/2025

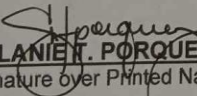
This is to attest that I, DR. MELANIE T. PORQUEZ, have agreed accepted to be the adviser of the following students with the Capstone Project entitled, INFORMAL SETTLER RELOCATION MANAGEMENT SYSTEM.

Group Name: SAMBAyanihan
Course/Year/Sec: BSIS 3B
Members:

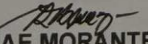
1. Alamis, John Paul C.
2. Doromal, Rovie Alfred S.
3. Villaruz, John Edwin B.

Furthermore, I agree to set schedule for the advising and consultation to help the students and ensure the success of the project.

Conforme:


MELANIE T. PORQUEZ, PhD
Signature over Printed Name of Adviser

Noted:


RUBY MAE MORANTE-BERMEJO, PhD
Capstone Project 1 Professor

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(034) 434 8148
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“Appendix E”

Sample Black Box Testing - Alpha Testing

BLACKBOX SAMPLE OF USER - REGISTRATION & AUTHENTICATION

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-001	User Registration	User details and facial recognition stored;	User registration successful; information stored in database	Pass	Fully functional
TC-002	User Login	User successfully logged in;	User successfully logged in and entered system interface	Pass	Fully functional
TC-003	User Forgot Password	Code sent to email; facial recognition verified;	Code sent via Gmail; facial recognition matched; password reset successful	Pass	Fully functional
TC-004	User Login with Invalid Credentials	Login rejected; error message displayed; remains on login page	Error message displayed; login failed	Pass	Fully functional
TC-005	User Logout	Session cleared; redirected to login;	Session terminated; redirected properly	Pass	Fully functional
TC-006	Password Strength Validation	Password requirements enforced;	Validation works correctly	Pass	Fully functional
TC-007	Session Timeout	Session expires after inactivity; redirected to login; secure logout	Session timeout works correctly	Pass	Fully functional



BLACKBOX SAMPLE OF USER - APPLICATION SUBMISSION

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-008	Submit Housing Application	Submitted successfully; status "Pending"; trackable	Submission successful; status recorded as Pending	Pass	Fully functional
TC-009	Housing Form Validation	Required fields enforced; errors displayed	Validation works correctly	Pass	Fully functional
TC-010	Document Upload Validation	Invalid files rejected; valid files accepted	Validation works correctly	Pass	Fully functional
TC-011	Cascading Location Dropdowns	Dropdowns cascade correctly; PSGC populated	Works correctly	Pass	Fully functional
TC-012	Submit Property Transfer	Submission successful; status Pending	Submission successful	Pass	UI layout fixed
TC-013	Property Transfer Validation	Required fields enforced; checkbox required	Validation works correctly	Pass	Fully functional
TC-014	Date Picker Validation	Future dates disabled; valid dates accepted	Works correctly	Pass	Fully functional
TC-015	Submit Utility Application	Submission successful; status Pending	Submission successful	Pass	UI layout fixed
TC-016	Utility Type Selection	Only one selectable; persists	Works correctly	Pass	Fully functional
TC-017	Utility Form Reset	All fields cleared; reset successful	Works correctly	Pass	Fully functional
TC-018	Submit Beneficiary Verification	Submission successful; status Applied	Status recorded as Applied	Failed	Not functional
TC-019	Verification Project Dropdown	Dropdown populated; selection saved	Works correctly	Pass	Fully functional



BLACKBOX SAMPLE OF ADMIN – HOUSING MANAGEMENT

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-025	View Housing Applications	Table loads; colored status badges	Loaded correctly	Pass	Fully functional
TC-026	Search Applications	Real-time search; partial match	Works correctly	Pass	Fixed & working
TC-027	Filter by Status	Filter shows selected status only	Works correctly	Pass	Fixed & working
TC-028	View Application Details	Modal shows full details	Works correctly	Pass	Fully functional
TC-029	Update Status	Status updates; remarks saved	Works correctly	Pass	Fixed & working
TC-030	Relocate Application	Status updates to Relocated	Does not work correctly	Failed	Not fully functional
TC-031	View Relocation History	History visible; accurate	Works correctly	Pass	Fully functional
TC-032	Housing Unit Validation	Prevents relocation to full units	Works correctly	Pass	Fully functional

BLACKBOX SAMPLE OF ADMIN – REPORTS & EXPORTS

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-050	View Interactive Map	Map loads; markers clickable	Works correctly	Pass	Fully functional
TC-051	View Map Statistics	Counts accurate; real-time updates	Works correctly	Pass	Fully functional
TC-052	Generate Custom Report	Report generates; downloadable PDF	Works correctly	Pass	Fixed & working
TC-053	Generate Certificate Forms	Correct PDF generated	Works correctly	Pass	Fully functional
TC-056	Export to Excel	Excel downloads correctly	Works correctly	Pass	Fully functional
TC-057	Export Beneficiaries to PDF	PDF generated properly	Works correctly	Pass	Fully functional



BLACKBOX SAMPLE OF SECURITY & VALIDATION

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-064	Role-Based Access Control	Unauthorized access blocked	Works correctly	Pass	Fully functional
TC-065	SQL Injection Prevention	Injection attempts fail	Works correctly	Pass	Fully functional
TC-066	XSS Prevention	Scripts executed	System Does not Prevent XSS	Fail	Not Functional
TC-067	Email Validation	Invalid emails rejected	Works correctly	Pass	Fully functional
TC-068	Phone Validation	Enforces 09 + 11 digits	Works correctly	Pass	Fully functional
TC-069	Required Field Validation	All required fields validated	Works correctly	Pass	Fully functional

BLACKBOX SAMPLE OF PERFORMANCE

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-076	Page Load Time	<3 seconds normal connection	Within limits	Pass	Fully functional
TC-077	Large Dataset Pagination	No lag; optimized queries	Works correctly	Pass	Fully functional
TC-078	Concurrent Users	Handles multiple sessions	Works correctly	Pass	Fully functional
TC-079	Query Optimization	Efficient queries; indexed	Optimized	Pass	Fully functional



BLACKBOX SAMPLE OF MONITORING SYSTEM

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-083	Access Monitoring Page	Only Relocated users allowed	Access control works	Pass	Fully functional
TC-084	Submit Monitoring Update	Photo + GPS captured	Works correctly	Pass	Fully functional
TC-085	View Monitoring History	History loads; photos viewable	Works correctly	Pass	Fully functional
TC-086	Admin Monitoring Dashboard	Stats accurate	Works correctly	Pass	Fully functional
TC-087	Admin Review Submission	Can Verify/Reject	Works correctly	Pass	Fully functional
TC-089	View Overdue Beneficiaries	Overdue calculated correctly	Works correctly	Pass	Fully functional
TC-090	Send Monitoring Warning	Email sent; logged	Works correctly	Pass	Fully functional
TC-094	Auto-Calculate Warning Level	Correct warning level auto-calculated	Does not correctly	Failed	Not functional



“Appendix F”

Sample Black Box Testing - Beta Testing

PHPUnit Testing Result:

TEST EXECUTION RESULTS

A. USER REGISTRATION & AUTHENTICATION

- ✓TC-001 User Registration
- ✓TC-002 User Login
- ✓TC-003 User Forgot Password
- ✓TC-004 User Login with Invalid Credentials
- ✓TC-005 User Logout
- ✓TC-006 Password Strength Validation
- ✓TC-007 Session Timeout

B. USER APPLICATION SUBMISSION

- ✓TC-008 User Submit Housing Application
- ✓TC-009 Housing Application Form Validation
- ✓TC-010 Housing Application Document Upload Validation
- ✓TC-011 Housing Application Cascading Location Dropdowns
- ✓TC-012 User Submit Property Transfer Application
- ✓TC-013 Property Transfer Form Validation
- ✓TC-014 Property Transfer Date Picker Validation
- ✓TC-015 User Submit Utility Connection Application
- ✓TC-016 Utility Connection Type Selection
- ✓TC-017 Utility Application Form Reset
- ✓TC-018 User Submit Beneficiary Verification Application
- ✓TC-019 Beneficiary Verification Project Type Dropdown
- ✓TC-020 Beneficiary Status Selection

C. USER APPLICATION STATUS TRACKING

- ✓TC-021 User Check Housing Application Status
- ✓TC-022 User Check Property Transfer Status
- ✓TC-023 User Check Utility Connection Status
- ✓TC-024 User Check Beneficiary Verification Status

D. ADMIN – HOUSING APPLICATIONS MANAGEMENT

- ✓TC-025 Admin View Housing Applications List
- ✓TC-026 Admin Search Housing Applications
- ✓TC-027 Admin Filter Housing Applications by Status
- ✓TC-028 Admin View Housing Application Details
- ✓TC-029 Admin Update Housing Application Status
- ✓TC-030 Admin Relocate Approved Application
- ✓TC-031 Admin View Relocation History
- ✓TC-032 Admin Relocate – Housing Unit Validation

E. ADMIN – PROPERTY TRANSFER MANAGEMENT

- ✓TC-033 Admin View Property Transfers List
- ✓TC-034 Admin Search Property Transfers
- ✓TC-035 Admin Update Property Transfer Status
- ✓TC-036 Admin Delete Property Transfer

F. ADMIN – UTILITY CONNECTION MANAGEMENT

- ✓TC-037 Admin View Utility Connections List
- ✓TC-038 Admin Filter Utility Connections by Status
- ✓TC-039 Admin Update Utility Connection Status
- ✓TC-040 Admin View Utility Connection Details



G. ADMIN – BENEFICIARY MANAGEMENT

- ✓TC-041 Admin View Beneficiary Verifications List
- ✓TC-042 Admin Search Beneficiary Verifications
- ✓TC-043 Admin Update Verification Status
- ✓TC-044 Admin View Beneficiaries List
- ✓TC-045 Admin Search and Filter Beneficiaries
- ✓TC-046 Admin Sort Beneficiaries Table

H. ADMIN – HOUSING UNIT MANAGEMENT

- ✓TC-047 Admin Add Housing Unit
- ✓TC-048 Admin Edit Housing Unit
- ✓TC-049 Admin Delete Housing Unit

I. ADMIN – MAPPING & REPORTS

- ✓TC-050 Admin View Interactive Map
- ✓TC-051 Admin View Map Statistics Panel
- ✓TC-052 Admin Generate Custom Report
- ✓TC-053 Admin Generate Certificate Forms
- ✓TC-054 Admin Generate Validation Form
- ✓TC-055 Admin Generate Certificate of Lot Assignment
- ✓TC-056 Export Housing Applications to Excel
- ✓TC-057 Export Beneficiaries to PDF

J. DOCUMENT VIEWER & FILE HANDLING

- ✓TC-058 Property Transfer Document Viewer
- ✓TC-059 File Upload Size Limit Validation
- ✓TC-060 File Upload Performance

K. NOTIFICATIONS & EMAILS

- ✓TC-061 Email Notification on Application Submission
- ✓TC-062 Email Notification on Status Change
- ✓TC-063 In-App Notification Bell

L. SECURITY & VALIDATION

- ✓TC-064 Role-Based Access Control – Admin Features
- ✓TC-065 SQL Injection Prevention
- ✓TC-066 XSS (Cross-Site Scripting) Prevention
- ✓TC-067 Email Format Validation
- ✓TC-068 Phone Number Format Validation
- ✓TC-069 Required Field Validation

M. UI/UX & NAVIGATION

- ✓TC-070 Sidebar Navigation – Expand/Collapse
- ✓TC-071 Submenu Expand/Collapse
- ✓TC-072 Responsive Design – Mobile View
- ✓TC-073 Breadcrumb Navigation
- ✓TC-074 Filter by Date Range
- ✓TC-075 Clear All Filters Button

N. PERFORMANCE & OPTIMIZATION

- ✓TC-076 Page Load Time
- ✓TC-077 Large Dataset Pagination
- ✓TC-078 Concurrent User Sessions
- ✓TC-079 Database Query Optimization



O. INTEGRATIONS & API

- ✓TC-080 Facial Recognition API Integration
- ✓TC-081 Mapbox API Integration
- ✓TC-082 Email Service Integration (SMTP)

P. HOUSING MONITORING

- ✓TC-083 User Access Monitoring Page
- ✓TC-084 User Submit Monitoring Update
- ✓TC-085 User View Monitoring History
- ✓TC-086 Admin View Monitoring Dashboard
- ✓TC-087 Admin Review Submission
- ✓TC-088 Admin Mark Active Owner
- ✓TC-089 Admin View Overdue Beneficiaries
- ✓TC-090 Admin Send Monitoring Warning
- ✓TC-091 User View Monitoring Warnings
- ✓TC-092 Admin Filter Monitoring Submissions
- ✓TC-093 Admin Verify GPS Location Map
- ✓TC-094 System Auto-Calculate Warning Level

OK (94 / 94 Test Cases Passed)

PS C:\Users\Rovie\Desktop> |



“Appendix G”

Sample White Box Testing - Alpha Testing

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result	Remarks
WB-001	actions/login.php	Login action file exists	basePath('actions/login.php')	expect(\$loginFile)->toBeFile()	File is a valid file	PASS	Validates that the login action file is present in the system.
WB-002	actions/login.php	Login requires POST method	file_get_contents(\$loginFile)	expect(\$content)->toContain("REQUEST_METHOD", "POST")	Content contains REQUEST_METHOD and POST	PASS	Ensures login form submissions use secure POST method.
WB-003	actions/login.php	Login includes recaptcha verification	file_get_contents(\$loginFile)	expect(\$content)->toContain('g-recaptcha-response', 'RECAPTCHA_SECRET_KEY')	Content contains g-recaptcha-response and RECAPTCHA_SECRET_KEY	PASS	Confirms reCAPTCHA integration for bot protection during login.
WB-004	actions/register.php	Register action file exists	basePath('actions/register.php')	expect(\$registerFile)->toBeFile()	File is a valid file	PASS	Validates that the registration action file is present in the system.
WB-005	actions/register.php	Register requires POST method	file_get_contents(\$registerFile)	expect(\$content)->toContain("REQUEST_METHOD", "POST")	Content contains REQUEST_METHOD and POST	PASS	Ensures registration form submissions use secure POST method.
WB-006	actions/register.php	Register includes email verification	file_get_contents(\$registerFile)	expect(\$content)->toContain('is_email_verified', 'email')	Content contains is_email_verified and email	PASS	Confirms email verification functionality is implemented for new users.



WB-007	actions/ register. php	Register includes facial recognition	file_get_ contents (\$regist erFile)	expect(\$c ontent)- >toContai n('faceDe scriptor', 'facial')	Content contains faceDescr iptor and facial	PASS	Verifies facial recognition biometric feature is integrated in registration.
WB-008	actions/ verify_e mail.php	Verify email action exists	basePat h('action s/verify_ email.ph p')	expect()- >toBeFile ()	File is a valid file	PASS	Confirms email verification handler file exists in the system.
WB-009	actions/ send_otp .php	Send OTP action exists	basePat h('action s/send_ otp.php')	expect()- >toBeFile ()	File is a valid file	PASS	Validates OTP sending functionality is available for two-factor authentication.
WB-010	actions/ verify_ otp.php	Verify OTP action exists	basePat h('action s/verify_ otp.php')	expect()- >toBeFile ()	File is a valid file	PASS	Confirms OTP verification handler exists for completing 2FA process.
WB-011	actions/ pagibig_ housing_ apply. php	Pagibig housing apply action exists	basePat h('action s/pagibi g_housi ng_appl y.php')	expect()- >toBeFile ()	File is a valid file	PASS	Validates Pag- IBIG housing application submission handler is present.
WB-012	actions/ get_appli cation_d etails.php	Get application details action exists	basePat h('action s/get_ap plication_ details. php')	expect()- >toBeFile ()	File is a valid file	PASS	Confirms application details retrieval functionality is available.
WB-013	actions/ update_a pplication _status. php	Update application status action exists	basePat h('action s/update_ applica tion_stat us.php')	expect()- >toBeFile ()	File is a valid file	PASS	Verifies application status update handler exists for workflow management.
WB-014	actions/ delete_a pplication .php	Delete application action exists	basePat h('action s/delete_ applica tion.php')	expect()- >toBeFile ()	File is a valid file	PASS	Confirms application deletion functionality is implemented.
WB-015	actions/r elocate_ applicant. php	Relocate applicant action exists	basePat h('action s/relocat e_applic ant.php')	expect()- >toBeFile ()	File is a valid file	PASS	Validates applicant relocation handler exists for housing reassignment.



WB-016	pages/ applications.php	Applications page exists	basePath('pages/ applications.php')	expect()->toBeFile()	File is a valid file	PASS	Confirms applications listing page is accessible in the system.
WB-017	pages/ pagibig_housing_ apply.php	Pagibig housing apply page exists	basePath('pages/ pagibig_housing_ apply.php')	expect()->toBeFile()	File is a valid file	PASS	Validates Pag- IBIG housing application form page is available.
WB-018	pages/ pagibig_housing_ status.php	Pagibig housing status page exists	basePath('pages/ pagibig_housing_ status.php')	expect()->toBeFile()	File is a valid file	PASS	Confirms application status tracking page is present for users.
WB-019	pages/ housing_relocation_ assignment.php	Housing relocation assignment page exists	basePath('pages/ housing_relocation_ assignment.php')	expect()->toBeFile()	File is a valid file	PASS	Verifies housing relocation assignment interface is available.
WB-020	pages/ applications.php	Applications page contains php code	file_get_contents(basePath('pages/ applications.php'))	expect(\$content)->toContain('<?php')	Content contains <?php tag	PASS	Ensures applications page has proper PHP code structure.
WB-021	actions/ beneficiary_verification.php	Beneficiary verification action exists	basePath('actions/ beneficiary_verification.php')	expect()->toBeFile()	File is a valid file	PASS	Validates beneficiary verification submission handler is present.
WB-022	actions/ get_beneficiary.php	Get beneficiary action exists	basePath('actions/ get_beneficiary.php')	expect()->toBeFile()	File is a valid file	PASS	Confirms beneficiary data retrieval functionality exists.
WB-023	actions/ get_beneficiary_verification_details.php	Get beneficiary verification details exists	basePath('actions/ get_beneficiary_verification_details.php')	expect()->toBeFile()	File is a valid file	PASS	Verifies beneficiary verification details retrieval is available.
WB-024	actions/ delete_beneficiary.php	Delete beneficiary action exists	basePath('actions/ delete')	expect()->toBeFile()	File is a valid file	PASS	Confirms beneficiary record deletion



WB-025	actions/delete_beneficiary_verification.php	Delete beneficiary verification exists	basePath('actions/delete_beneficiary_verification.php')	expect()->toBeFile()	File is a valid file	PASS	Validates beneficiary verification deletion handler is present.
WB-026	actions/update_beneficiary_status.php	Update beneficiary status action exists	basePath('actions/update_beneficiary_status.php')	expect()->toBeFile()	File is a valid file	PASS	Ensures beneficiary status update functionality exists for workflow.
WB-027	pages/beneficiaries.php	Beneficiaries page exists	basePath('pages/beneficiaries.php')	expect()->toBeFile()	File is a valid file	PASS	Confirms beneficiaries listing page is accessible.
WB-028	pages/beneficiary_verification.php	Beneficiary verification page exists	basePath('pages/beneficiary_verification.php')	expect()->toBeFile()	File is a valid file	PASS	Validates beneficiary verification form page is available.
WB-029	pages/beneficiary_verification_status.php	Beneficiary verification status page exists	basePath('pages/beneficiary_verification_status.php')	expect()->toBeFile()	File is a valid file	PASS	Confirms verification status tracking page exists for users.
WB-030	pages/manage_beneficiary_verifications.php	Manage beneficiary verifications page exists	basePath('pages/manage_beneficiary_verifications.php')	expect()->toBeFile()	File is a valid file	PASS	Verifies admin interface for managing verifications is present.
WB-031	actions/add_user.php	Add user action exists	basePath('actions/add_user.php')	expect()->toBeFile()	File is a valid file	PASS	Validates user creation functionality is implemented in the system.
WB-032	actions/edit_user.php	Edit user action exists	basePath('actions/edit_user.php')	expect()->toBeFile()	File is a valid file	PASS	Confirms user profile editing functionality is available.
WB-033	actions/delete_user.php	Delete user action exists	basePath('actions/delete_user.php')	expect()->toBeFile()	File is a valid file	PASS	Ensures user account deletion handler exists for admin operations.



WB-034	actions/ get_user s.php	Get users action exists	basePat h('action s/get_us ers.php')	expect()- >toBeFile ()	File is a valid file	PASS	Validates user list retrieval functionality is present.
WB-035	pages/us er_mana gement.p hp	User manage page exists	basePat h('pages /user_m anagem ent.php')	expect()- >toBeFile ()	File is a valid file	PASS	Confirms admin user management interface is accessible.
WB-036	pages/us er_home. php	User home page exists	basePat h('pages /user_h ome.ph p')	expect()- >toBeFile ()	File is a valid file	PASS	Validates user dashboard/hom e page is available after login.
WB-037	includes/ access_c ontrol. php	Access control file exists	basePat h('includ es/acce ss_contr ol.php')	expect()- >toBeFile ()	File is a valid file	PASS	Ensures access control and permission system is implemented.
WB-038	includes/ check_ad min.php	Check admin file exists	basePat h('includ es/chec k_admin .php')	expect()- >toBeFile ()	File is a valid file	PASS	Confirms admin privilege verification mechanism is present.
WB-039	actions/w ater_elec trical_con nection. php	Water electrical connection action exists	basePat h('action s/water_ electrica l_conne ction.ph p')	expect()- >toBeFile ()	File is a valid file	PASS	Validates utility connection request submission handler exists.
WB-040	actions/g et_utility_ details.ph p	Get utility details action exists	basePat h('action s/get_uti lity_deta ils.php')	expect()- >toBeFile ()	File is a valid file	PASS	Confirms utility connection details retrieval is available.
WB-041	actions/ update_u tility_stat us.php	Update utility status action exists	basePat h('action s/update _utility_ status.p hp')	expect()- >toBeFile ()	File is a valid file	PASS	Ensures utility connection status update functionality exists.
WB-042	actions/ delete_ut ility_conn ection. php	Delete utility connection action exists	basePat h('action s/delete _utility_ connecti on.php')	expect()- >toBeFile ()	File is a valid file	PASS	Validates utility connection deletion handler is implemented.
WB-043	pages/ water_el ectrical_c onnectio n.php	Water electrical connection page exists	basePat h('pages /water_e lectrica l_conne ction.php')	expect()- >toBeFile ()	File is a valid file	PASS	Confirms utility connection request form page is accessible.



WB-044	pages/water_electrical_connection_status.php	Water electrical connection status exists	basePath('pages/water_electrical_connection')	expect()->toBeFile()	File is a valid file	PASS	Validates utility connection status tracking page is present.
WB-045	pages/manage_utility_connections.php	Manage utility connections page exists	basePath('pages/manage_utility_connections.php')	expect()->toBeFile()	File is a valid file	PASS	Ensures admin interface for managing utility requests exists.
WB-046	includes/helper_functions.php	GetUtilityPhase returns correct phases	getUtilityPhase('Under Review'), 'getUtilityPhase('Processing'), getUtilityPhase('Approved'), getUtilityPhase('Connected')	expect()->toBe(1), expect()->toBe(2), expect()->toBe(3), expect()->toBe(4)	Returns 1, 2, 3, 4 respectively	PASS	Verifies utility workflow phase tracking returns correct numeric values.
WB-047	actions/assign_housing.php	Assign housing action exists	basePath('actions/assign_housing.php')	expect()->toBeFile()	File is a valid file	PASS	Validates housing unit assignment handler is implemented.
WB-048	actions/get_available_lots.php	Get available lots action exists	basePath('actions/get_available_lots.php')	expect()->toBeFile()	File is a valid file	PASS	Confirms available housing lots retrieval functionality exists.
WB-049	actions/get_lots_with_location.php	Get lots with location action exists	basePath('actions/get_lots_with_location.php')	expect()->toBeFile()	File is a valid file	PASS	Ensures lots with location data retrieval is available.
WB-050	actions/update_lot_info.php	Update lot info action exists	basePath('actions/update_lot_info.php')	expect()->toBeFile()	File is a valid file	PASS	Validates housing lot information update handler is present.



WB-051	actions/save_lot_boundary.php	Save lot boundary action exists	basePath('actions/save_lot_boundary.php')	expect()->toBeFile()	File is a valid file	PASS	Confirms lot boundary mapping save functionality is implemented.
WB-052	actions/delete_lot_boundary.php	Delete lot boundary action exists	basePath('actions/delete_lot_boundary.php')	expect()->toBeFile()	File is a valid file	PASS	Ensures lot boundary deletion handler exists for map updates.
WB-053	actions/property_transfer_current.php	Property transfer current action exists	basePath('actions/property_transfer_current.php')	expect()->toBeFile()	File is a valid file	Fail	Property transfer submission handler is not available.
WB-054	actions/get_property_transfer_details.php	Get property transfer details exists	basePath('actions/get_property_transfer_details.php')	expect()->toBeFile()	File is a valid file	PASS	Confirms property transfer details retrieval is implemented.
WB-055	actions/update_property_transfer_status.php	Update property transfer status exists	basePath('actions/update_property_transfer_status.php')	expect()->toBeFile()	File is a valid file	PASS	Ensures property transfer status update functionality exists.
WB-056	actions/delete_property_transfer.php	Delete property transfer action exists	basePath('actions/delete_property_transfer.php')	expect()->toBeFile()	File is a valid file	PASS	Validates property transfer record deletion handler is present.
WB-057	actions/generate_report.php	Generate report action exists	basePath('actions/generate_report.php')	expect()->toBeFile()	File is a valid file	Fail	Report generation functionality is not implemented in the system.
WB-058	actions/delete_report.php	Delete report action exists	basePath('actions/delete_report.php')	expect()->toBeFile()	File is a valid file	PASS	Validates generated report deletion handler is available.
WB-059	actions/get_chart_data.php	Get chart data action exists	basePath('actions/get_chart_data.php')	expect()->toBeFile()	File is a valid file	PASS	Ensures chart data retrieval for dashboard analytics exists.



WB-060	actions/generate_certificate.php	Generate certificate action exists	basePath('actions/generate_certificate.php')	expect()->toBeFile()	File is a valid file	PASS	Confirms certificate generation functionality is implemented.
WB-061	actions/generate_form_action.php	Generate form action exists	basePath('actions/generate_form_action.php')	expect()->toBeFile()	File is a valid file	PASS	Validates form generation handler is present for document creation.
WB-062	pages/reports.php	Reports page exists	basePath('pages/reports.php')	expect()->toBeFile()	File is a valid file	PASS	Ensures reports dashboard page is accessible to users.
WB-063	config.php	Config file exists and is readable	basePath('config.php'), is_readable(\$configFile)	expect(\$configFile)->toBeFile(), expect()->toBeTrue()	File exists and is readable	PASS	Validates main configuration file exists and has proper read permissions.
WB-064	config.php	Database configuration defined	defined('DB_HOST'), defined('DB_USER'), defined('DB_PASSWORD'), defined('DB_NAME')	expect()->toBeTrue() for each	All constants return true	PASS	Confirms all required database connection constants are defined.
WB-065	config.php	Recaptcha configuration defined	defined('RECAPTCHA_SITE_KEY'), defined('RECAPTCHA_SECRET_KEY')	expect()->toBeTrue() for each	Both constants return true	PASS	Ensures reCAPTCHA API keys are properly configured in the system.
WB-066	config.php	SMS API configuration defined	defined('SMS_API_TOKEN'), defined('SMS_API_URL')	expect()->toBeTrue() for each	Both constants return true	PASS	Validates SMS API credentials are configured for OTP functionality.



WB-067	Directory	Critical directories exist	is_dir(basePath('actions')), is_dir(basePath('includes')), is_dir(basePath('pages')), is_dir(basePath('tests'))	expect()->toBeTrue() for each	All directories return true	PASS	Confirms essential system directories are present with correct structure.
WB-068	Directory	Asset directories exist	is_dir(basePath('css')), is_dir(basePath('js')), is_dir(basePath('uploads'))	expect()->toBeTrue() for each	All directories return true	PASS	Ensures frontend asset and file upload directories exist.
WB-069	index.php	Index file exists	basePath('index.php')	expect()->toBeFile()	File is a valid file	PASS	Validates application entry point file is present in root directory.
WB-070	composer	Composer configuration exists	basePath('composer.json')	expect()->toBeFile()	File is a valid file	PASS	Confirms PHP dependency management configuration is available.
WB-071	includes/db_connect.php	Database connection file exists	basePath('includes/db_connect.php')	expect()->toBeFile()	File is a valid file	PASS	Ensures database connection handler file is present in the system.
WB-072	includes/helper_functions.php	Helper functions file exists	basePath('includes/helper_functions.php')	expect()->toBeFile()	File is a valid file	PASS	Validates utility helper functions file is available for reuse.
WB-088	pages/admin_home.php	Admin home page exists	basePath('pages/admin_home.php')	expect()->toBeFile()	File is a valid file	PASS	Validates admin dashboard landing page is accessible.
WB-089	pages/dashboard.php	Dashboard page exists	basePath('pages/dash')	expect()->toBeFile()	File is a valid file	PASS	Confirms main dashboard page file is present.



WB-090	pages/dashboard.php	Dashboard page contains php code	file_get_contents(basePath('pages/dashboard.php'))	expect(\$content)->toContain('<?php')	Content contains <?php tag	PASS	Ensures dashboard has proper PHP code structure for dynamic content.
WB-091	pages/housing_units.php	Housing units page exists	basePath('pages/housing_units.php')	expect()->toBeFile()	File is a valid file	PASS	Validates housing units management page is available.
WB-092	pages/manage_housing_lots.php	Manage housing lots page exists	basePath('pages/manage_housing_lots.php')	expect()->toBeFile()	File is a valid file	PASS	Confirms housing lots administration
WB-093	pages/property_transfer_status.php	Property transfer status page exists	basePath('pages/property_transfer_status.php')	expect()->toBeFile()	File is a valid file	PASS	Ensures property transfer tracking page is present for users.
WB-094	pages/manage_property_transfers.php	Manage property transfers page exists	basePath('pages/manage_property_transfers.php')	expect()->toBeFile()	File is a valid file	PASS	Validates admin interface for managing transfers is available.
WB-095	pages/visitor_logs.php	Visitor logs page exists	basePath('pages/visitor_logs.php')	expect()->toBeFile()	File is a valid file	PASS	Confirms visitor activity tracking page is accessible.
WB-096	pages/map.php	Map page exists	basePath('pages/map.php')	expect()->toBeFile()	File is a valid file	PASS	Ensures interactive map interface page is present in the system.
WB-097	pages/generate_forms.php	Generate forms page exists	basePath('pages/generate_forms.php')	expect()->toBeFile()	File is a valid file	Fail	Form generation interface page is not available to users.



“Appendix H”

Sample White Box Testing - Beta Testing

PestPHP Testing Results:

```
PS C:\xampp\htdocs\current-working\isrms> ./vendor/bin/pest --display-errors

PASS Tests\Feature\ApplicationTest 0.01s
✓ Application - Files → pagibig housing apply action exists
✓ Application - Files → get application details action exists
✓ Application - Files → update application status action exists
✓ Application - Files → delete application action exists
✓ Application - Files → relocate applicant action exists
✓ Application - Pages → applications page exists
✓ Application - Pages → pagibig housing apply page exists
✓ Application - Pages → pagibig housing status page exists
✓ Application - Pages → housing relocation assignment page exists
✓ Application - Pages → applications page contains php code

PASS Tests\Feature\AuthenticationTest 0.02s
✓ Authentication - Login → login action file exists
✓ Authentication - Login → login requires POST method
✓ Authentication - Login → login includes recaptcha verification
✓ Authentication - Registration → register action file exists
✓ Authentication - Registration → register requires POST method
✓ Authentication - Registration → register includes email verification
✓ Authentication - Registration → register includes facial recognition
✓ Authentication - Email & OTP → verify email action exists
✓ Authentication - Email & OTP → send OTP action exists
✓ Authentication - Email & OTP → verify OTP action exists

PASS Tests\Feature\BeneficiaryTest 0.01s
✓ Beneficiary - Actions → beneficiary verification action exists
✓ Beneficiary - Actions → get beneficiary action exists
✓ Beneficiary - Actions → get beneficiary verification details action exists
✓ Beneficiary - Actions → delete beneficiary action exists
✓ Beneficiary - Actions → delete beneficiary verification action exists
✓ Beneficiary - Actions → update beneficiary status action exists
✓ Beneficiary - Pages → beneficiaries page exists
✓ Beneficiary - Pages → beneficiary verification page exists
✓ Beneficiary - Pages → beneficiary verification status page exists
✓ Beneficiary - Pages → manage beneficiary verifications page exists

PASS Tests\Feature\ConfigurationTest 0.01s
✓ Configuration - Files → config file exists and is readable
✓ Configuration - Files → database configuration defined
✓ Configuration - Files → recaptcha configuration defined
✓ Configuration - Files → sms api configuration defined
✓ Configuration - Directories → critical directories exist
✓ Configuration - Directories → asset directories exist
✓ Configuration - Core Files → index file exists
✓ Configuration - Core Files → composer configuration exists
✓ Configuration - Core Files → database connection file exists
✓ Configuration - Core Files → helper functions file exists
```



```
PASS Tests\Feature\DatabaseTest
✓ Database - Connection → can connect to database 0.02s
✓ Database - Connection → uses mysql driver
✓ Database - Connection → has error mode set to exception 0.01s
✓ Database - Core Tables → users table exists 0.02s
✓ Database - Core Tables → pagibig applications table exists 0.02s
✓ Database - Core Tables → beneficiaries table exists
✓ Database - Core Tables → housing units table exists 0.01s
✓ Database - Core Tables → utility connections table exists 0.02s
✓ Database - Core Tables → property transfers table exists 0.02s
✓ Database - Table Structure → users table has required columns 0.03s
✓ Database - Table Structure → pagibig applications table has required columns 0.01s
✓ Database - Table Structure → beneficiaries table has required columns 0.02s
✓ Database - Operations → can prepare and execute statements 0.01s
✓ Database - Operations → can fetch results
✓ Database - Operations → supports transactions 0.01s
```

```
PASS Tests\Feature\HousingTest
✓ Housing - Actions → assign housing action exists 0.02s
✓ Housing - Actions → get available lots action exists
✓ Housing - Actions → get lots with location action exists
✓ Housing - Actions → update lot info action exists
✓ Housing - Actions → save lot boundary action exists
✓ Housing - Actions → delete lot boundary action exists
✓ Housing - Property Transfer → property transfer current action exists
✓ Housing - Property Transfer → get property transfer details action exists
✓ Housing - Property Transfer → update property transfer status action exists
✓ Housing - Property Transfer → delete property transfer action exists
```

```
PASS Tests\Feature\PageTest
✓ Pages - Admin → admin home page exists 0.01s
✓ Pages - Admin → dashboard page exists
✓ Pages - Admin → dashboard page contains php code
✓ Pages - Housing & Property → housing units page exists
✓ Pages - Housing & Property → manage housing lots page exists
✓ Pages - Housing & Property → property transfer status page exists
✓ Pages - Housing & Property → manage property transfers page exists
✓ Pages - Other → visitor logs page exists
✓ Pages - Other → map page exists
✓ Pages - Other → generate forms page exists
```

```
PASS Tests\Feature\ReportTest
✓ Report - Actions → generate report action exists 0.01s
✓ Report - Actions → delete report action exists
✓ Report - Actions → get chart data action exists
✓ Report - Certificate & Forms → generate certificate action exists
✓ Report - Certificate & Forms → generate form action exists
✓ Report - Certificate & Forms → reports page exists
```

```
PASS Tests\Feature\UserTest
✓ User - CRUD Actions → add user action exists 0.01s
✓ User - CRUD Actions → edit user action exists
✓ User - CRUD Actions → delete user action exists
✓ User - CRUD Actions → get users action exists
✓ User - Pages & Security → user management page exists
✓ User - Pages & Security → user home page exists
✓ User - Pages & Security → access control file exists
✓ User - Pages & Security → check admin file exists
```

```
PASS Tests\Feature\UtilityTest
✓ Utility - Actions → water electrical connection action exists 0.01s
✓ Utility - Actions → get utility details action exists
✓ Utility - Actions → update utility status action exists
✓ Utility - Actions → delete utility connection action exists
✓ Utility - Pages → water electrical connection page exists
✓ Utility - Pages → water electrical connection status page exists
✓ Utility - Pages → manage utility connections page exists
✓ Utility - Helper Functions → getUtilityPhase returns correct phases
```

Tests: 97 passed (127 assertions)
Duration: 0.51s



“Appendix I”

Grammarian Certificate



Carlos Hilado Memorial State University

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(Good governance, Research-oriented, Extension-driven, Education for Sustainable Development, and Nation-building)

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CERTIFICATION

This is to certify that the generated capstone output of **John Paul C. Alamis, Rovie Alfred S. Doromal, John Edwin B. Villaruz** with the research title **“Informal Settler Relocation Management System”** has gone through Grammarly and Plagiarism Check with **2%** rating.

Attaching herewith is the original copy of the manuscript with the following comments and findings issued by the Grammarly software.

This certification is issued for whatever legal purpose it may serve them best.

Issued this 22nd Day of January 2026.


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GREEN CHMSU ExCELSIOR

Excellence • Compassion • Environmentalism • Love of Country • Social Responsibility • Integrity • Openness • Resilience

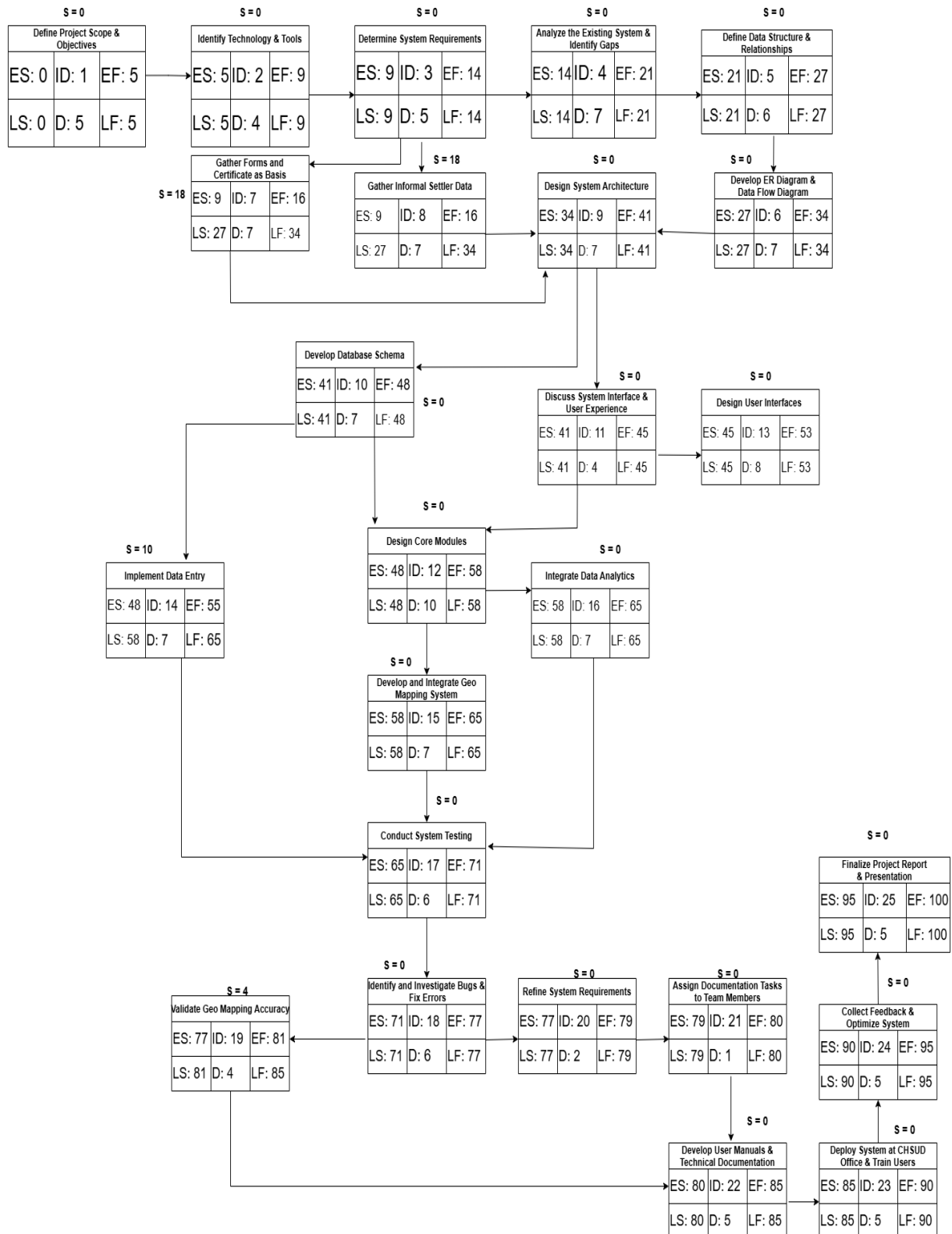


```

graph TD
    T1[Define Project Scope & Objectives  
S: 1 ID: 1  
F: 5 D: 5] --> T2[Identify Technology & Tools  
S: 6 ID: 2  
F: 9 Du: 4]
    T2 --> T3[Determine System Requirements  
S: 10 ID: 3  
F: 14 Du: 5]
    T3 --> T4[Analyze the System & Identify Gaps  
S: 15 ID: 4  
F: 21 Du: 7]
    T4 --> T5[Define Data Structure & Relationships  
S: 22 ID: 5  
F: 27 Du: 6]
    T5 --> T6[Develop ER Diagram & Data Flow Diagram  
S: 28 ID: 6  
F: 34 Du: 7]
    T5 --> T7[Gather Informal Settler Data  
S: 42 ID: 8  
F: 48 Du: 7]
    T7 --> T8[Design System Architecture  
S: 49 ID: 9  
F: 55 Du: 7]
    T8 --> T9[Design User Interfaces  
S: 77 ID: 13  
F: 84 Du: 8]
    T8 --> T10[Discuss System Interface & User Experience  
S: 63 ID: 11  
F: 66 Du: 4]
    T10 --> T9
    T10 --> T11[Integrate Visualization Tool  
S: 99 ID: 16  
F: 105 Du: 7]
    T9 --> T11
    T11 --> T12[Conduct System Testing  
S: 106 ID: 17  
F: 111 Du: 6]
    T6 --> T13[Develop Database Schema  
S: 56 ID: 10  
F: 62 Du: 7]
    T13 --> T14[Implement Data Entry  
S: 85 ID: 14  
F: 91 Du: 7]
    T14 --> T12
    T12 --> T15[Identify Bugs & Fix Errors  
S: 112 ID: 18  
F: 117 Du: 6]
    T15 --> T16[Validate Geo Mapping Accuracy  
S: 118 ID: 19  
F: 121 Du: 4]
    T16 --> T17[Deploy System at CHSUD Office & Train Users  
S: 130 ID: 23  
F: 134 Du: 5]
    T17 --> T18[Collect Feedback & Optimize System  
S: 135 ID: 24  
F: 134 Du: 5]
    T18 --> T19[Finalize Project Report & Presentation  
S: 135 ID: 25  
F: 139 Du: 5]
    T12 --> T20[Assign Documentation Tasks to Team Members  
S: 124 ID: 21  
F: 124 Du: 1]
    T20 --> T21[Develop User Manuals & Technical Documentation  
S: 125 ID: 22  
F: 129 Du: 5]
    T21 --> T22[Refine System Requirements  
S: 122 ID: 20  
F: 123 Du: 2]
    
```



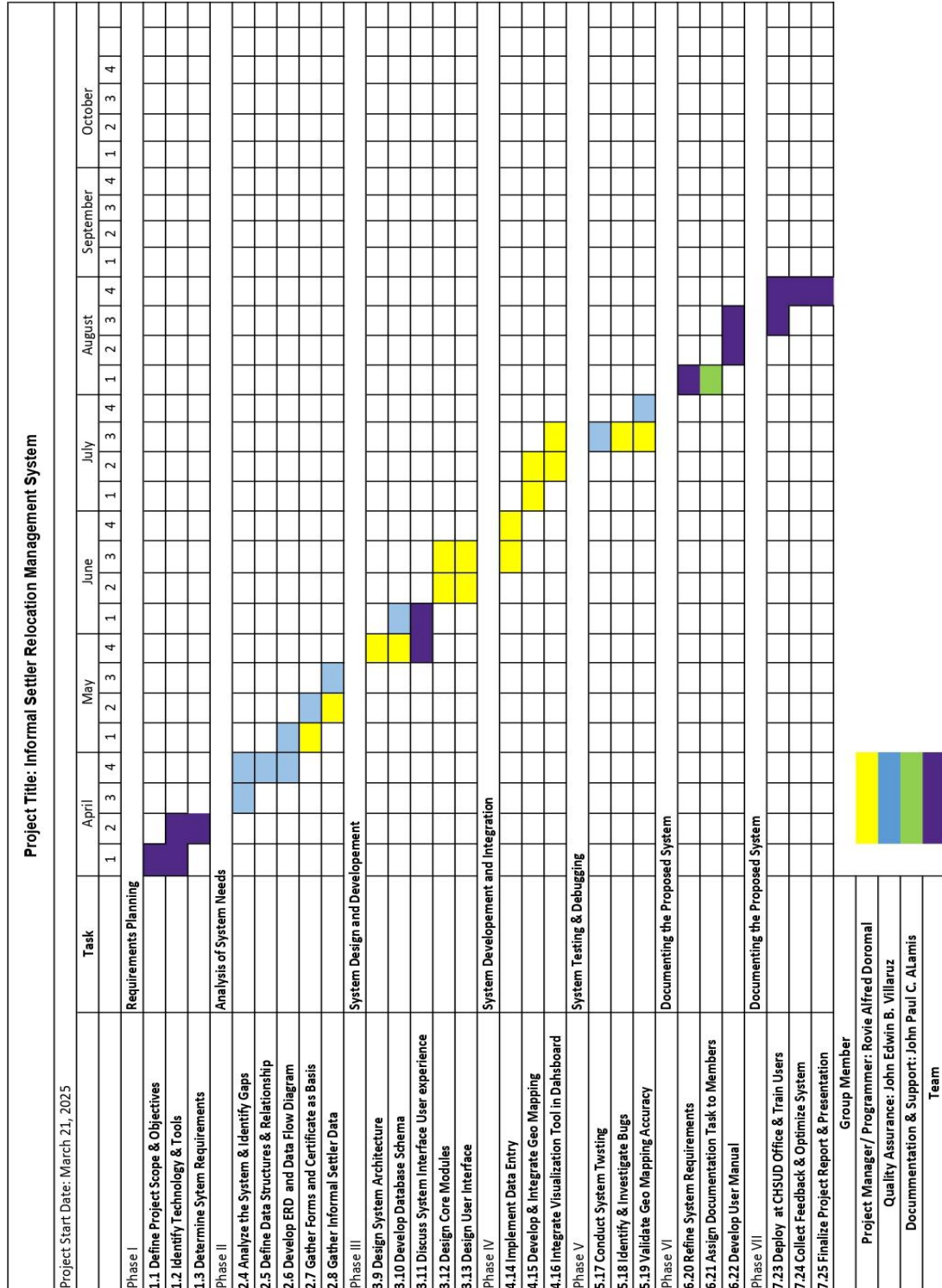
Critical Path Method





“Appendix K”

Gantt Chart





“Appendix L”

User's Guide

Informal Settler Relocation Management System

User's Guide





USER / ADMIN

1. Login Form

- a. Go to isrms.chsud.online.
- b. Click Register Here to create an account.
- c. Enter the registered username or email and password.
- d. Complete the CAPTCHA verification.
- e. Click the Sign In button.
- f. Continue by signing in with a Google account.



1.1 Data Privacy Notice Form

Data Privacy Notice

CHSUD (Informal Settler Relocation Management System) is committed to protecting your privacy and ensuring the security of your personal information in compliance with Republic Act No. 10173, also known as the Data Privacy Act of 2012.

Information We Collect

By registering and using this system, you consent to the collection and processing of the following information. We collect personal information including your full name (first name, middle name, last name, suffix), contact details (email address, phone number), residential address (region, province, city, barangay, street address), username and encrypted password, and government-issued identification documents.

We also collect sensitive personal information, most notably biometric data in the form of facial recognition data points captured through your device camera for identity verification and account security purposes. Additionally, we collect information regarding your marital status and family composition, housing and relocation information, income and financial information for program eligibility assessment, and previous housing records and displacement history.

The system also generates and maintains certain information automatically, including login timestamps and activity logs, IP addresses and device information, application status and transaction history, as well as uploaded documents and form submissions.

Purpose of Data Collection

a → X Decline I Agree and Proceed to Registration ← b

a. Click the Decline button to refuse the agreement.

b. Click the Agree button to proceed with registration.

1.3 Registration Form

Create Account

First Name Middle Name Last Name Suffix

a → None

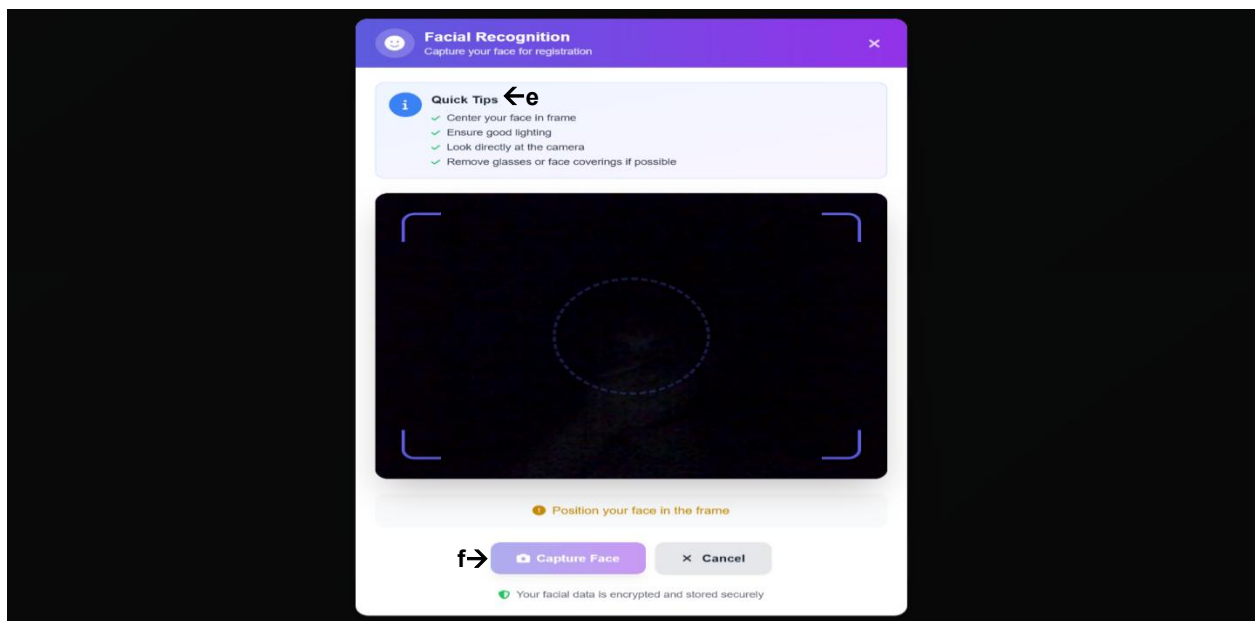
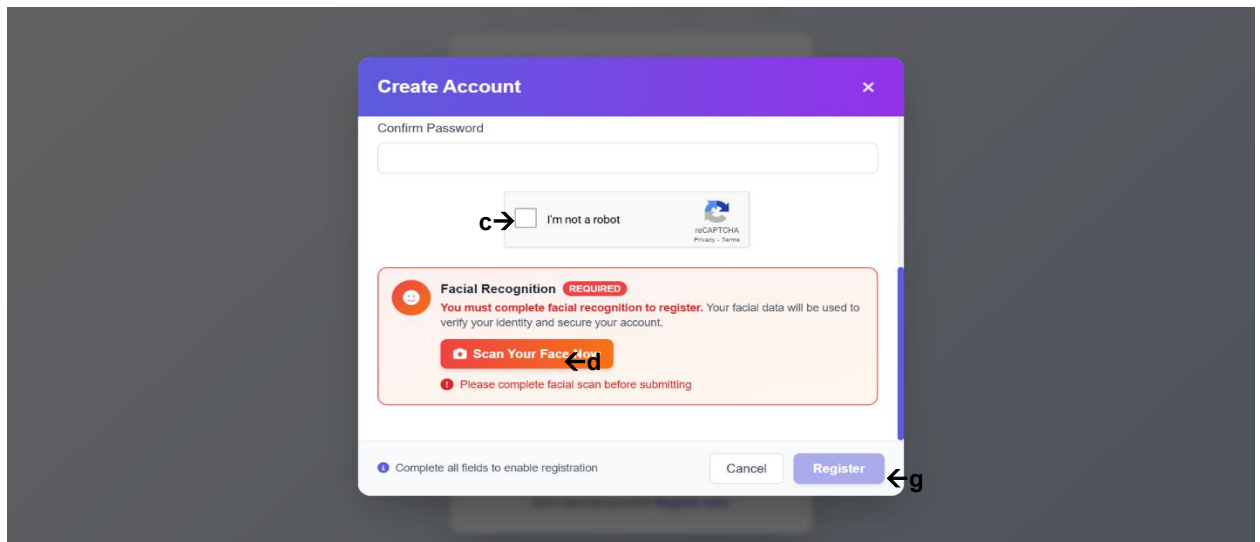
Username

Email Verify ← b

Password

Confirm Password

Complete all fields to enable registration Cancel Register



- a. Fill out all required input fields.
- b. Click Verify to complete Gmail account verification.
- c. Complete the CAPTCHA verification.
- d. Click Scan Your Face Now to begin facial recognition.
- e. Follow the on-screen tips for accurate facial recognition.
- f. Click Capture Face to record facial data.
- g. Click the Register button to complete the registration process.



USER:

2. User Dashboard

The dashboard interface includes a sidebar menu with the following services:

- Home (selected)
- Application Services
 - Housing Application
 - Property Transfer
 - Water & Electrical
 - Beneficiary Verification
 - Housing Monitoring

The main content area displays a welcome message and two primary service cards:

- Housing Application:** Apply for affordable housing units and check your application status. Buttons: Apply Now, Check Status.
- Property Transfer:** Transfer property ownership and track your transfer requests. Buttons: Apply Now, Check Status.

The notifications dropdown menu shows the following updates:

- Housing Application Status Updated: Your housing application status has been updated to: Approved. Just now.
- Housing Application Status Updated: Your housing application status has been updated to: Processing. Just now.
- Housing Application Status Updated: Your housing application status has been updated to: Verification. Just now.
- Housing Application Status Updated: (truncated)

- Click Home to display the dashboard.
- Click the Bell icon to view notifications.
- Click Housing Application to submit an application.
- Click Property Transfer to apply for a transfer request.
- Click Water and Electrical to submit utility-related applications.



f. Click Beneficiary Verification to apply for verification.

g. Click Housing Monitoring to submit a housing monitoring report.

USER

3. Housing Application

ISRMS

Services

- Home
- Application Services
 - Housing Application
 - Apply**
 - Check Status
 - Property Transfer
 - Water & Electrical
 - Beneficiary Verification
 - Housing Monitoring

Housing Application

About this Service

CHSUD, in coordination with DHSUD, Pag-IBIG fund, and other relevant national and local housing agencies, endeavors to provide government employees and Pag-IBIG members who belong to informal settler families (ISFs).

Requirements

- Certificate of No Real Property
- Certificate of Net Pay OR any other proof of income
- Pag-IBIG MID Number
- Photocopy of valid ID with specimen signature

First Name: PABLITO Middle Name: CONSTANTINO DOROMAL Last Name: JR. Suffix: Jr.

Contact Number: 09XXXXXXXXX **Send OTP**

Date of Birth: mm/dd/yyyy

Region: Province:

Region: NEGROS ISLAND REGION Province: NEGROS OCCIDENTAL

City/Municipality: VICTORIAS CITY Barangay: BARANGAY 13

Current Address (Street/House Number): ENTER STREET ADDRESS OR HOUSE NUMBER **Get Location**

☐ I am a Pag-IBIG member Employment Status: Select Status

Monthly Income:

Please upload copies of the required documents (Images or PDF only, max 5MB):

Certificate of No Real Property: **Choose File** No file chosen

Proof of Income: **Choose File** No file chosen



ISRMS

ENTER STREET ADDRESS OR HOUSE NUMBER [Get Location](#)

☐ I am a Pag-IBIG member

Employment Status
[Select Status](#)

Monthly Income

Please upload copies of the required documents (Images or PDF only, max 5MB):

Certificate of No Real Property
[Choose File](#) No file chosen

Proof of Income
[Choose File](#) No file chosen

Valid ID with Specimen Signature
[Choose File](#) No file chosen

☐ I hereby certify that all information provided are true and correct to the best of my knowledge.

[Reset](#) [Review Application](#)

- Fill out all required input fields.
- Click Send OTP to verify the phone number.
- Click Get Location to automatically retrieve the location.
- Click Choose File to upload the required documents.
- Check the Certify checkbox to confirm the information provided.
- Click Reset to clear all entered data.
- Click Review Application to proceed with reviewing the submitted information.



USER:

4. Property Transfer

ISRMS

Services

- Home
- Application Services
 - Housing Application
 - Property Transfer **Apply**
 - Check Status
- Water & Electrical
- Beneficiary Verification
- Housing Monitoring

Property Transfer - Current Beneficiaries

About this Service
This service facilitates the transfer of property rights for current beneficiaries of housing projects.

Requirements

- Proof of Beneficiary Status
- Photocopy of Valid ID
- Lot Assignment Document

First Name: PABLITO Middle Name: CONSTANTINO DOROMAL Last Name: JR. Suffix: None

Contact Number: 09924341723

Lot Number: Site 1, Phase 1, Block 1, Lot 10 Lot Assignment Date: 12/01/2025

Please upload copies of the required documents (Images or PDF only, max 5MB):

Proof of Beneficiary Status

a → No file chosen

Valid ID

No file chosen

Lot Assignment Document

No file chosen

b → ☐ I hereby certify that all information provided are true and correct to the best of my knowledge

c → ← d

- Click Choose File and select the required documents.
- Check the Certify checkbox to confirm the information provided.
- Click Reset to clear all current input data.
- Click Review Application to proceed with reviewing the application.



USER:

5. Water & Electrical

ISRMS

Services

- Home
- Application Services
 - Housing Application
 - Property Transfer
 - Apply
 - Check Status
 - Water & Electrical**
 - Apply**
 - Check Status
 - Beneficiary Verification
 - Housing Monitoring

Water & Electrical Connection Application

About this Service
Apply for water and electrical connections for your housing unit.

Requirements

- Proof of Residence
- Photocopy of Valid ID
- Housing Unit Assignment Document

First Name: PABLITO Middle Name: CONSTANTINO DOROMAL Last Name: JR. Suffix: None

Contact Number: 09924341723

Housing Unit Address: VILLA VICTORIAS, BARANGAY XIII, VICTORIAS CITY

Connection Type: Select Type Block Number: 1 Lot Number: 10

Please upload copies of the required documents (Images or PDF only, max 5MB):

Proof of Residence: Choose File No file chosen

Valid ID: Choose File No file chosen

Housing Unit Assignment Document: Choose File No file chosen

☐ I hereby certify that all information provided are true and correct to the best of my knowledge

Reset Review Application

a. Fill out all required input fields.

b. Click Choose File and select the required documents.

c. Check the Certify checkbox to confirm the information provided.

d. Click Reset to clear all current input data.

e. Click Review Application to proceed with submission.



USER:

6. Beneficiary Verification

Services

- Home
- Application Services
 - Housing Application
 - Property Transfer
 - Water & Electrical
 - Beneficiary Verification**
 - Apply**
 - Check Status
 - Housing Monitoring

Beneficiary Verification

About this Service
Verify your status as a beneficiary of a housing project or report concerns.

Requirements

- Proof of Identity
- Housing Project Details

First Name: PABLITO Middle Name: CONSTANTINO DOROMAL Last Name: JR. Suffix: None

Contact Number: 09924341723 Current Address: SAMPLE

Block Number: 1 Lot Number: 10

Housing Project Type: Select Type Beneficiary Status: Select Status

Details of Concern (if any)

Please upload copies of the required documents (Images or PDF only, max 5MB):

Proof of Identity: Choose File No file chosen

Housing Project Details: Choose File No file chosen

☒ I hereby certify that all information provided are true and correct to the best of my knowledge.

Reset Review Verification Request

a. Fill out all required input fields.

b. Click Choose File and select the required documents.

c. Check the Certify checkbox to confirm the information provided.

d. Click Reset to clear all current input data.

e. Click Review Application to proceed with the submission.



USER:

7. Housing Monitoring

The image displays two screenshots of a web application interface for Housing Unit Monitoring. The top screenshot shows the dashboard with a sidebar menu and a main content area. The bottom screenshot shows the 'Submit Monitoring Update' form with a camera view and a 'Capture Photo' button.

Top Screenshot: Housing Unit Monitoring Dashboard

Services

- Home
- Application Services
 - Housing Application
 - Property Transfer
 - Water & Electrical
 - Beneficiary Verification
- Apply
- Check Status
- Housing Monitoring**

Housing Unit Monitoring
Submit monitoring updates every 3 months to maintain your housing unit

Your Property
Villa Victorias
Site 1, Phase 1, Block 1, Lot 10

Next Update
OVERDUE
No submissions yet

Total Submissions
0
Verified: 0

Owner Status
Monitoring
Pending Verification

Submit Monitoring Update
Take Photo of Your Housing Unit *

Start Camera **Capture Photo**

Bottom Screenshot: Submit Monitoring Update Form

Submit Monitoring Update
Take Photo of Your Housing Unit *

a → **Start Camera** **Capture Photo** ← **b**

GPS Location Required: Your location will be automatically captured when you submit this form to verify you are at your housing unit.
📶 Detecting location...

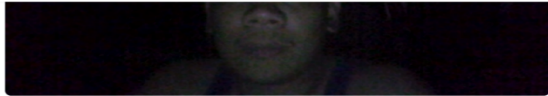


☰ **ISRMS**   **PABLITO CONSTANTINO DOROMAL JR.** 

Services

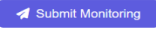
- Home
- Application Services 
 - Housing Application 
 - Property Transfer 
 - Water & Electrical 
 - Beneficiary Verification 
 - Apply
 - Check Status
 - Housing Monitoring** 

Monitoring Form

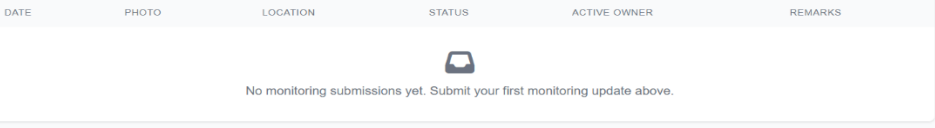


 **GPS Location Required:** Your location will be automatically captured when you submit this form to verify you are at your housing unit.

 Detecting location...

  **c**

Monitoring History

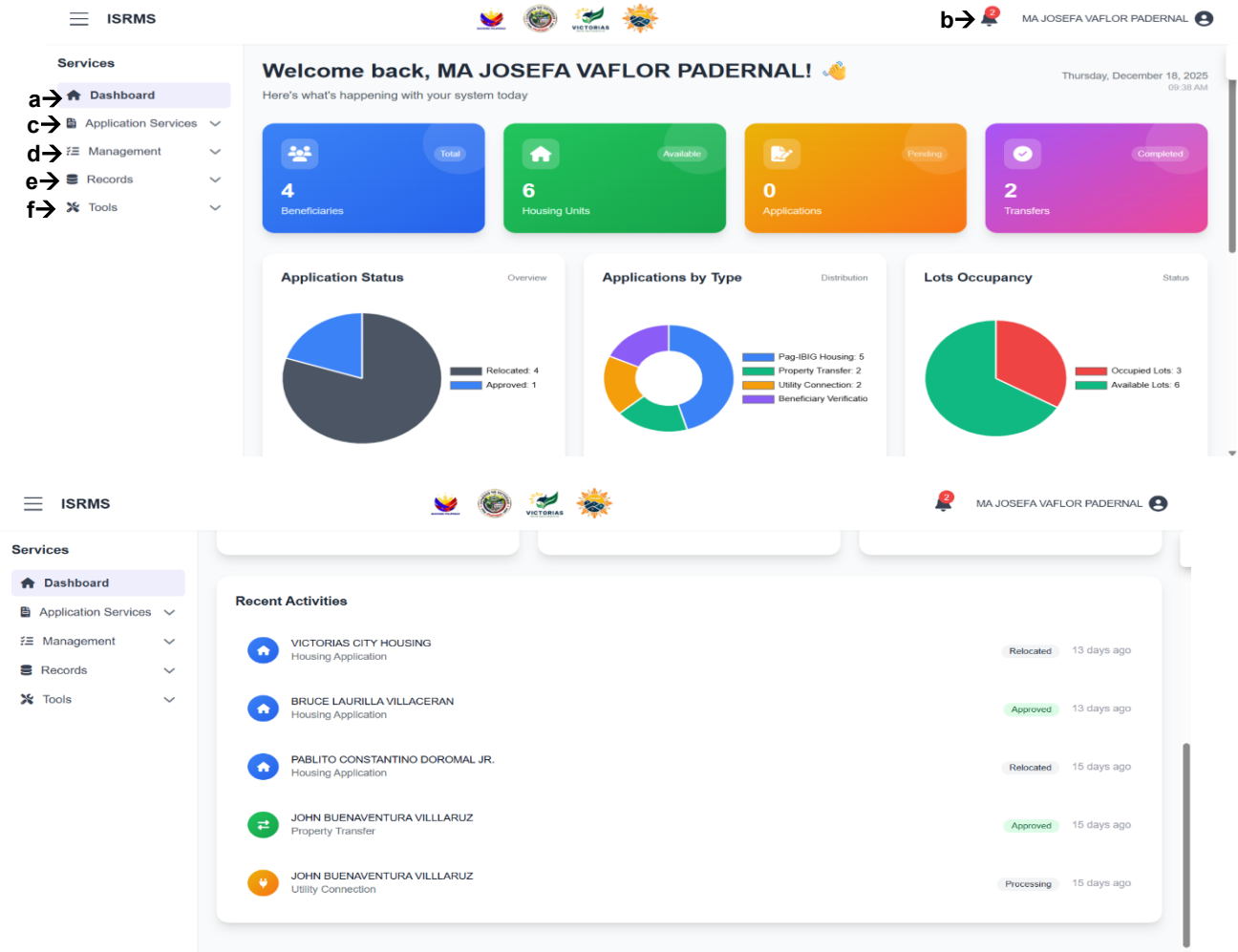
DATE	PHOTO	LOCATION	STATUS	ACTIVE OWNER	REMARKS
					

- Click Start Camera to begin facial recognition.
- Click Capture Photo to take the required image.
- Click Submit Monitoring to complete the process.



ADMIN:

1. Admin Dashboard



- Click Home to display the dashboard.
- Click the Bell icon to view notifications.
- Click Application Services to access available services.
- Click Management to view and manage relevant tasks.
- Click Records to view submitted or stored records.
- Click Tools to access additional system utilities.



2. Management Housing Applications

ISRMS

Services

- Dashboard
- Application Services
- Management
 - Housing Applications**
 - Relocation Assignment
 - Utilities
 - Verifications
 - Property Transfers
 - User Management
 - Housing Monitoring

Housing Applications

Search by name or ID... All Status

ID	NAME	STATUS	DATE	ACTIONS
#35	JOHN BUENAVENTURA VILLARUZ jbvillaruz.chmsu@gmail.com	Relocated	Nov 30, 2025	b → View Update ← c
#37	MHIEL TOMELDEN PORQUEZ melanie.porquez@chmsu.edu.ph	Relocated	Dec 2, 2025	View Update Delete
#38	PABLITO CONSTANTINO DOROMAL JR. rsdoromal.chmsu@gmail.com	Relocated	Dec 3, 2025	View Update Delete
#50	BRUCE LAURILLA VILLACERAN brucevillaceran@gmail.com	Approved	Dec 4, 2025	View Update Delete
#51	VICTORIAS CITY HOUSING chsud.victorias.demo@gmail.com	Relocated	Dec 4, 2025	View Update Delete

Show 10 entries Showing 1 to 5 of 5 entries Previous 1 Next

- Navigate to Housing Applications.
- Click View Text to see detailed information.
- Click Update Text to modify the status.

2.1 Housing Application Details

ISRMS

Services

- Dashboard
- Application Services
- Management
 - Housing Applications**
 - Relocation Assignment
 - Utilities
 - Verifications
 - Property Transfers
 - User Management
 - Housing Monitoring

Housing Applications

Search by name or ID... All Status

ID	NAME	STATUS	DATE	ACTIONS
#35	JOHN BUENAVENTURA VILLARUZ jbvillaruz.chmsu@gmail.com	Relocated	Nov 30, 2025	View Update Delete
#37	MHIEL TOMELDEN PORQUEZ melanie.porquez@chmsu.edu.ph	Relocated	Dec 2, 2025	View Update Delete
#38	PABLITO CONSTANTINO DOROMAL JR. rsdoromal.chmsu@gmail.com	Relocated	Dec 3, 2025	View Update Delete
#50	BRUCE LAURILLA VILLACERAN brucevillaceran@gmail.com	Approved	Dec 4, 2025	View Update Delete
#51	VICTORIAS CITY HOUSING chsud.victorias.demo@gmail.com	Relocated	Dec 4, 2025	View Update Delete

Show 10 entries Showing 1 to 5 of 5 entries Previous 1 Next

Personal Information

Full Name: VICTORIAS CITY HOUSING

Date of Birth: 2003-02-17

Contact Number: 09567800659

Pag-IBIG MID: N/A

Address Information

Street Address: MALIHAO STREET, ESTRELLA VILLAGE

Region: NEGROS ISLAND REGION

City/Municipality: VICTORIAS CITY

Province: NEGROS OCCIDENTAL

Barangay: BARANGAY 13

Documents

Certificate of No Real Property **Proof of Income** **Valid ID**

View Document **View Document** **View Document**

- Click View Document Text to open and view the submitted document.



2.2 Update Housing Application Status

The screenshot shows the ISRMS interface. On the left is a sidebar with 'Services' and 'Management' sections. The 'Housing Applications' link is highlighted. The main area displays a table of housing applications. A modal titled 'Update Housing Application Status' is open over the table. The modal contains a 'Current Phase' dropdown set to 'Relocated', a 'New Phase' dropdown, a 'Remarks' text area, and 'Cancel' and 'Update Status' buttons. Arrows 'a', 'b', and 'c' point to the 'Select phase...', 'Add any notes or remarks...', and 'Update Status' buttons respectively.

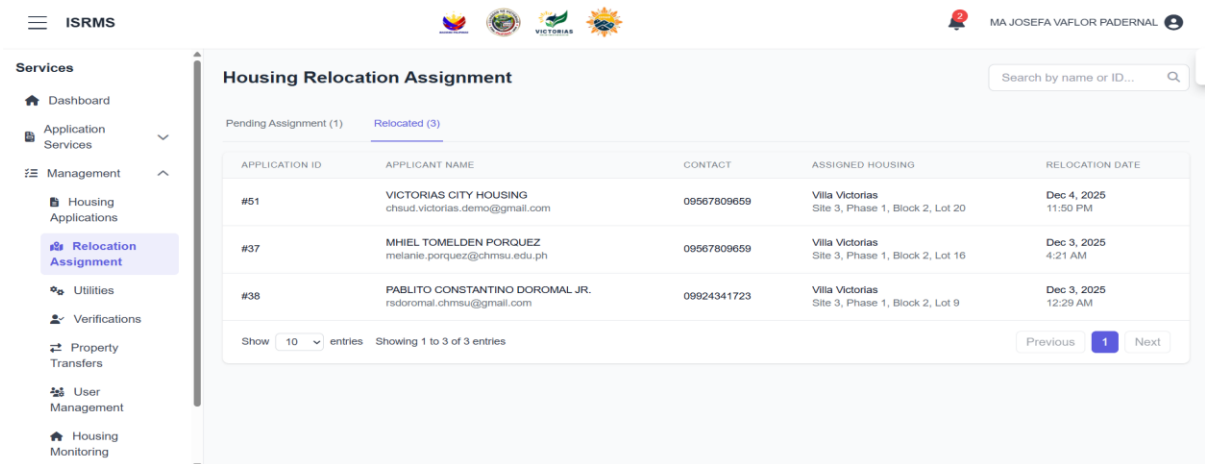
ID	NAME	DATE	ACTIONS
#35	JOHN BUENAVENTURA jbvillaruz.chmsu@gmail.com	Nov 30, 2025	View Update Delete
#37	MHIEL TOMELDEN P melanie.porquez@chmsu.edu.ph	Dec 2, 2025	View Update Delete
#38	PABLITO CONSTANTINO rsdoromal.chmsu@gmail.com	Dec 3, 2025	View Update Delete
#50	BRUCE LAURILLA VILLACERAN brucevillaceran@gmail.com	Dec 4, 2025	View Update Delete
#51	VICTORIAS CITY HOUSING chsud.victorias.demo@gmail.com	Dec 4, 2025	View Update Delete

- Select the desired Status.
- Add Remarks if necessary.
- Click Update Status to save the changes.

3. Relocation Assignment Management

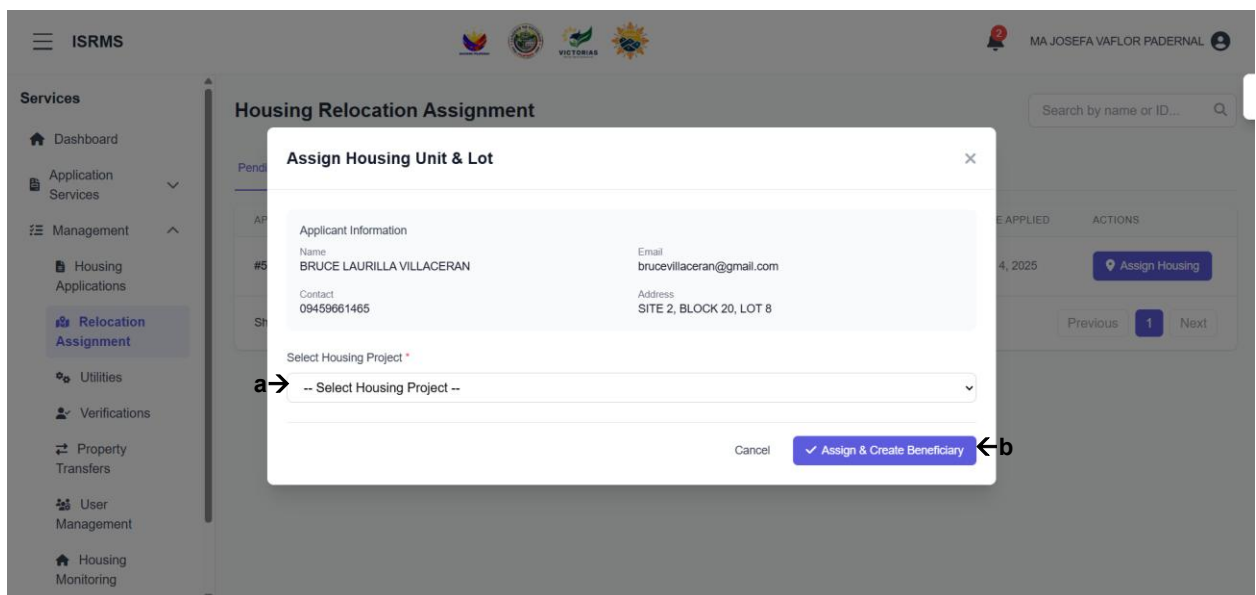
The screenshot shows the ISRMS interface with the 'Housing Relocation Assignment' page. The left sidebar has 'Relocation Assignment' highlighted. The main area shows a table of relocation assignments. A modal is open over the table. Arrows 'a', 'b', and 'c' point to the 'Relocation Assignment' link in the sidebar, the 'Relocated (3)' tab, and the 'Assign Housing' button respectively.

APPLICATION ID	APPLICANT NAME	CONTACT	LOCATION	DATE APPLIED	ACTIONS
#50	BRUCE LAURILLA VILLACERAN brucevillaceran@gmail.com	09459661465	VICTORIAS CITY, NEGROS OCCIDENTAL	Dec 4, 2025	Assign Housing



- Navigate to Relocation Assignment.
- Click Relocated to view relocated applicants.
- Click Assign Housing to assign housing to selected applicants.

3.1 Assign Housing Unit & Lot



- Select the desired Available Housing Projects.
- Click the Save button to apply the changes.



4. Utility Connections Management

ISRMS

Services

- Dashboard
- Application Services
- Management
 - Housing Applications
 - Relocation Assignment
 - Utilities**
 - Verifications
 - Property Transfers
 - User Management
 - Housing Monitoring

Manage Utility Connections

Search connections... All Status

APPLICANT	TYPE	ADDRESS	STATUS	DATE	ACTIONS
JOHN BUENAVENTURA VILLARUZ jbvillaruz.chmsu@gmail.com	Both	Block 2, Lot 9 VILLA VICTORIAS, BARANGAY XIII, VICTORIAS CITY	Processing	Dec 3, 2025	View Update Delete
PABLITO CONSTANTINO DOROMAL JR. rsdoromal.chmsu@gmail.com	Both	Block 2, Lot 15 VILLA VICTORIAS, BARANGAY XIII, VICTORIAS CITY	Pending	Dec 3, 2025	View Update Delete

Show 10 entries Showing 1 to 2 of 2 entries

Previous 1 Next

- Navigate to Utilities.
- Click View to see the full details of the selected record.
- Click Update to modify the current status.

4.1 Utility Connection Details

Utility Connection Details

Applicant Information

Full Name: JOHN BUENAVENTURA VILLARUZ
Email: jbvillaruz.chmsu@gmail.com
Contact Number: 09158751248

Connection Details

Type: both
Block/Lot: Block 2, Lot 9
Address: VILLA VICTORIAS, BARANGAY XIII, VICTORIAS CITY
Status: Processing
Date Submitted: 12/3/2025

Documents

- Proof of Residence [View Document](#)
- Valid ID [View Document](#)
- Assignment Document [View Document](#)

Remarks: None

- Click View Document to open and view the uploaded document.



4.2 Update Utility Connection

Update Utility Connection Status

Current Status: **Processing**

New Status
a → Select status...

Remarks (Optional)
b → Add any notes or remarks...

Cancel Update Status ← c

ID	APPLICANT	HOUSING UNIT	STATUS	DATE	ACTIONS
#14	JOHN BUENAVENTURA VILLARUZ jbvillaruz.chmsu@gmail.com	Block 2, Lot 15 pagibig	Processing	Dec 3, 2025	View Update Delete
#13	PABLITO CONSTANTINO DOROMAL JR. rsdoromal.chmsu@gmail.com	Block 1, Lot 3 pagibig	Pending	Dec 3, 2025	View Update Delete

- Select the desired Status.
- Enter any Remarks if applicable.
- Click Update Status to save the changes.

5. Beneficiary Verifications Management

Manage Beneficiary Verifications

a → Verifications

ID	APPLICANT	HOUSING UNIT	STATUS	DATE	ACTIONS
#14	PABLITO CONSTANTINO DOROMAL JR. rsdoromal.chmsu@gmail.com	Block 2, Lot 15 pagibig	Pending	Dec 3, 2025	b → View Update ← c
#13	JOHN EDWIN BUENAVENTURA VILLARUZ jbvillaruz.chmsu@gmail.com	Block 1, Lot 3 pagibig	Verified	Nov 30, 2025	View Update Delete

- Navigate to Verifications.
- Click View Text to see detailed information.



c. Click Update Text to modify the status.

5.1 Beneficiary Details

The screenshot shows the ISRMS interface with the 'Manage Beneficiary Verifications' page. A modal window titled 'Beneficiary Verification Details' is open, displaying the following information:

- Personal Information:**
 - Full Name: PABLITO CONSTANTINO DOROMAL JR.
 - Email: rsdoromal.chmsu@gmail.com
 - Contact Number: 09924341723
- Verification Details:**
 - Verification ID: #14
 - Status: Pending
 - Date Submitted: 12/3/2025
- Housing Unit Information:**
 - Block Number: 2
 - Lot Number: 15
 - Housing Project Type: pagibig
- Documents:**
 - Proof of Identity: [View Document](#)
 - Housing Project Details: [View Document](#)

The modal is labeled with 'a' pointing to the 'View Document' link.

a. Click View Document to open and view the uploaded document.

5.2 Update Beneficiary Status

The screenshot shows the ISRMS interface with the 'Manage Beneficiary Verifications' page. A modal window titled 'Update Verification Status' is open, displaying the following information:

- Current Status:** Pending
- New Status:** [Select status...](#)
- Remarks (Optional):** [Add any notes or remarks...](#)
- Buttons:** Cancel, [Update Status](#)

The modal is labeled with 'a' pointing to the 'Select status...' dropdown, 'b' pointing to the 'Remarks (Optional)' text area, and 'c' pointing to the 'Update Status' button.

a. Select the desired Status.

b. Enter any Remarks if applicable.

c. Click Update Status to save the changes.



5. Property Transfers Management

ISRMS

Services

- Dashboard
- Application Services
- Management
 - Housing Applications
 - Relocation Assignment
 - Utilities
 - Verifications
 - Property Transfers**
 - User Management
 - Housing Monitoring

Manage Property Transfers

Search transfers... All Status

Current Transfers (2) Transfer Property (4 Beneficiaries) ←b

ID	APPLICANT	LOT NUMBER	ASSIGNMENT DATE	STATUS	DATE	ACTIONS
#15	JOHN BUENAVENTURA VILLARUZ jbvillaruz.chmsu@gmail.com	Site 3, Phase 1, Block 2, Lot 9	Dec 3, 2025	Approved	Dec 3, 2025	c→ View Update ←d
#12	PABLITO CONSTANTINO DOROMAL JR. rsdoromal.chmsu@gmail.com	Site 3, Phase 1, Block 2, Lot 15	Dec 3, 2025	Approved	Dec 3, 2025	View Update Delete

Show 10 entries Showing 1 to 2 of 2 entries

Previous 1 Next

- Navigate to Property Transfers.
- Click Transfer Property to view property transfer details.
- Click View Text to see more detailed information.
- Click Update Text to modify the current status.

5.1 Property Transfer Details

Services

- Dashboard
- Application Services
- Management
- Housing Applications
- Relocation Assignment
- Utilities
- Verifications
- Property Transfers**
- User Management
- Housing Monitoring

MA JOSEFA VAFLOP PADERNAL

Manage Property Transfers

Current Transfers (2)

ID	APPLICANT	Status	Date	Actions
#15	JOHN BUENAVENTURA VILLARUZ jbvillaruz@gmail.com	Approved	Dec 3, 2025	View Update Delete
#12	PABLO TORRES DOMESTICO rsdoromestico@gmail.com	Approved	Dec 3, 2025	View Update Delete

Property Transfer Details

Applicant Information

Full Name: JOHN BUENAVENTURA VILLARUZ

Email: jbvillaruz.chmsu@gmail.com

Contact Number: 09158751248

Documents

- Beneficiary Proof [View Document](#)
- Valid ID [View Document](#)

Remarks

None

Transfer Details

Lot Number: Site 3, Phase 1, Block 2, Lot 9

Assignment Date: 12/3/2025

Status: Approved

Date Submitted: 12/3/2025

[View Document](#)

- a.** Click View Document to open and view the uploaded document.

5.2 Update Property Transfer Status

ISRMS

MA JOSEFA VAFLOR PADERNAL

Services

Dashboard

Application Services

Management

Housing Applications

Relocation Assignment

Utilities

Verifications

Property Transfers

User Management

Housing Monitoring

Manage Property Transfers

Current Transfers (2)

Transfer Properties

ID	APPLICANT	STATUS	DATE	ACTIONS
#15	JOHN BUENAVENTURA V. Jbvillaruz.chmsu@gmail.com	Approved	Dec 3, 2025	View Update Delete
#12	PABLITO CONSTANTINO I. rsdoromal.chmsu@gmail.com	Approved	Dec 3, 2025	View Update Delete

Show 10 entries Showing 1 to 2 of 2

Update Property Transfer Status

Current Status: Approved

New Status

Select status...

Remarks (Optional)

Add any notes or remarks...

These remarks will be visible to the applicant.

Cancel

Update

- a.** Select the desired Status.
- b.** Enter any Remarks if applicable.
- c.** Click Update Status to save the changes.



5.3 Transfer Property

ISRMS

Services

- Dashboard
- Application Services
- Management
 - Housing Applications
 - Relocation Assignment
 - Utilities
 - Verifications
 - Property Transfers**
 - User Management
 - Housing Monitoring

Manage Property Transfers

Search transfers... All Status

Current Transfers (2) Transfer Property (4 Beneficiaries)

BENEFICIARY ID	NAME	CONTACT	CURRENT PROPERTY	ASSIGNED DATE	ACTIONS
#BEN-19	VICTORIAS CITY HOUSING	09567809659	Villa Victorias Site 3, Phase 1, Block 2, Lot 20	Dec 4, 2025	Transfer Property
#BEN-18	MHIEL TOMELDEN PORQUEZ	09567809659	Villa Victorias Site 3, Phase 1, Block 2, Lot 16	Dec 3, 2025	Transfer Property
#BEN-17	PABLITO CONSTANTINO DOROMAL JR.	09924341723	Villa Victorias Site 3, Phase 1, Block 2, Lot 9	Dec 3, 2025	Transfer Property
#BEN-16	JOHN BUENAVENTURA VILLARUZ	09158751248	N/A Site N/A, Phase N/A, Block N/A, Lot N/A	Dec 3, 2025	Transfer Property

Show 10 entries Showing 1 to 4 of 4 entries

Previous 1 Next

a. Click the Transfer Property button.

5.4 Transfer Property to Another Beneficiary

ISRMS

Manage Property Transfers

Transfer Property to Another Beneficiary

From Beneficiary (Current Owner)

Name: PABLITO CONSTANTINO DOROMAL JR. Contact: 09924341723

Current Property: Villa Victorias - Site 3, Phase 1, Block 2, Lot 9

To Beneficiary (New Owner)

Search Beneficiary *

Type to search by name or contact number...

Select New Beneficiary *

Use the search box above to find a beneficiary

Reason for Transfer *

Enter the reason for this property transfer...



ISRMS

Beneficiary Verification

Management

Housing Applications

Relocation Assignment

Utilities

Verifications

Property Transfers

User Management

Housing Monitoring

Records

Tools

To Beneficiary (New Owner)

Search Beneficiary *

Type to search by name or contact number...

Select New Beneficiary *

Type at least 2 characters to search

Reason for Transfer *

Enter the reason for this property transfer... Please fill out this field.

Additional Remarks (Optional)

c → Any additional notes or comments...

Cancel Transfer Property ← d

MA JOSEFA VAFLOP PADERNAL

All Status

ED DATE ACTIONS

2025 Transfer Property

2025 Transfer Property

2025 Transfer Property

2025 Transfer Property

Previous 1 Next

- Search and select the Beneficiary to transfer the property.
- Enter the Reason for Transfer.
- Add Remarks if applicable.
- Click the Transfer Property button again to save the changes.



6. User Management

ISRMS

Beneficiary Verification

Management

Housing Applications

Relocation Assignment

Utilities

Verifications

Property Transfers

a → User Management

Housing Monitoring

Records

Tools

ISRMS

MA JOSEFA VAFLOP PADERNAL

User Management

Manage system users and their roles

b → [Add New User](#)

Search by name, username, or email...

All Roles All Status

USER	USERNAME	EMAIL	ROLE	STATUS	REGISTER	
J	JASMIN FLORES BAGAHANSOL	jassy	jfbagahansol.chmsu@gmail.com	User	Active	12/4/2025
B	BRUCE LAURILLA VILLACERAN	bruce	brucevillaceran@gmail.com	User	Active	12/4/2025
J	JOHN PAUL COLIPAPA ALAMIS	It'sme_Jp	jcalamis.chmsu@gmail.com	User	Active	12/4/2025
V	VICTORIAS CITY HOUSING	user	chsud.victorias.demo@gmail.com	User	Active	12/4/2025
M	MA JOSEFA VAFLOP PADERNAL	admin	vchousing2022@outlook.com	Admin	Active	12/4/2025

ISRMS

Dashboard

Application Services

Management

Housing Applications

Relocation Assignment

Utilities

Verifications

Property Transfers

User Management

Housing Monitoring

Records

User Management

Manage system users and their roles

[Add New User](#)

Search by name, username, or email...

All Roles All Status

	USERNAME	EMAIL	ROLE	STATUS	REGISTERED	ACTIONS
J	JASMIN FLORES BAGAHANSOL	jassy	jfbagahansol.chmsu@gmail.com	User	Active	12/4/2025 c → [Edit] [Delete]
B	BRUCE LAURILLA VILLACERAN	bruce	brucevillaceran@gmail.com	User	Active	12/4/2025 [Edit] [Delete]
J	JOHN PAUL COLIPAPA ALAMIS	It'sme_Jp	jcalamis.chmsu@gmail.com	User	Active	12/4/2025 [Edit] [Delete]
V	VICTORIAS CITY HOUSING	user	chsud.victorias.demo@gmail.com	User	Active	12/4/2025 [Edit] [Delete]
M	MA JOSEFA VAFLOP PADERNAL	admin	vchousing2022@outlook.com	Admin	Active	12/4/2025 [Edit] [Delete]

- Navigate to User Management.
- Click the Add button to add a new user.
- Click the Edit icon to modify an existing user.



6.1 Add New User Account

The screenshot shows the 'Add New User' modal form in the ISRMS system. The form is titled 'Add New User' and has a close button (X) in the top right corner. It contains the following fields:

- First Name * (text input)
- Middle Name (text input)
- Last Name * (text input)
- Suffix (text input with placeholder 'Jr., Sr., III, etc.')
- Username * (text input)
- Email * (text input)
- Password * (text input)
- Confirm Password * (text input)
- Role * (dropdown menu with 'User' selected)

At the bottom of the modal, there are two buttons: 'Cancel' and 'Add User'. A red arrow labeled 'c' points to the 'Add User' button. In the background, the 'User Management' table is visible, showing a list of users with columns for USER, ROLE, STATUS, and REGISTER. A red arrow labeled 'a' points to the 'First Name' field, and a red arrow labeled 'b' points to the 'Role' dropdown menu.

- Fill in all required input details.
- Select the Role of the user.
- Click the Add User button to save the changes.

6.2 Edit Existing User Account

The screenshot shows the 'Edit User' modal form in the ISRMS system. The form is titled 'Edit User' and has a close button (X) in the top right corner. It contains the following fields:

- First Name * (text input with value 'JASMIN')
- Middle Name (text input with value 'FLORES')
- Last Name * (text input with value 'BAGAHANSOL')
- Suffix (text input with placeholder 'Jr., Sr., III, etc.')
- Username * (text input with value 'jassy')
- Email * (text input with value 'jfbagahansol.chmsu@gmail.com')
- New Password (text input with placeholder 'Leave blank to keep current')
- Role * (dropdown menu)

At the bottom of the modal, there are two buttons: 'Cancel' and 'Update User'. A red arrow labeled 'c' points to the 'Update User' button. In the background, the 'User Management' table is visible, showing a list of users with columns for USER, ROLE, STATUS, and REGISTER. A red arrow labeled 'a' points to the 'Unlock account if locked' checkbox, and a red arrow labeled 'b' points to the 'Update User' button.

- Click the Checkbox if the account is locked.
- Click the Update button to save the changes.



7. Housing Monitoring Mangement

ISRMS

Beneficiary Verification

Management

- Housing Applications
- Relocation Assignment
- Utilities
- Verifications
- Property Transfers
- User Management

a → **Housing Monitoring**

Records

Tools

Housing Unit Monitoring Management

Review and verify beneficiary monitoring submissions

b → **Overdue Beneficiaries (0)**

Total Submissions: 9

Active Beneficiaries: 3

Pending Review: 9

Verified: 0

Active Beneficiaries: 0

Overdue: 0

Search: Name, email, phone...

Status: All Status

Active Beneficiary: All

Filter

SUBMISSION	BENEFICIARY	PROPERTY	PHOTO	LOCATION	STATUS	LAST UPDATE	ACTIONS
Dec 07, 2025 05:51 AM 9 days ago	VICTORIAS CITY HOUSING choud.victorias.demo@gmail.com 09567809659	Villa Victorias Site 3, Ph 1, Blk 2, Lot 20	Monitorin Photo	10.9062, 123.0729 Verified	Pending	9 days	c → Review
Dec 04, 2025 11:58 PM 12 days ago	VICTORIAS CITY HOUSING choud.victorias.demo@gmail.com 09567809659	Villa Victorias Site 3, Ph 1, Blk 2, Lot 20	Monitorin Photo	10.9058, 123.0732 Verified	Pending	9 days	Review
Dec 04, 2025 11:58 PM 12 days ago	VICTORIAS CITY HOUSING choud.victorias.demo@gmail.com 09567809659	Villa Victorias Site 3, Ph 1, Blk 2, Lot 20	Monitorin Photo	10.9058, 123.0732 Verified	Pending	9 days	Review

- Navigate to Housing Monitoring.
- Click the Overdue button to view overdue items.
- Click the Review button to see detailed information.



7.1 Review Monitoring Submission

Review Monitoring Submission

Beneficiary Information

Name: VICTORIAS CITY HOUSING Contact: 09567809659
Email: chsud.victorias.demo@gmail.com Property: Villa Victorias
Location: Site 3, Phase 1, Block 2, Lot 20

Submission Details

Submitted: Dec 07, 2025 05:51 AM Status: Pending

Housing Unit Photo

a → Monitoring Photo

GPS Location

Coordinates: 10.906212, 123.072915 ← b
Verified: Yes

Review Action

Submission Status: Pending c →
d → ☐ Mark as Active Beneficiary
Check this to mark the beneficiary as an active beneficiary (in good standing)
Admin Remarks: e → Add remarks about this submission...
f → Save Changes

- Click the Monitoring Photo to view the image.
- Click the Icon to view the GPS location.
- Update the Submission Status as needed.
- Check the Mark as Active Beneficiary checkbox if applicable.
- Enter any Remarks.
- Click the Save Changes button to save all updates.



8. Beneficiaries Records

Beneficiaries

Search by name or ID

dd/mm/yyyy dd/mm/yyyy Filter

ID	NAME	LOCATION	CONTACT	DATE RELOCATED	ACTIONS
16	JOHN BUENAVENTURA VILLARUZ	Site , Phase , Block , Lot	09158751248	Dec 03, 2025	View Delete
17	PABLITO CONSTANTINO DOROMAL JR.	Site 3, Phase 1, Block 2, Lot 9	09924341723	Dec 03, 2025	View Delete
18	MHIEL TOMELDEN PORQUEZ	Site 3, Phase 1, Block 2, Lot 16	09567809659	Dec 03, 2025	View Delete
19	VICTORIAS CITY HOUSING	Site 3, Phase 1, Block 2, Lot 20	09567809659	Dec 04, 2025	View Delete

a. Navigate to Beneficiaries

b. Click View Text to see detailed information.

8.1 Beneficiary Details

Beneficiary Details

Full Name: JOHN BUENAVENTURA VILLARUZ
Contact Number: 09158751248
Region: NEGROS ISLAND REGION
Province: NEGROS OCCIDENTAL
City/Municipality: VICTORIAS CITY
Barangay: BARANGAY 13
Complete Address: STREET 1
Site Number: null
Phase Number: null
Block Number: null
Lot Number: null
Date Relocated: 12/3/2025

[Generate Certificate of Lot Assignment](#)

a. Click Generate Certificate of Lot Assignment to create the certificate.



9. Housing Units Records

Add Housing Unit

Region: NEGROS ISLAND REGION Province: NEGROS OCCIDENTAL

City/Municipality: VICTORIAS CITY Barangay: BARANGAY 13

Name: Total Units:

Add Unit

Search by name or location

E	REGION	PROVINCE	CITY/MUNICIPALITY	BARANGAY	TOTAL UNITS	AVAILABLE UNITS	ACTIONS
Victorias	NEGROS ISLAND REGION	NEGROS OCCIDENTAL	VICTORIAS CITY	BARANGAY 13	10	6	Manage Lots Edit Delete

- Navigate to Housing Units.
- Fill in all required input details.
- Click Add Unit to create a new housing unit.
- Click Manage Lots to manage individual lots within the unit.
- Click Edit to modify an existing housing unit.

9.1 Edit Housing Unit

Edit Housing Unit

Region: NEGROS ISLAND REGION Province: NEGROS OCCIDENTAL

City/Municipality: VICTORIAS CITY Barangay: BARANGAY 13

Name: VILLA VICTORIAS Total Units: 10

Save

- Fill in all required input details.
- Click the Save button to save all updates.



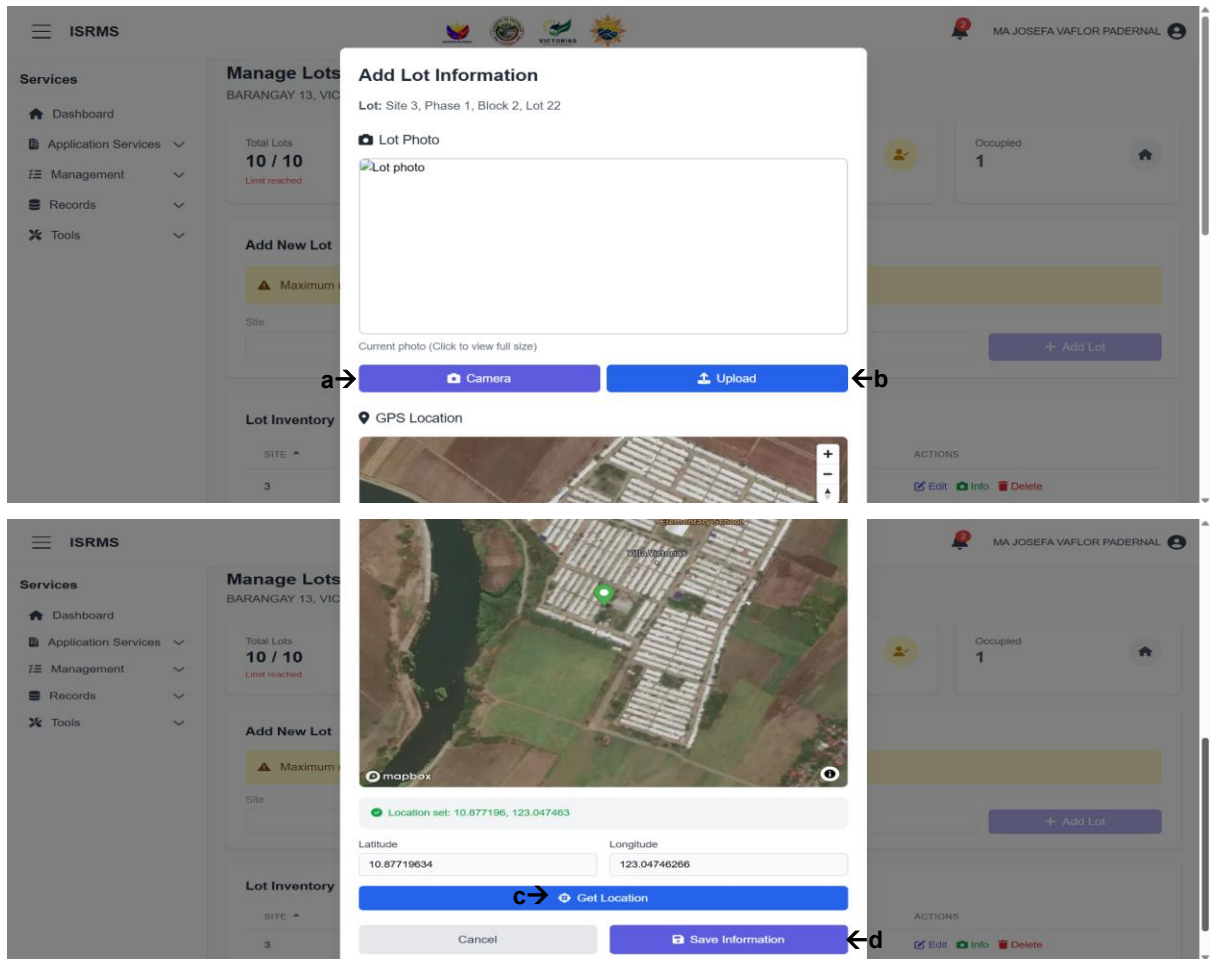
9.2 Manage Lots

- Fill in all required input details.
- Click the Add Lot button to save the new lot.
- Click Edit Text to modify an existing lot.
- Click Info to view detailed lot information.

9.2 Edit Lot

- Fill in all required input details.
- Click the Save button to save changes.

9.3 Add Lot Information



- a. Click Camera to take a photo of the lot.
- b. Click Upload to upload a photo of the lot.
- c. Click Get Location to automatically capture the lot's location.
- d. Click Save Information to save all changes.



10. Visitor Logs Records

- Navigate to Visitor Logs.
- Click the Add Visitor Log button to create a new entry.
- Click the Eye Icon to view visitor details.

10.1 Add Visitor Log

- Fill in all required input details.
- Click Start Camera to begin facial recognition.
- Click Take Photo to capture the visitor's image.
- Click Save Visitor Log to save the entry.



11. Property Transfer Records

ISRMS

Services

- Dashboard
- Application Services
- Management
- Records
 - Beneficiaries
 - Housing Units
 - Visitor Logs
 - a → Property Transfer Records**
- Tools

Property Transfer Records

View all property transfer history

Total Transfers: 1

This Month: 1

This Year: 1

Search by beneficiary name, housing unit...

dd/mm/yyyy dd/mm/yyyy

TRANSFER ID	FROM BENEFICIARY	TO BENEFICIARY	PROPERTY DETAILS	TRANSFER DATE	ACTIONS
#PTR-1	JOHN BUENAVENTURA VILLARUZ 09158751248	PABLITO CONSTANTINO DOROMAL JR. 09924341723	Villa Victorias Site 3, Phase 1, Block 2, Lot 9	Dec 3, 2025, 03:00 AM	View ← b

Property Transfer Details

Transfer Information

Transfer ID: #PTR-1

Transfer Date: Dec 3, 2025, 03:00 AM

Processed By: ROVIE ALFRED DOROMAL

Property Details

Housing Unit: Villa Victorias

Location: BARANGAY 13, VICTORIAS CITY, NEGROS OCCIDENTAL, NEGROS ISLAND REGION

Site/Phase/Block/Lot: Site 3, Phase 1, Block 2, Lot 9

From Beneficiary

Name: JOHN BUENAVENTURA VILLARUZ

Contact Number: 09158751248

To Beneficiary

Name: PABLITO CONSTANTINO DOROMAL JR.

Contact Number: 09924341723

Reason for Transfer

sample

Remarks

sample

a. Navigate to Property Transfer Records.

b. Click the Eye Icon to view detailed information.



12. Map

The interface displays a map of the area around Talisay, Negros Occidental, Philippines. The map is titled "Housing Units & Applications Map". The left sidebar contains a "Services" menu with options: Dashboard, Application Services, Management, Records, Beneficiaries, Housing Units, Visitor Logs, Property Transfer Records, Tools, and Map (selected). The top right shows the user "MA JOSEFA VAFLOP PADERNAL".

Top Screenshot: The map shows a tooltip for "VICTORIAS HOUSING" with details: Status: Relocated, Location: BARANGAY 13, VICTORIAS CITY, NEGROS OCCIDENTAL, Philippines. The right sidebar shows "Lot Information" for "Villa Victorias", which is "Available". It includes a "Lot photo" placeholder and "Lot Details" for "Site 3, Phase 1, Block 3, Lot 4". The location is "BARANGAY 13, VICTORIAS CITY, NEGROS OCCIDENTAL, NEGROS ISLAND REGION". GPS coordinates are provided.

Bottom Screenshot: The map shows a "Housing Overview" panel with "Total Capacity 30 Lots" and "Available Inventory 26 Units". It includes an "Occupancy Distribution" bar chart showing 13% utilized, with 1 occupied, 3 assigned, and 7 available units. The "Map Guide & Legend" panel provides instructions for map markers, interactive markers, map views, layer controls, and boundary management.

- Navigate to Map.
- Use Filters to identify specific housing units and applicants.
- Choose between Standard and Satellite views based on preference.
- Use the Mouse Scroll to zoom in and out and navigate within the map.



13. Reports

ISRMS

Services

- Dashboard
- Application Services
- Management
- Records
 - Beneficiaries
 - Housing Units
 - Visitor Logs
 - Property Transfer Records
- Tools
- Map
- Reports**

Reports

Generate Report
Create custom reports based on service type, project, and date range.

Report Name:

Report Type:

Project Filter:

Export Format:

Start Date:

End Date:

c→

Generated Reports
View and download previously generated reports.

REPORT NAME	TYPE	FORMAT	DATE GENERATED	ACTION
SAMPLE REPORT	Pag-IBIG Applications	PDF	Dec 03, 2025 04:39 AM	d→ Download Delete
sample	Property Transfers	PDF	Dec 03, 2025 03:01 AM	Download Delete
SAMPLE 2	Non-PagIBIG Applications	PDF	Dec 02, 2025 06:39 PM	Download Delete
SAMPLE	Pag-IBIG Applications	PDF	Dec 02, 2025 06:39 PM	Download Delete

a. Navigate to Reports.

b. Fill in all required input details.

c. Click the Generate Report button to create a specific type of report.

d. Click Download Text to download the generated report.



“Appendix M”

Curriculum Vitae



JOHN PAUL C. ALAMIS

📍 **Address:** Doña Enrica Street, Brgy. Zone 4-A, Talisay City,
Negros Occidental

📞 **Contact Number:** 09984101590

✉ **Email:** johnpaulalamis587@gmail.com

Summary of Qualifications

An Information Systems student with a strong academic record as a Dean's Lister and hands-on experience in system documentation and quality assurance. Skilled in analyzing system requirements, creating detailed documentation, and supporting team coordination to ensure smooth project operations. Collaborative and detail-oriented, with a focus on delivering accurate and efficient solutions while maintaining compliance with project standards.

Skills

- Critical thinking
- Time management
- Effective communication & Leading
- MS Office Suite
- Recording and documentation
- Programming – PHP, MYSQL, HTML, CSS,

References

- 👤 **Melanie Porquez, PhD**
College of Computer Studies
e-mail: melanie.porquez@chmsu.edu.ph

Education

Carlos Hilado Memorial State University **2022 - 2026**
Bachelor of Science in Information Systems
Domingo Lizares Street, Talisay City, Negros Occidental

- 4th Year Student
- Dean's Lister

Experiences

Documentation and Support

- Developed project and system documentation while offering technical and user support to ensure smooth system operation and team coordination.

Quality Assurance

- Collaborated with developers in analyzing system requirements, creating documentation, and executing quality assurance tests to verify system performance and compliance with project standards.

Projects

Informal Settler Relocation Management System

- Created a management system related to the process of transferring informal settlers for the City Human Settlements and Urban Development Office in our local government of Victorias City.

Barangay Help App

- Developed and implemented core features of a mobile reporting system, including user interfaces, data handling, and real-time notification, to streamline the communication between residents and officials.

Certifications

Cybersecurity Foundation	Jan 2025
MySQL database	Jan 2025
Business Analysis & Process Management	Nov 2024



ROVIE ALFRED S. DOROMAL

📍 **Address:** Malihao Street, Estrella Village, Brgy. 6,

Victorias City, Negros Occidental

☎ **Contact Number:** 09606186221

✉ **Email:** roviedoromal21703@gmail.com

Summary of Qualifications

An Information Systems student with a strong academic foundation on technical and documentation experience. Passionate about creating and delivering exceptional systems. Adept in collaborating with colleagues, adaptable to dynamic work environments, eager to apply professional skills to meet organizational needs, and committed to embracing learning opportunities.

Skills

- Critical thinking
- Time management
- Effective communication & Leading
- MS Office Suite
- Recording and documentation
- Programming – PHP, MYSQL, Python, HTML, CSS, JAVASCRIPT, DJANGO
- OS – Windows, Linux (Ubuntu, Arch)
- Bash Scripting

References

👤 **Melanie Porquez, PhD**
College of Computer Studies
e-mail: melanie.porquez@chmsu.edu.ph

Education

Carlos Hilado Memorial State University **2022 - 2026**
Bachelor of Science in Information Systems
Domingo Lizares Street, Talisay City, Negros Occidental

- 4th Year Student
- Dean's Lister

Experiences

Programming Assistant

- Collaborated with faculty members at my university to design and program a Sentiment Analysis System tailored to assess sentiment feedback from students for curriculum enhancement.

Projects

Informal Settler Relocation Management System

- Created a management system related to the process of transferring informal settlers for the City Human Settlements and Urban Development Office in our local government of Victorias City.

Domestic Worker Incident Reporting System

- Created a reporting system for domestic workers with live chat and video call functionality using the Django framework.

Pet Health Monitoring System with Predictive Analytics

- Created a pet health system with inventory management and predictive analytics using SARIMA.

Sentiment Analysis System with Distil BERT

- Created a system that analyzes sentiment data from a spreadsheet and is integrated with Distil BERT AI model from HuggingFace.

Certifications

Cybersecurity Essentials **Jan 2025**
The Linux Foundation

Cloud Computing **Jan 2025**
Cybrary

Cryptography **Jan 2025**
Cybrary



JOHN EDWIN B. VILLARUZ

📍 **Address:** Carmela Valley Executive Village, Zone 12-A

Talisay City, Negros Occidental

📞 **Contact Number:** 09158751248

✉ **Email:** johnedwin.villaruz@chmsc.edu.ph

Summary of Qualifications

An Information Systems student with a standard academic foundation and practical exposure to system analysis, documentation, and quality assurance. Has contributed to the development and validation of business and government-oriented information systems, ensuring functionality and alignment with project requirements. Adaptable, collaborative, and motivated to continuously improve through hands-on experience in real-world system

Skills

- Planning
- Critical Thinking
- Problem Solving
- Documentation
- Communication
- Adaptability
- Testing
- Programming – PHP, MYSQL, HTML, CSS, JAVASCRIPT

References

- 👤 **Melanie Porquez, PhD**
College of Computer Studies
e-mail:
melanie.porquez@chmsu.edu.ph

Education

Carlos Hilado Memorial State University **2022 - 2026**
Bachelor of Science in Information Systems
Domingo Lizares Street, Talisay City, Negros Occidental

- 4th Year Student

Experiences

System Analyst & Quality Assurance

- Collaborated with the development team to perform requirements analysis, system documentation, and quality assurance testing, ensuring system functionality and compliance with defined requirements.

Projects

Informal Settler Relocation Management System

- Conducted requirements analysis, documentation, and system validation for a local government relocation management system supporting the City Human Settlements and Urban Development Office of Victorias City.

Animal Supplies Sales and Stocks Management System for Smarter Business Decision

- Designed and developed sales and inventory management system to support data-driven business decisions for a local business in Talisay City.

Certifications

BUS250 **Nov 2025**
Introduction to Business Intelligence and Analytics

CS105 **Oct 2025**
Introduction to Python

Intellectual Property Management Office **Feb 2025**
IP Awareness and Patent Search Training for Information System Students